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A COMPUTER CODE FOR THREE-DIMENSIONAL INCOMPRESSIBLE FLOWS USING NONORTHOGONAL BODY-FITTED COORDINATE SYSTEMS

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16. ABSTRACT

In this report, a numerical method for solving the equations of motion of three-dimensional incompressible flows in nonorthogonal body-fitted coordinate (BFC) systems has been developed. The equations of motion are transformed to a generalized curvilinear coordinate system from which the transformed equations are discretized using finite difference approximations in the transformed domain. The hybrid scheme is used to approximate the convection terms in the governing equations. Solutions of the finite difference equations are obtained iteratively by using a pressure-velocity correction algorithm (SIMPLE-C). Numerical examples of two- and three-dimensional, laminar and turbulent flow problems are employed to evaluate the accuracy and efficiency of the present computer code. The user's guide and computer program listing of the present code are also included.

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NOMENCLATURE

Symbol	Definition
A	link coefficient of finite difference equation
A_p	link coefficient of the node at the center of a control volume
A^u	link coefficient for the u-equation
A^v	link coefficient for the v-equation
A^w	link coefficient for the w-equation
A^o	link coefficient for time marching scheme
C_v	specific heat constant at constant volume
C_1	turbulence model constant, = 1.44
C_2	turbulence model constant, = 1.92
C_μ	turbulence model constant, = 0.09
D	diffusion coefficient for the pressure correction equation
e	internal energy per unit mass (jour/kg)
J	Jacobian of the metric transformation
k	turbulence kinetic energy (m^2/S^2)
k'	thermal conductivity of the fluid
P	pressure in the fluid (N/m^2)
P_r	production term for the turbulent kinetic energy
Q	energy added per unit volume ($jour/m^3$)
S	source term
S_u	source term of the u-equation
S_v	source term of the v-equation
S_w	source term of the w-equation
T	temperature (°K)
t	time (sec)
u	velocity in x direction

v	velocity in y direction
w	velocity in z direction
X	X-coordinate (m)
Y	Y-coordinate (m)
Z	Z-coordinate (m)

Greek

Γ	diffusion coefficient
ϵ	turbulent kinetic energy dissipation rate (m^2/s^3)
Δ	difference operator
ϕ	variable of general transport equation
ϕ^o	solution at the previous time level
σ_k	turbulence model constant, = 1.0
σ_ϵ	turbulence model constant, = 1.3
ξ	curvilinear coordinate
ζ	curvilinear coordinate
μ	molecular viscosity ($N\text{-}S/m^2$)
μ_t	turbulent eddy viscosity ($N\text{-}S/m^2$)
μ_{eff}	effective viscosity ($N\text{-}S/m^2$)
ρ	density (kg/m^3)
Σ_i	summation over all values around a grid node P
η	curvilinear coordinate

Subscript

i	index of all possible values
ref	reference value
max	maximum quantity

Superscript

o	previous time level solution
*	current solution
'	correction quantity

CONTRACTOR REPORT

A COMPUTER CODE FOR THREE-DIMENSIONAL INCOMPRESSIBLE FLOWS USING NONORTHOGONAL BODY-FITTED COORDINATE SYSTEMS

INTRODUCTION

With the currently increasing computer capability and various flow solvers developed, numerical simulations of three-dimensional incompressible flow problems using Reynolds-average Navier-Stokes equations are now becoming more feasible in many engineering design and analysis applications. In many real world flow problems, the boundary geometries are complex such that it is more accurate to describe the geometries using body-fitted coordinate (BFC) systems. Especially for internal flow problems with complex geometries such as those of the hot gas manifold (HGM) of the Space Shuttle Main Engine (SSME), the use of nonorthogonal BFC systems for numerical solutions can be beneficial in many aspects. It is not only the boundary geometries that can be represented more closely using BFC systems, but also grid-refined solutions can be obtained without increasing an excessive amount in computer memory. In addition, once a particular flow problem has been set up, the redesign or optimization process of the boundary shapes can be performed very easily using BFC systems.

Several numerical methods [1,2,3,4,5,6] has been developed for solving the incompressible Navier-Stokes equations in 3-D BFC systems. The main difference between these methods lies in the way of finding a pressure field such that the flowfield can be as close to divergence-free as possible (i.e. to satisfy the mass conservation equation). This is the main feature and difficulty of solving the incompressible flow problems. Numerical methods of References 1, 2 and 3, for instance, have employed the pseudocompressibility approach and time-iterative scheme to generate the pressure field so that the continuity equation is satisfied when a steady state solution is reached. In these methods, artificial smoothing techniques must be used to obtain a strong coupling between the velocity and pressure fields. Methods of References 4, 5 and 6, on the other hand, have utilized a successive pressure-velocity correction scheme by using a Poisson's equation for pressure correction derived approximately form the continuity and momentum equations. For these latter methods, grid staggering between the velocity vectors and the pressure nodes must be used to ensure stability of the numerical solutions.

There are several possible methods of grid staggering associated with different features in solving the pressure correction equation. These grid staggering methods were discussed in Reference 6, from which one of the methods was shown to be the most promising arrangement (i.e. with the velocity vectors located at the faces of a volume which contains the pressure and other scalars at its center). But, this method has one drawback, that the velocity components are solved using different control volumes. It is for this reason that a grid staggering system similar to the one used by Vanka et al. [4] is developed in the present study. The present method of grid staggering and pressure correction equation that was described by Vanka [4[and Maliska [6]. Also, using the present method, the same control volume is used for the velocity components and scalar quantities.

In the following sections, basic elements for establishing the present computer code for solving the curvilinear Navier-Stokes equations in three-dimensional space (CNS3D) will be described. These are followed by a series of standard numerical examples used to evaluate the accuracy and efficiency of the present numerical method. The numerical examples include laminar flow driven-cavity problem, cases of laminar and turbulent flows over backward-facing steps, and 3-D laminar flows inside a 90-deg-bend square duct. Applications of the present code to the internal flow problems of SSME will be included in future publications.

A user's guide to the present CNS3D code is provided in Appendix A. Appendix B contains a list and definitions of all the major fortran symbols used in the computer program which is listed in Appendix C.

TRANSFORMATION OF THE EQUATIONS OF MOTION

For incompressible Newtonian fluid, the continuity, momentum and energy equations can be written as:

$$U_t + E_x + F_y + G_z = S \tag{1}$$

where (x,y,z) represent the Cartesian coordinates, and

$$U = \begin{Bmatrix} \rho \\ \rho u \\ \rho v \\ \rho w \\ \rho e - Q \end{Bmatrix} \qquad E = \begin{Bmatrix} \rho u \\ \rho uu - \mu u_x \\ \rho uv - \mu v_x \\ \rho uw - \mu w_x \\ \rho ue - k' T_x \end{Bmatrix}$$

$$F = \begin{Bmatrix} \rho v \\ \rho vu - \mu u_y \\ \rho vu - \mu v_y \\ \rho vw - \mu w_y \\ \rho ve - k' T_y \end{Bmatrix} \qquad G = \begin{Bmatrix} \rho w \\ \rho wu - \mu u_z \\ \rho wv - \mu w_z \\ \rho ww - \mu w_z \\ \rho we - k' T_z \end{Bmatrix}$$

$$S = \begin{Bmatrix} 0 \\ (\mu u_x)_x + (\mu v_x)_y + (\mu w_x)_z - P_x \\ (\mu u_y)_x + (\mu v_y)_y + (\mu w_y)_z - P_y \\ (\mu u_z)_x + (\mu v_z)_y + (\mu w_z)_z - P_z \\ \mu[2(u_x^2 + v_y^2 + w_z^2) + (v_x + u_y)^2 + (w_y + v_z)^2 + (u_z + w_x)^2 \\ \qquad - \frac{2}{3}(u_x + v_y + w_z)^2] \end{Bmatrix}$$

e = the internal energy per unit mass = $C_v T$ for perfect gas

Q = energy added per unit volume

k' = thermal conductivity of the fluid .

Equation (1) is transformed to a general curvilinear coordinate system (ξ, η, ζ), which results in equation (2).

$$U_t + E_\xi \xi_x + E_\eta \eta_x + E_\zeta \zeta_x + F_\xi \xi_y + F_\eta \eta_y + F_\zeta \zeta_y + G_\xi \xi_z + G_\eta \eta_z + G_\zeta \zeta_z = S \qquad (2)$$

where

$$\xi_x = J(y_\eta z_\zeta - y_\zeta z_\eta)$$

$$\xi_y = -J(x_\eta z_\zeta - x_\zeta z_\eta)$$

$$\xi_z = J(x_\eta y_\zeta - x_\zeta y_\eta)$$

$$\eta_x = -J(y_\xi z_\zeta - y_\zeta z_\xi)$$

$$\eta_y = J(x_\xi z_\zeta - x_\zeta z_\xi)$$

$$\eta_z = -J(x_\xi y_\zeta - x_\zeta y_\xi)$$

$$\zeta_x = J(y_\xi z_\eta - y_\eta z_\xi)$$

$$\zeta_y = -J(x_\xi z_\eta - x_\eta z_\xi)$$

$$\zeta_z = J(x_\xi y_\eta - x_\eta y_\xi)$$

$$J = 1/[x_\xi(y_\eta z_\zeta - y_\zeta z_\eta) - x_\eta(y_\xi z_\zeta - y_\zeta z_\xi) + x_\zeta(y_\xi z_\eta - y_\eta z_\xi)] \quad .$$

The transformation coefficients, ξ_x, ξ_y, ξ_z, η_x, η_y, η_z, ζ_x, ζ_y, and ζ_z, are computed numerically using second order central differencing. In the transformed domain, the grid sizes (i.e., $\Delta\xi$, $\Delta\eta$, and $\Delta\zeta$) are set to be unity. This simplifies the calculation of the transformation coefficients.

For turbulent flow computations, the present code has employed the standard k-ε turbulence model [7] to provide the turbulent eddy viscosity μ_t. The standard k-ε turbulence model (which consists of a turbulent kinetic energy equation, k-equation, and a turbulent kinetic energy dissipation rate equation, ε-equation) is given as:

$$(\rho k)_t + \left(\rho u_i k - \frac{\mu_{eff}}{\sigma_k} k_{x_i}\right)_{x_i} = \rho(P_r - \varepsilon) \tag{3}$$

$$(\rho \varepsilon)_t + \left(\rho u_i k - \frac{\mu_{eff}}{\sigma_\varepsilon} \varepsilon_{x_i}\right)_{x_i} = \rho \frac{\varepsilon}{k} (C_1 P_r - C_2 \varepsilon) \tag{4}$$

where the effective viscosity μ_{eff} is calculated from:

$$\mu_{eff} = \mu + \mu_t = \mu + \rho C_\mu k^2/\varepsilon$$

and the turbulent kinetic energy production term, P_r, is defined as:

$$P_r = C_\mu \frac{k^2}{\varepsilon} [(u_y + v_x)^2 + (v_z + w_y)^2 + (w_x + u_z)^2 + 2(u_x^2 + v_y^2 + w_z^2)]$$

The turbulence model constants are:

$$C_\mu = 0.09 \quad , \quad \sigma_k = 1.0 \quad , \quad \sigma_\varepsilon = 1.3$$

$$C_1 = 1.44 \quad , \quad C_2 = 1.92 \quad .$$

Also, the molecular viscosity μ in equation (1) is replaced by the effective viscosity μ_{eff} for turbulent flow cases.

In order to save the computational efforts, the widely used wall function approach [8] is employed to provide the near wall boundary conditions for the momentum and energy equations and the k-ε turbulence model. This approach avoids the requirement of integrating the governing equations up to the wall which requires a large number of additional grid points near the wall.

Equations (2), (3), and (4) form a closed set of nonlinear partial differential equations governing the fluid motion. This set of equations are to be solved by means of finite difference approximations which are performed in the transformed domain. For treating the convection terms, the hybrid scheme [9] is employed for simplicity (although other more elaborate schemes such as central differencing plus artificial dissipation scheme, QUICK scheme, or skew upwind differencing scheme, etc. can be implemented [10]). These are described in the following sections.

DISCRETIZATION OF THE EQUATIONS OF MOTION

In this section, finite difference approximations are used to discretize the governing equations, equations (2), (3), and (4). Second-order central differencing is used for the diffusion terms and the source terms. The hybrid differencing scheme [9] is employed to approximate the convection terms in the governing equations. The finite difference discretizations are performed in the transformed domain. The solution procedure for the discretized equations using a velocity-pressure correction algorithm (SIMPLE-C) of References 11 and 12 will be described in the next section.

The governing equations of motion can be represented by the following model transport equation in which ϕ denotes all the dependent variables respectively and Γ is the diffusion coefficient..

$$
\begin{aligned}
&(\rho\phi)_t + [\rho u\phi - \Gamma(\phi_\xi\,\xi_x + \phi_\eta\,\eta_x + \phi_\zeta\,\zeta_x)]_\xi\,\xi_x \\
&\quad + [\rho u\phi - \Gamma(\phi_\xi\,\xi_x + \phi_\eta\,\eta_x + \phi_\zeta\,\zeta_x)]_\eta\,\eta_x \\
&\quad + [\rho u\phi - \Gamma(\phi_\xi\,\xi_x + \phi_\eta\,\eta_x + \phi_\zeta\,\zeta_x)]_\zeta\,\zeta_x \\
&\quad + [\rho v\phi - \Gamma(\phi_\xi\,\xi_y + \phi_\eta\,\eta_y + \phi_\zeta\,\zeta_y)]_\xi\,\xi_y \\
&\quad + [\rho v\phi - \Gamma(\phi_\xi\,\xi_y + \phi_\eta\,\eta_y + \phi_\zeta\,\zeta_y)]_\eta\,\eta_y \\
&\quad + [\rho v\phi - \Gamma(\phi_\xi\,\xi_y + \phi_\eta\,\eta_y + \phi_\zeta\,\zeta_y)]_\zeta\,\zeta_y \\
&\quad + [\rho w\phi - \Gamma(\phi_\xi\,\xi_z + \phi_\eta\,\eta_z + \phi_\zeta\,\zeta_z)]_\xi\,\xi_z \\
&\quad + [\rho w\phi - \Gamma(\phi_\xi\,\xi_z + \phi_\eta\,\eta_z + \phi_\zeta\,\zeta_z)]_\eta\,\eta_z \\
&\quad + [\rho w\phi - \Gamma(\phi_\xi\,\xi_z + \phi_\eta\,\eta_z + \phi_\zeta\,\zeta_z)]_\zeta\,\zeta_z = S \quad . \qquad (5)
\end{aligned}
$$

Discretization of equation (5) is performed using finite difference approximations in the transformed domain. The second order central differencing is used for approximating the diffusion terms. For the convection terms, the hybrid differencing scheme [9] is employed (i.e., using central differencing for cell Peclet number less than or equal to 2 and switching to upwind differencing when the cell Peclet number is greater than 2). The finite difference equation is arranged by collecting terms according to the grid nodes around a control volume as shown in Figure 1. The final expression is given by equation (6) in which A represents the link coefficients between grid nodes P, E, W, N, S, T, B, NE, NW, NT, NB, SE, SW, ST, SB, ET, EB, WT, and WB as shown in Figure 1.

$$A_P \phi_P = A_E \phi_E + A_W \phi_W + A_N \phi_N + A_S \phi_S + A_T \phi_T + A_B \phi_B + S_1 \qquad (6)$$

where

$$S_1 = S + A_P^\circ \phi_P^\circ + A_{NE} \phi_{NE} + A_{NW} \phi_{NW} + A_{NT} \phi_{NT} + A_{NB} \phi_{NB}$$
$$+ A_{SE} \phi_{SE} + A_{SW} \phi_{SW} + A_{ST} \phi_{ST} + A_{SB} \phi_{SB}$$
$$+ A_{ET} \phi_{ET} + A_{EB} \phi_{EB} + A_{WT} \phi_{WT} + A_{WB} \phi_{WB}$$

$$A_P = A_E + A_W + A_N + A_S + A_T + A_B + A_P^\circ$$

$$A_P^\circ = \rho_P^\circ / \Delta t \quad .$$

The subscript $^\circ$ denotes the solution at the previous time level. A fully implicit formulation is employed for solving the time dependent transient problems.

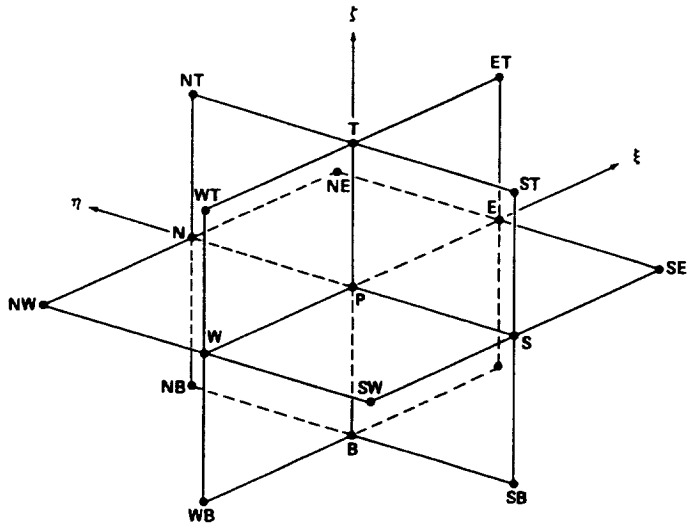


Figure 1. Three-dimensional grid structure and labeling around a grid node P.

Thus, the nonlinear equations of motion are approximated by a system of linear algebraic equations which have the form of equation (6). Only one program subroutine is designed to calculate the link coefficients and the source terms. The number of algebraic equations depends on the number of interior grid points. For a grid size of 10 x 10 x 10 the number of algebraic equations to be solved would be around 512. This large system of equations are preferred to be solved by some iterative methods, such as Gauss-Seidel iteration, line-underrelaxation method [13] or Stone's method [14], etc., rather than using direct methods such as Gaussian elimination method. Only a few (6 to 10) iterations through the whole computational domain are needed and a complete convergence of the system of algebraic equations is not required. Since equation (6) is only a linearized version of the governing equations which are nonlinear and coupled in nature, solutions of the equations of motion must be obtained through global iterations among the equations. A tentative solution to equation (6) will not affect the final results significantly. On the other hand, if too many iterations are used to get a better solution of equation (6), then a great deal of computing time would be virtually wasted. However, the above argument can not be applied when the pressure correction equation (which will be derived in the next section) is solved. Since during each global iteration it is desirable to retain a divergence-free velocity field, better solution of the pressure correction equation would in effect promote the convergence of the whole numerical scheme. Therefore, more iterations are usually used to solve the pressure correction equation.

SOLUTION PROCEDURES

The governing equations used in the present analysis are nonlinear and strongly coupled. Iterative procedures are employed to drive the equations to a converged solution. It is particularly important for incompressible flow to make the flow field satisfy the continuity equation and the momentum equations at the same time. This requires a correct pressure field associated with a divergence-free velocity field. A velocity-pressure correction procedure is developed in the present study to drive the pressure field and the velocity field to be divergence free. This kind of procedure requires grid staggering between the velocity components and the locations where the pressure is estimated and stored such that the velocity field and the pressure field will not be uncoupled.

In the present study, staggering grid systems as shown in Figure 2 (for 2-D case) are used. The velocity components, u and v, are solved and stored at the grid nodes and the pressure, p, is located at the corners of the control volume of u and v. In this way, solutions of u and v can be solved using the same control volume and coupling between u, v and p can also be enforced. To estimate the pressure field, a pressure correction equation is derived approximately from the discretized momentum and continuity equations. The velocity and pressure fields are then corrected using the solutions of the pressure correction equation.

First, the finite difference momentum equations (for u, v and w) can be written as:

$$A_P^u u_P^* = \sum_i A_i^u u_i^* = P_x^* + S_u \qquad (8a)$$

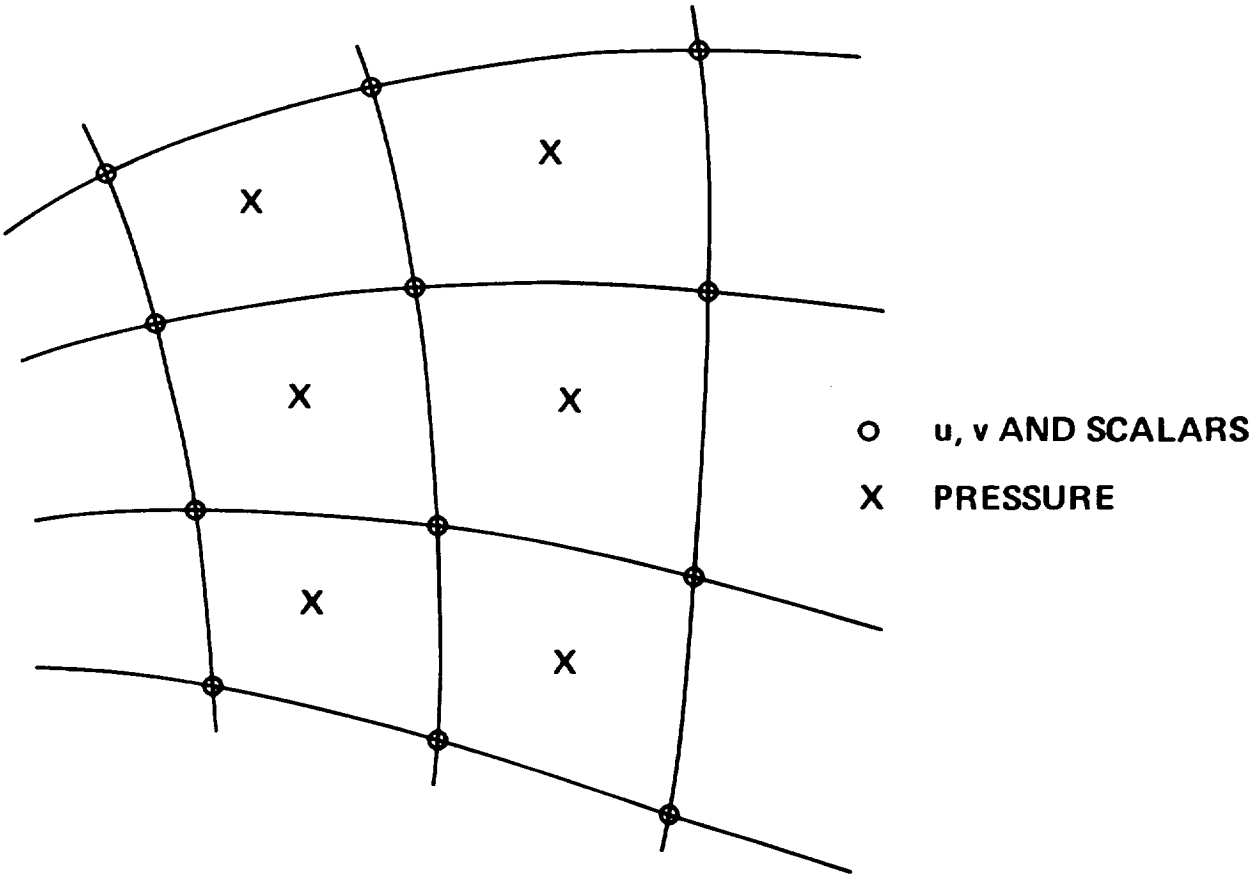


Figure 2. Locations where the variables are stored (staggering grids are used).

$$A_P^{\,v}\, v_P^{\,*} = \sum_i A_i^{\,v}\, v_i^{\,*} = P_y^{\,*} + S_v \tag{8b}$$

$$A_P^{\,w}\, w_P^{\,*} = \sum_i A_i^{\,w}\, w_i^{\,*} - P_z^{\,*} + S_w \tag{8c}$$

where u*, v*, w*, and p* represent the solutions of equations (8a) and (8b). To satisfy the continuity equation the velocities and pressure are corrected according to the following relations:

$$u = u^* + u' \tag{9a}$$

$$v = v^* + v' \tag{9b}$$

$$w = w^* + w' \tag{9c}$$

$$P = P^* + P' \tag{9d}$$

A new set of momentum equations can be constructed approximately using the divergence-free flow field, u, v, w, and p:

$$A_P^u u_P = \sum_i A_i^u u_i - P_x + S_u \tag{10a}$$

$$A_P^v v_P = \sum_i A_i^v v_i - P_y + S_v \tag{10b}$$

$$A_P^w w_P = \sum_i A_i^w w_i - P_z + S_w \tag{10c}$$

By subtracting equations (8a) through (8c) from equations (10a) through (10c), respectively, the following equations result:

$$A_P^u u_P' = \sum_i A_i^u u_i' - P_x' \tag{11a}$$

$$A_P^v v_P' = \sum_i A_i^v v_i' - P_y' \tag{11b}$$

$$A_P^w w_P' = \sum_i A_i^w w_i' - P_z' \tag{11c}$$

According to SIMPLE-C algorithm [11], equations (11a) through (11c) are rearranged to be:

$$(A_P^u - \sum_i A_i^u)\, u_P' = \sum_i A_i^u (u_i' - u_P') - P_x' \tag{12a}$$

$$(A_P^v - \sum_i A_i^v)\, v_P' = \sum_i A_i^v (v_i' - v_P') - P_y' \tag{12b}$$

$$(A_P^w - \sum_i A_i^w)\, w_P' = \sum_i A_i^w (w_i' - w_P') - P_z' \tag{12c}$$

The first terms on the right-hand side of equations (12a) through (12c) are neglected to simplify the formulation. Thus,

$$u_P' = -\left(\frac{1}{A_P^u - \sum_i A_i^u}\right) P_x' = -D_u P_x' \qquad (13a)$$

$$v_P' = -\left(\frac{1}{A_P^v - \sum_i A_i^v}\right) P_y' = -D_v P_y' \qquad (13b)$$

$$w_P' = -\left(\frac{1}{A_P^w - \sum_i A_i^w}\right) P_z' = -D_w P_z' \quad . \qquad (13c)$$

Using the decompositions of equations (9a) through (9c), the continuity equation can be written as:

$$u_x + v_y + w_z = (u_x^* + v_y^* + w_z^*) + (u_x' + v_y' + w_z') = 0 \quad . \qquad (14)$$

Substituting equations (13a) through (13c) into equation (14), the following pressure correction equation can be obtained:

$$-[(D_u P_x')_x + (D_v P_y')_y + (D_w P_z')_z] = -(u_x^* + v_y^* + w_z^*) \quad . \qquad (15)$$

Equation (15) is a Poisson's equation with the source term equal to the local divergence of the flow field. To enforce the coupling between the velocity and pressure fields, the source term of equation (15) is first evaluated at the control volumes centered between the velocity nodes as shown in Figure 3. An averaged source term is then calculated at the cell center of p node for solving equation (15). In this way, the difficulties in solving the pressure correction equation, as described by Vanka [4] and Maliska [6], are eliminated. Coupling between the velocity and the pressure field is also assured.

According to the above analyses, the present numerical method contains the following solution steps:

1) Guess initial velocity and pressure field.

2) Solve for the velocity field using equations (8a) through (8c).

3) Solve for other scalar transport equations.

4) Solve the pressure correction equation, equation (15).

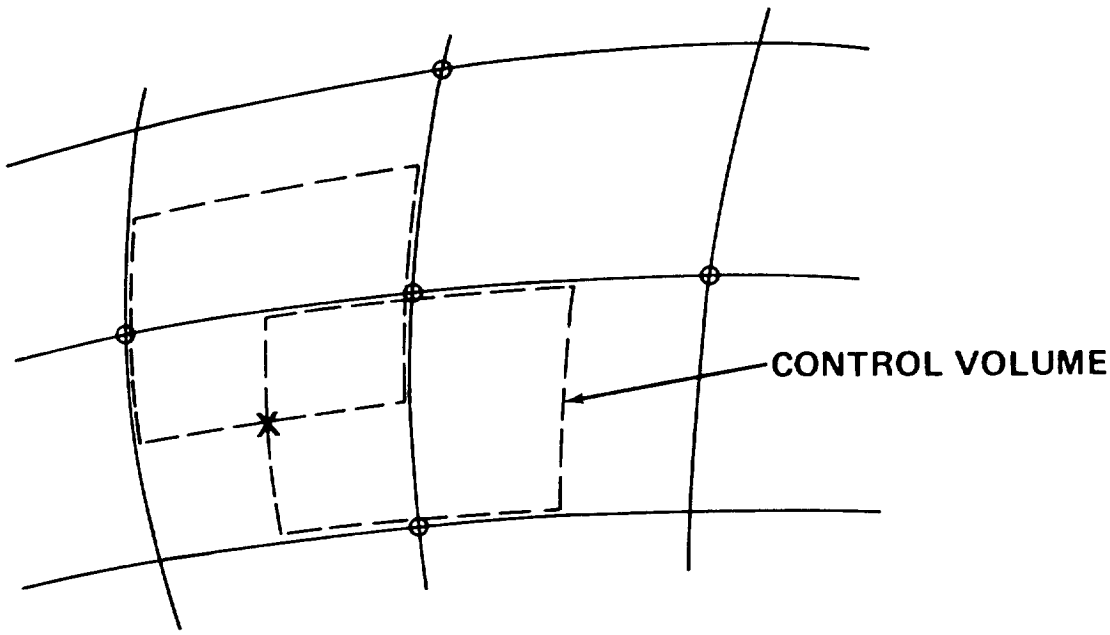


Figure 3. Control volumes where the mass conservation is evaluated for solving the pressure correction equation.

5) Correct the velocity and pressure fields using equations (13a) through (13c) and equation (9d).

6) Go back to step (2) until solution converges.

A converged solution is obtained when the following criterion is met:

$$\text{Error} \equiv \left(|\Delta u|_{max} + |\Delta v|_{max} + |\Delta w|_{max}\right) / U_{ref} + |P'|_{max}/\rho U_{ref}^2$$

$$\leq 3 \times 10^{-4} \quad ,$$

where Δu, Δv, and Δw represent velocity changes during each iteration due to the solutions of the momentum equations.

In solving the momentum equations in step (b) above, underrelaxation factor of about 0.6 is recommended. With this, A_P's in equations (8a) through (8c) are modified according to the underrelaxation factor. For the correction of velocity field, no underrelaxation is required. But the correction of pressure field should be underrelaxed slightly (around 0.9) when the grid nonorthogonality is strong. This is different from that suggested by References 11 and 12 (which recommend no underrelaxation for pressure correction).

NUMERICAL EXAMPLES

In this section, several numerical examples are employed to demonstrate the efficiency and accuracy of the present numerical method. To serve this purpose, 2-D and 3-D, laminar and turbulent flow cases are included. These cases are: (a) 2-D laminar driven square-cavity flows; (b) 2-D laminar flows over a backward-facing step; (c) 2-D turbulent flows over a backward-facing step; (d) 3-D developing laminar flow inside a 90-deg-bend square duct. Detailed descriptions and results of the computation of the above cases are included as follows.

A. 2-D Laminar Driven Square-Cavity Flows

The first test case is concerning laminar recirculating flows inside a square cavity. Only one side of the walls is moving at a constant speed tangent to that wall. This case has been studied extensively by Burggraf [15] and has often been used as one of the standard testing cases for numerical methods in solving the incompressible Navier-Stokes equations. Physical geometry and wall boundary conditions are illustrated in Figure 4. Reynolds number of the flow (based on the cavity size and the moving wall velocity) studied in the present analysis is 400. Two different mesh systems, as shown in Figure 5, are used to study the effect of grid non-orthogonality on the accuracy of the present method. The grid system of Figure 5(a) is uniform and orthogonal while the grid system of Figure 5(b) is non-uniform and non-orthogonal.

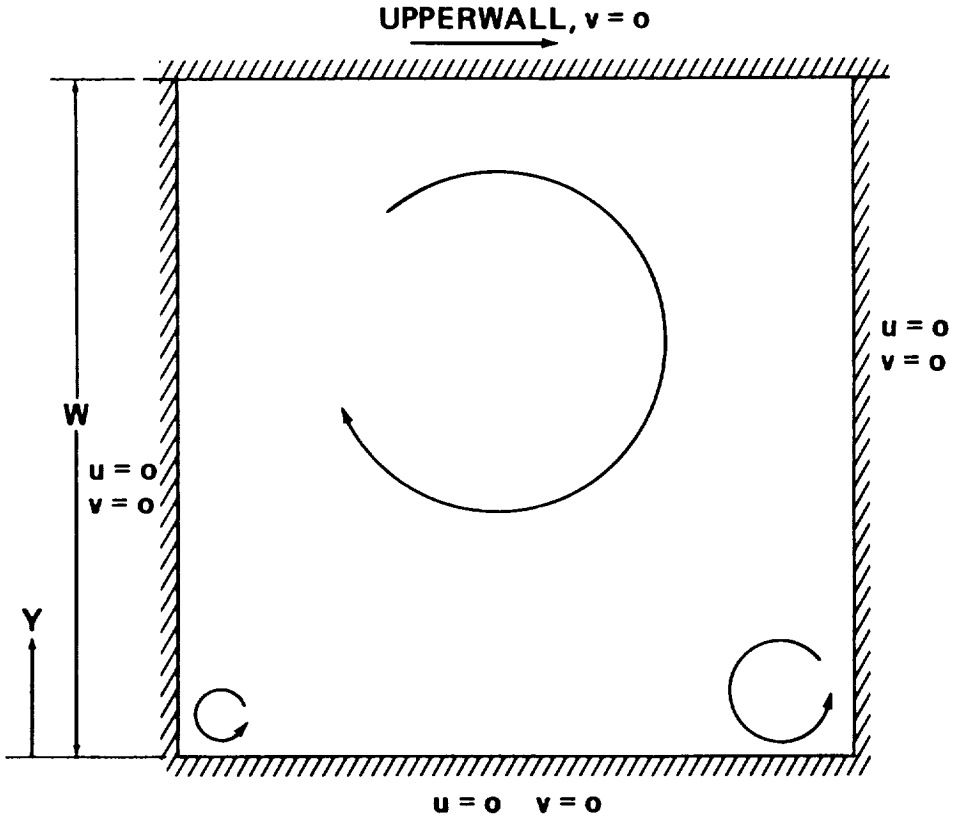


Figure 4. Physical geometry and wall boundary conditions for laminar flows inside a wall-driven square cavity.

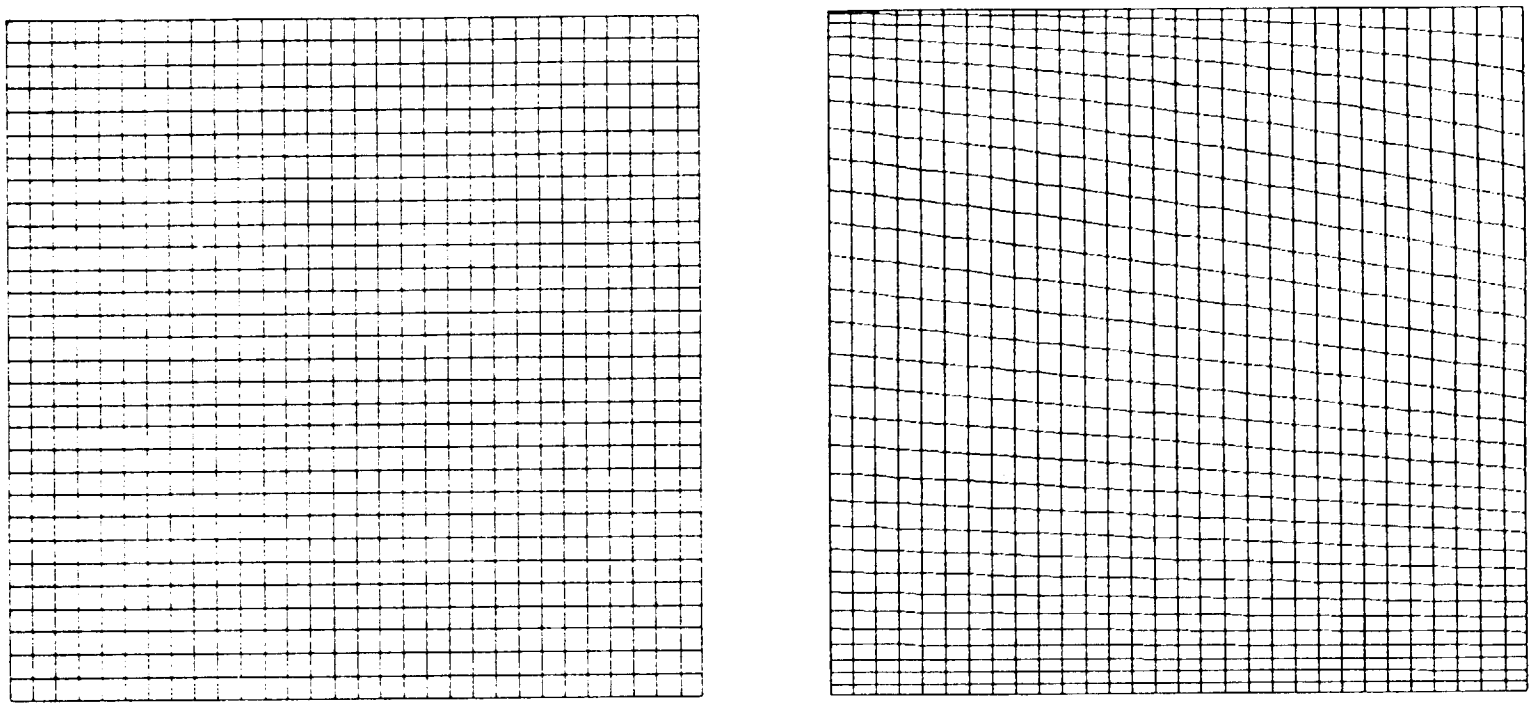


Figure 5. Mesh systems used for driven cavity problem. (a) Uniform and orthogonal grid. (b) Nonuniform and nonorthogonal grid.

Results of the computations are shown in Figures 6 and 7. Velocity vector plots of the predicted flow fields are compared in Figure 6 for the mesh systems shown in Figure 5. Detailed comparisons of the predicted velocity profiles along the mid-section of the cavity are illustrated in Figure 7. Predicted results of Burggraf [15] are also included. Good agreements between the present calculations and those

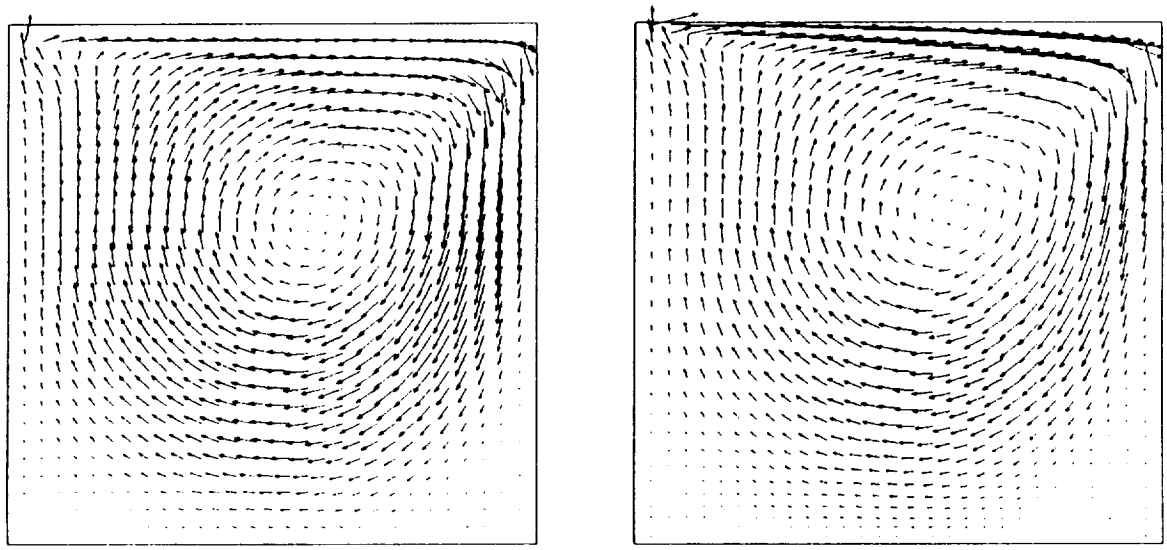


Figure 6. Velocity vector plots. (a) Orthogonal grid. (b) Nonorthogonal grid.

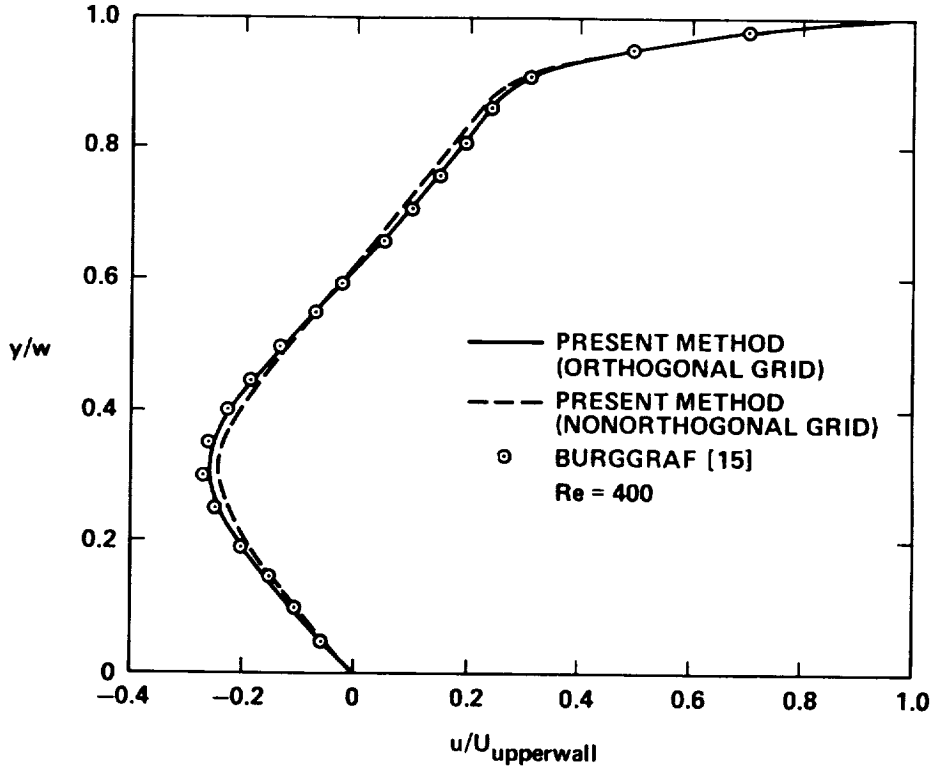


Figure 7. Comparisons of velocity profiles along the mid-section of the square cavity.

of Burggraf [15] are also included. Good agreements between the present calculations and those of Burggraf are shown in Figure 7. Discrepancies between the present predictions and Burggraf's results are mainly due to the hybrid differencing scheme used in the present method. The upwind part of the hybrid scheme produces large numerical diffusion which tends to reduce the strength of the vorticity inside the cavity. Effects of differencing schemes in approximating the convection terms on the predicted results will be studied in the next test case.

Convergence history of the computation of the present case using uniform grids is given in Figure 8 which shows that the present numerical method is quite different. Almost identical convergence rates were found for the non-orthogonal case.

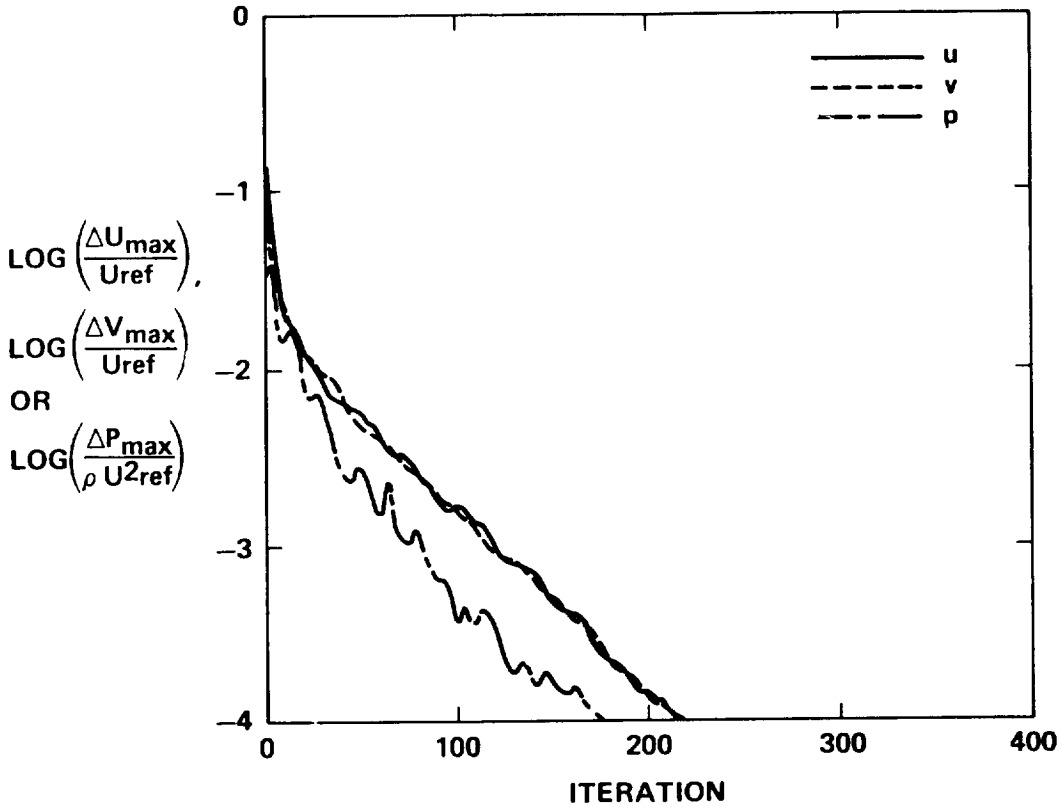


Figure 8. Convergence history for the driven cavity problem, Re = 400.

B. 2-D Laminar Flows Over a Backward-facing Step

This test case concerns 2-D laminar recirculating flows over a backward-facing step with 1:2 expansion ratio. The dependence of the size of the recirculation region (characterized by the reattachment length) on the Reynolds number (based on the inlet bulk velocity and twice of the inlet channel width) of the flow is of major concern. The physical domain and boundary conditions are illustrated in Figure 9 in which a fully developed laminar flow velocity profile is imposed at the flow entrance. A non-uniform grid of 45 x 45 was used for numerical computations. Several cases with different Reynolds numbers from 100 to 800 have been studied. An experimental and theoretical study about this problem, which results will be used as the basis of data comparisons, has been provided by Amaly et al. [16].

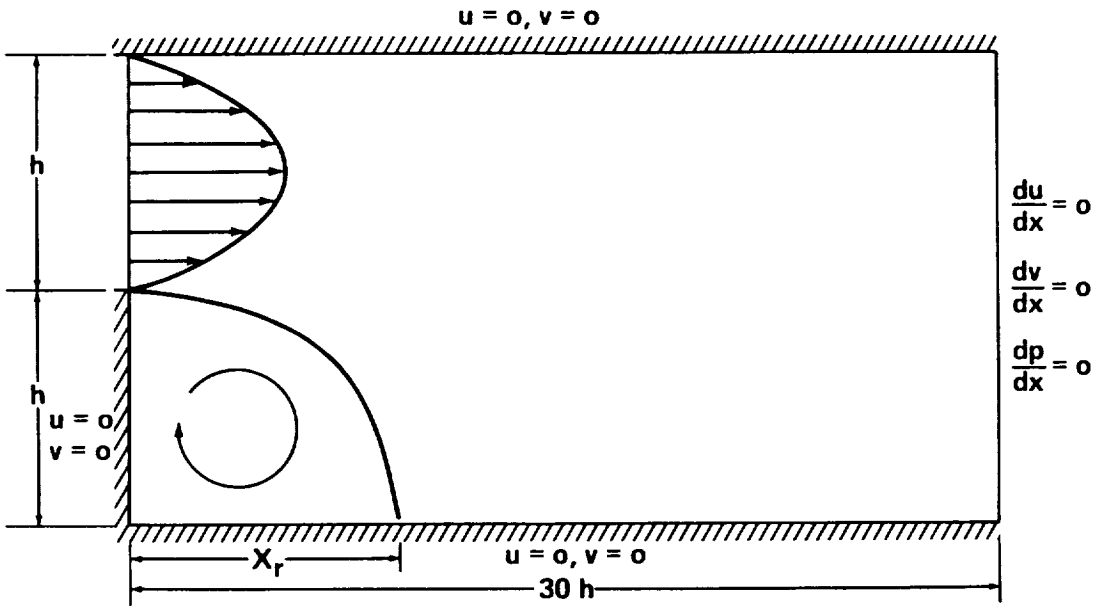


Figure 9. Physical geometry and boundary conditions of laminar flows over a backward-facing step (1:2 expansion).

To save computational efforts, the solution of one case with Reynolds number 100 is obtained in the first run. Then, a series of cases with increasing Reynolds numbers (i.e., 100, 200, 300, 400, 600, and 800) are calculated using the preceding results of lower Reynolds number as the initial guesses of the flow field. In this way, an average of 500 iterations for each case were needed to obtain converged solutions.

Two different differencing schemes in approximating the convection terms are employed to demonstrate the effects of the differencing schemes on the predictions. One of the schemes is the widely used hybrid scheme [9]. The other scheme employs the central differencing scheme plus an artificial dissipation term used to stabilize the solution which is similar to the one used by Rhie [17]. The artificial dissipation term becomes effective only when the cell Peclet number (or cell Reynolds number) exceeds 10.

Results of the present predictions using two different differencing schemes are compared with the experimental measurements [16] and other predictions as shown in Figure 10. It can be seen clearly from Figure 10 that the present method with hybrid scheme gives results similar to those predicted by TEACH code [16] while the present method with central differencing and artificial dissipation reveals predictions close to those predicted by INS3D [18] and the method of Kim and Moin [19]. This is reasonable since the TEACH code and the present method (with the first scheme) use the hybrid scheme which introduces large numerical dissipation by its upwind part (for cell Peclet number greater than 2). This tends to reduce the reattachment length for Reynolds number greater than 400. The second scheme, which is similar to the ones used in INS3D and the method of Kim and Moin, has the numerical accuracy close to second order by setting the artificial dissipation to be as small as the solution stability permits such that better accuracy of the predictions is expected.

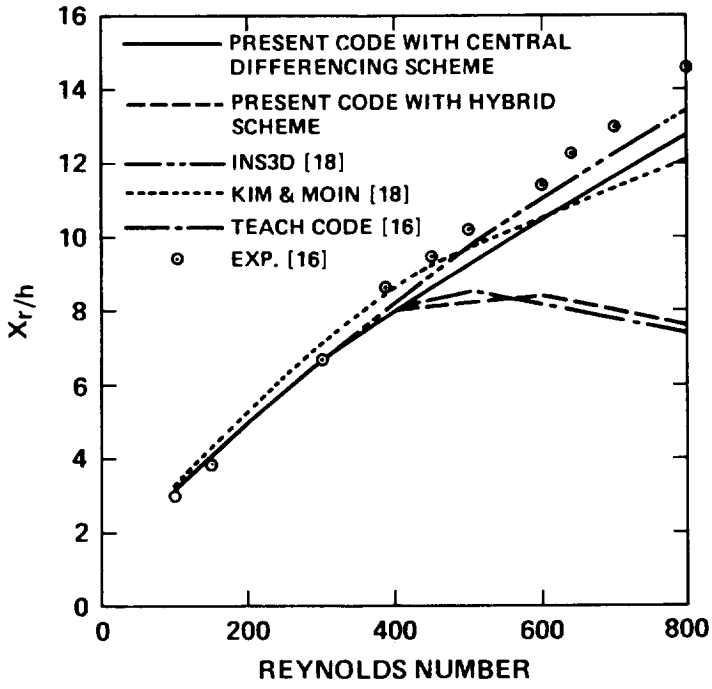


Figure 10. Reattachment length versus Reynolds number for laminar flows over a backward-facing step (1:2 expansion).

Stream function plots of the predictions using the two differencing schemes for Reynolds number 600 are compared in Figure 11. It is shown in Figure 11 that the second scheme gives a smooth shape of the recirculation zone while the hybrid scheme gives a sudden change in the shape of the recirculation region upstream of the reattachment point. Also, larger sizes of the separation regions on the step side wall and along the upper wall are predicted using the second scheme.

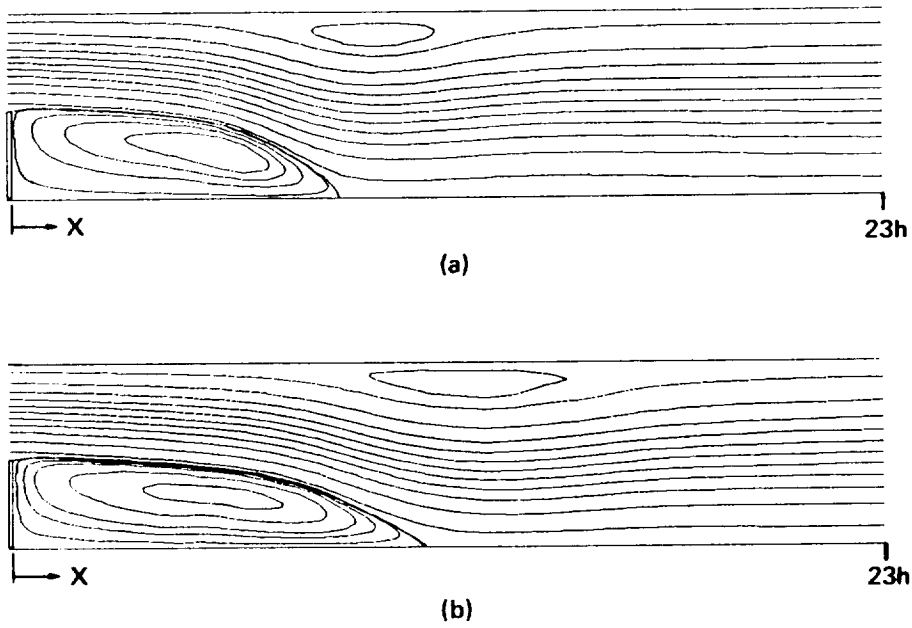


Figure 11. Streamline plots for laminar flow over a backward-facing step (1:2 expansion). (a) Hybrid Scheme. (b) Central differencing plus artificial dissipation scheme.

C. 2-D Turbulent Flows Over a Backward-Facing Step

In order to demonstrate the applicability of the present method to turbulent flow case, one of the standard test cases presented in the Stanford Conference [20] is selected here (i.e., turbulent flow over a 2:3 expansion backward-facing step). The standard k-ε turbulence model was used to provide the eddy viscosity for the transport equations. The physical geometry and boundary conditions imposed are shown in Figure 12. The calculation domain extends upstream of the expansion plane by 4 step heights and downstream of the expansion plane by 30 step heights to assure a fully developed velocity profile at the exit. A uniform velocity profile is located at the inlet plane. A 45 x 42 grid was used in the computation. 300 iterations were required to obtain converged solutions. Only hybrid differencing schemes were used in this case.

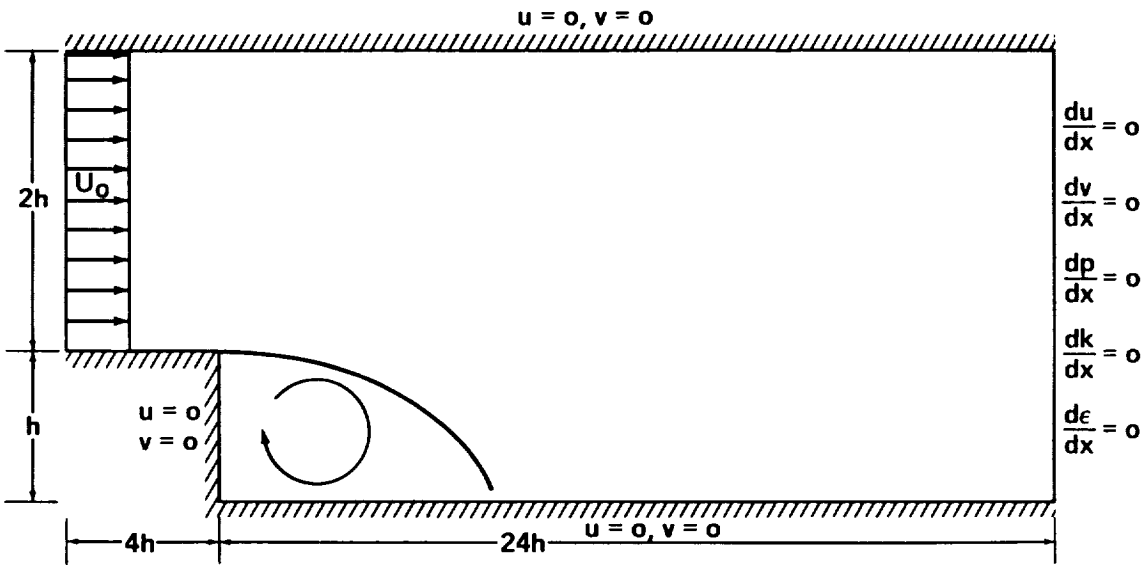


Figure 12. Physical geometry and boundary conditions of turbulent flows over a backward-facing step (2:3 expansion).

Results of the computation are shown in Figures 13, 14, and 15. These results are compared with the experimental measurements [20]. The under-prediction of the reattachment length is mainly due to the fast development of the mixing layer downstream of the expansion plane which is the characteristics of the standard k-ε turbulence model. Numerical diffusion provided by the hybrid scheme also contributes some part to the discrepancies between the predictions and measurements.

D. Developing Laminar Flow Inside a 90-Deg-Bend Square Duct

This test case simulates a three-dimensional developing laminar flow inside a 90-deg-bend square duct as illustrated in Figure 16(a). The symmetry plane is located at z = 0 where the symmetric boundary conditions are imposed. A fully developed velocity profile of laminar flow inside a straight square duct is prescribed at the entrance which is 2.8 duct widths upstream of the bend, A zero pressure

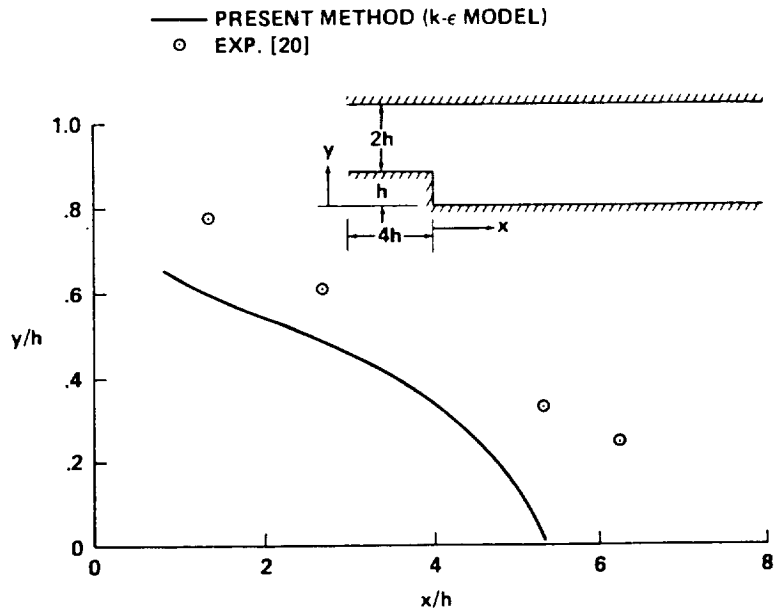


Figure 13. Locus of flow reversal inside the recirculation region for turbulent flow over a backward-facing step (2:3 expansion).

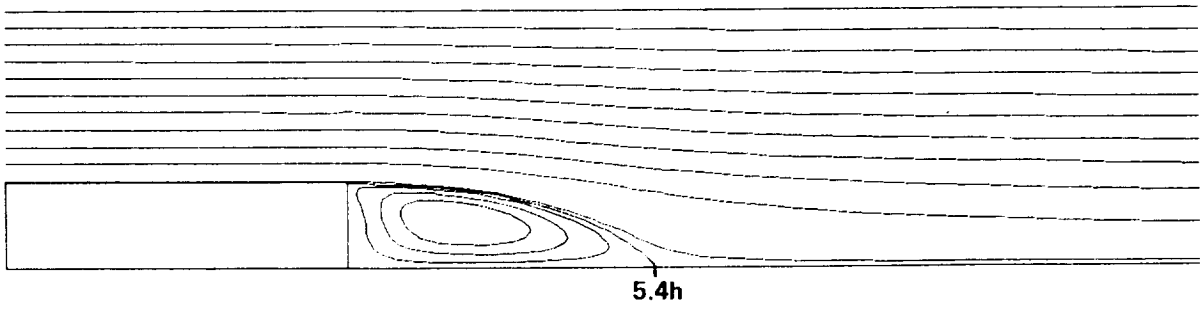


Figure 14. Stream line pattern of turbulent flow over a backward-facing step with 2:3 expansion ratio.

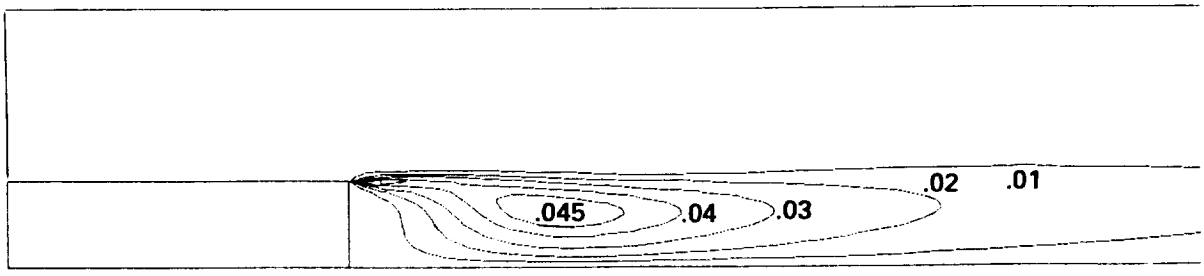


Figure 15. Contours of turbulent kinetic energy (k/U_o^2) of turbulent flow over a backward-facing step with 2:3 expansion ratio.

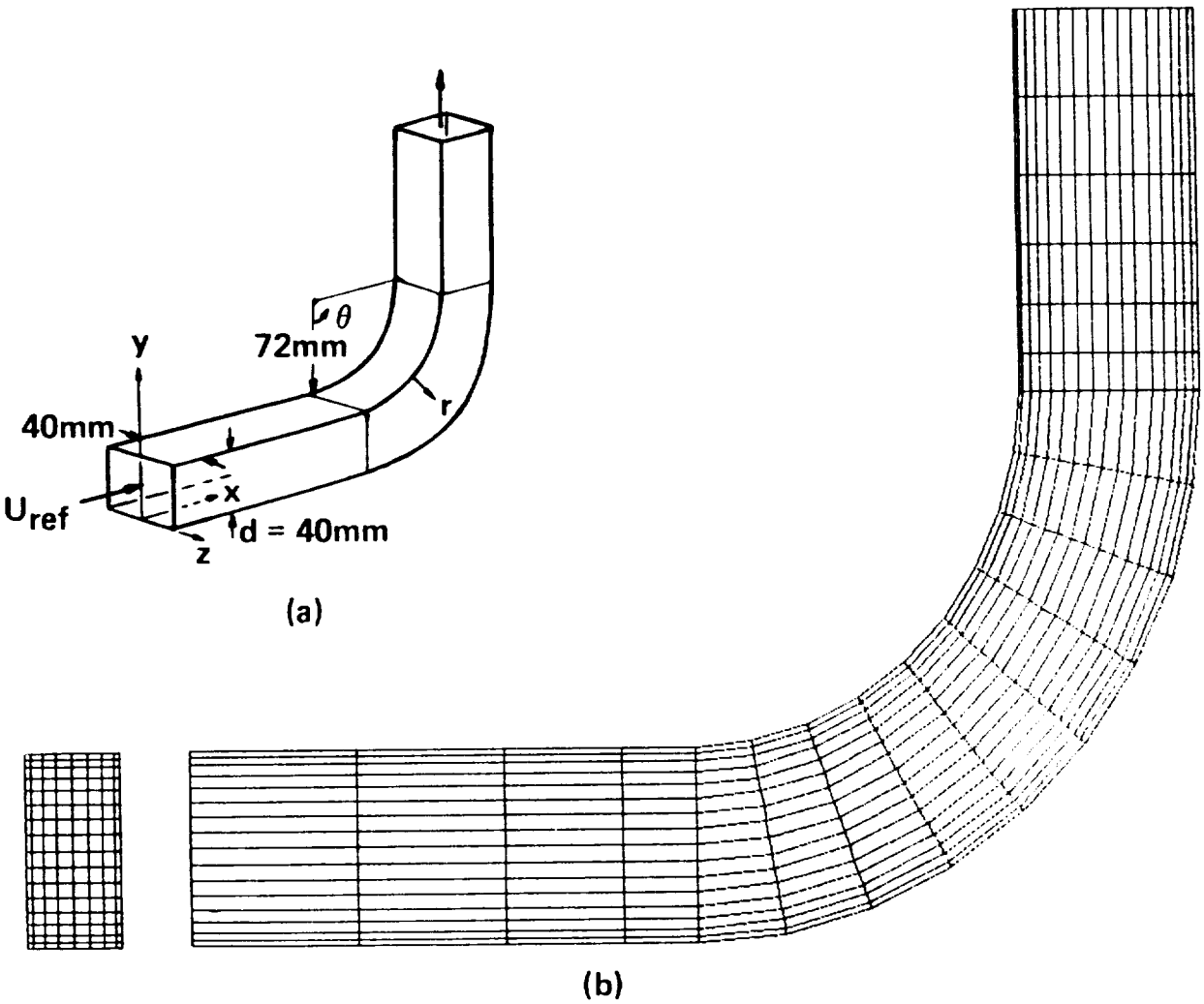


Figure 16. Geometry and mesh system of a 90-deg-bend square duct developing laminar flow problem.

gradient exit (which is 4.5 duct widths downstream of the bend) boundary condition is imposed. The Reynolds number of the flow (based on the duct hydrolic diameter and the inlet bulk velocity) is 790. A 21 x 18 x 10 grid was used for numerical computations. The front view and side view of the mesh system are illustrated in Figure 16(b). Experimental measurements of Humphrey et al. [21] are used for data comparisons.

Velocity vector plots on three sections along the main flow directions (i.e., on x-y plane) are shown in Figure 17. Secondary flow patterns at several stations across the bend are illustrated in Figure 18. These results are very similar to those obtained by Vanka [22] and Rhie [23]. Grid sizes of 50 x 22 x 15 and 58 x 15 x 11 were used by Rhie and Vanka, respectively. The present investigation, using only less than half of their grid numbers, gives highly encouraging results. Detailed comparisons between the measured and the predicted main velocity profiles are given in Figure 19.

With the above successful numerical simulations, it is believed that the present numerical method can be applied to general fluid dynamics problems with good numerical accuracy and efficiency.

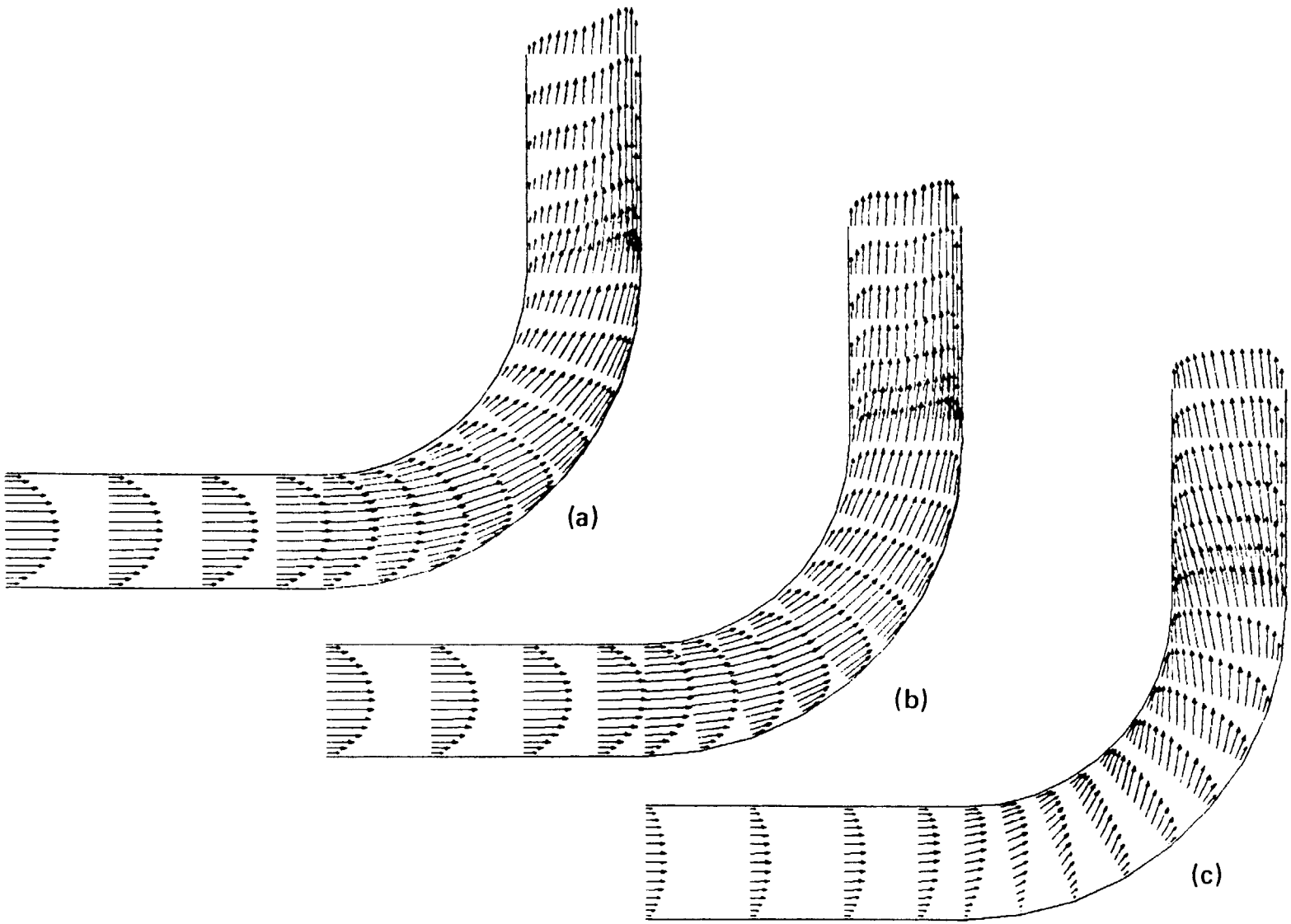


Figure 17. Primary velocity patterns of laminar flow inside a 90-deg-bend square duct. (a) z/d = 0.0. (b) z/d = 0.25. (c) z/d = 0.48.

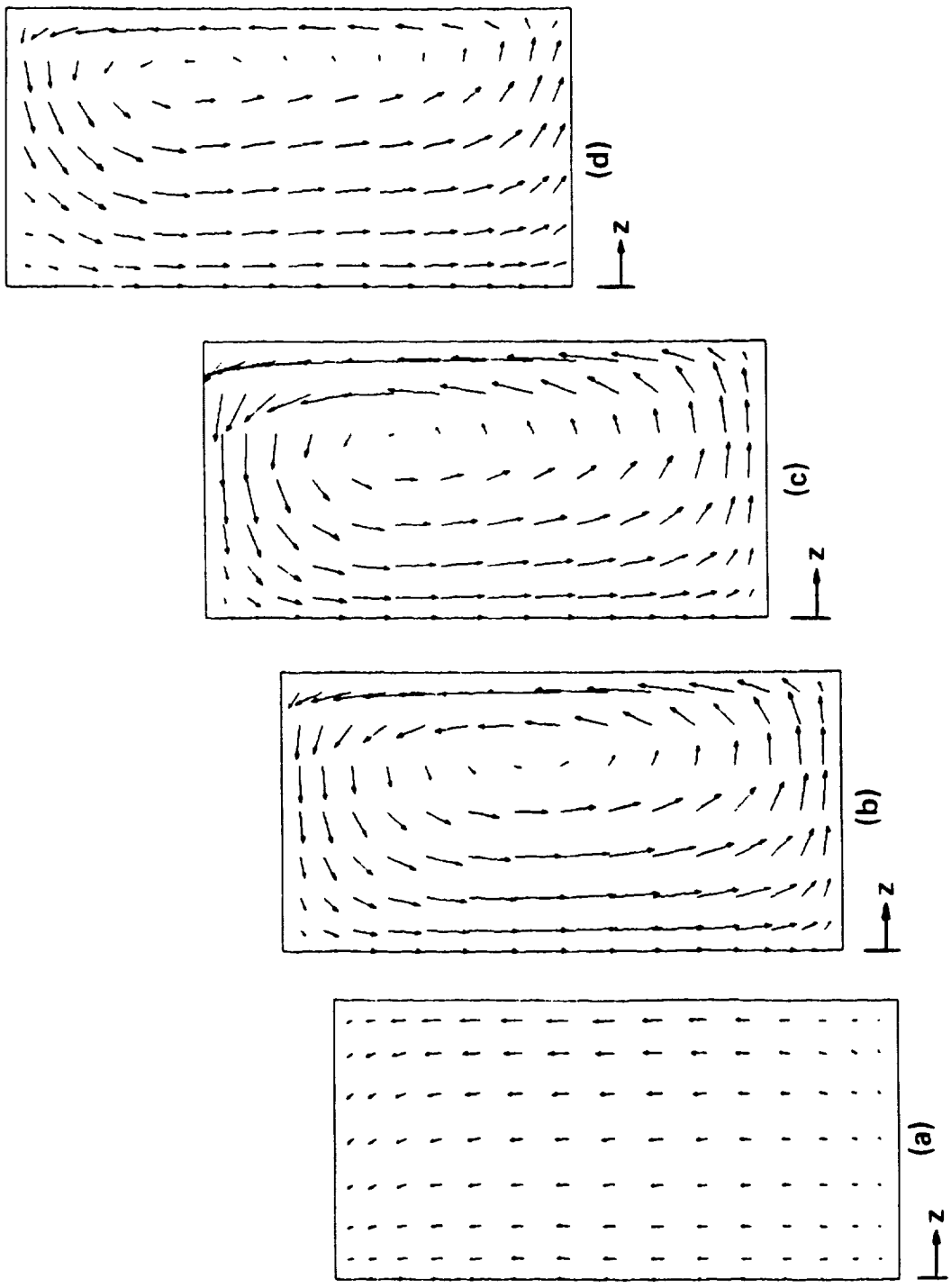


Figure 18. Secondary velocity patterns of laminar flow inside a 90-deg bend. (a) θ = 0 deg. (b) θ = 30 deg. (c) θ = 60 deg. (d) θ = 90 deg.

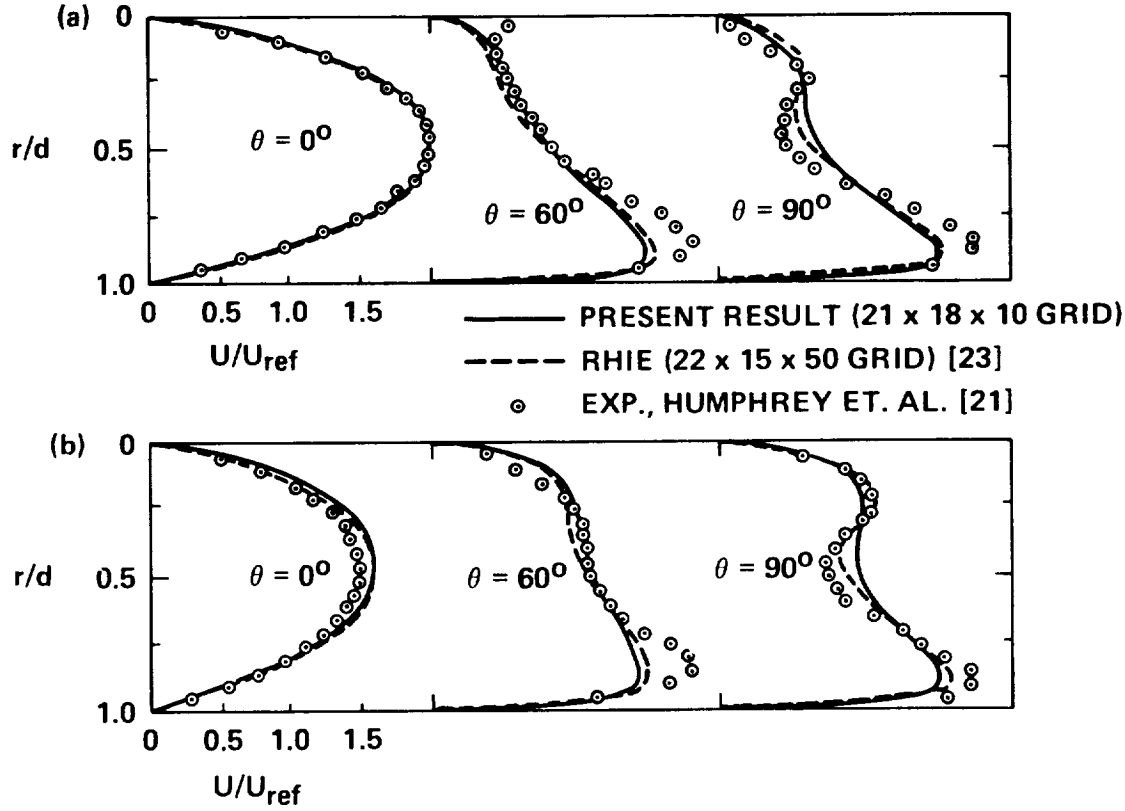


Figure 19. Primary velocity profiles for a 3-D 90-deg-bend square duct. (a) z/d = 0.0. (b) z/d = 0.25.

CONCLUSIONS

A numerical method for solving the steady or transient incompressible Navier-Stokes equations in three-dimensional body-fitted coordinate systems has been developed. In the present paper, the basic numerical algorithms and grid arrangements have been described in detail. A brief user's guide to the present computer code (CNS3D) has been included in Appendix A. A program listing has also been attached in Appendix C.

Several numerical testing examples of 2-D and 3-D, laminar and turbulent flow problems included in the present work have demonstrated that the present computer code is efficient and robust, and can be used as a reliable tool for engineering design and analysis applications. Applications of the present code to the internal turbulent flow problems of the SSME will be presented in the future publications.

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APPENDIX A

COMPUTER CODE STRUCTURES AND USER'S GUIDE

The global structure of the present computer code (CNS3D) can be represented by a flow chart, shown in Figure A-1. The user is referred to Appendix C for detailed information. First, the program requests inputs, from logic unit 5 (LU = 5), of program control parameters that specify the maximum number of iterations, the type of flow (i.e., laminar or turbulent), number of iterations for solving the pressure correction equation (typically 10), and underrelaxation factors for solving the transport equations, etc. This is followed by the definitions of all the program constants including turbulence model constants (these constants are subject to change according to the user's specific flow problem). Next, the program asks for inputs of the initial flow field guess from a restart file (LU = 8) which contains the grid system coordinates and flow field data that may be created by the user (including grid generation) or obtained from the previous solutions. Format of this data file is also subject to change according to the user's preference. Next, wall boundary control parameters, boundary grid normal distance to the wall, and wall boundary direction cosine are calculated in subroutine DIRCOS. Subroutine TRANF is then invoked to obtain the grid transformation coefficients. Before the solution procedure starts, the inlet mass flow rate is calculated which will be used to control the outlet mass flow rate to enhance mass conservation. The solution procedures consist of a series of subroutine calls to SOLVEQ starting from the solutions of the velocity vectors, u, v, and w, and then the solutions of scalar quantities (including the energy equation and the turbulence model equations) and finally the solution of the pressure correction equation to update the velocity and pressure field such that a divergence-free flow field can be retained.

After each global iteration of the solution procedures, the numerical of iterations and the maximum flow field corrections are checked with the initial settings. If the convergence criterion is satisfied or the number of iterations reaches the prescribed value then the solution procedures stop and the flow field solutions will be written on the pre-assigned disc file (LU = 7).

For instance, if a steady-state laminar flow problem (Reynolds number of 600) is of interest and a converged solution is expected within 300 iterations and the number of iterations for solving the pressure correction equation is 10 and the underrelaxation factors are 0.5 and 0.95 for transport equations and pressure correction equation, respectively, the first inputs from LU = 5 would be:

Line								
1.	300	1	10	1				
2.	0.5	0.5	0.5	0.95	0.5	0.5	0.5	0.5
3.	600.	0.0						

In the second input sequence (i.e., from restart file), the program reads in L x M x N lines of data records. See Figure A-2 for grid structures. Notice that the program requires variable dimensions of (L+1, M+1, N+1) for solving the pressure correction equation. It is important to check the COMMON table for proper variable dimensions.

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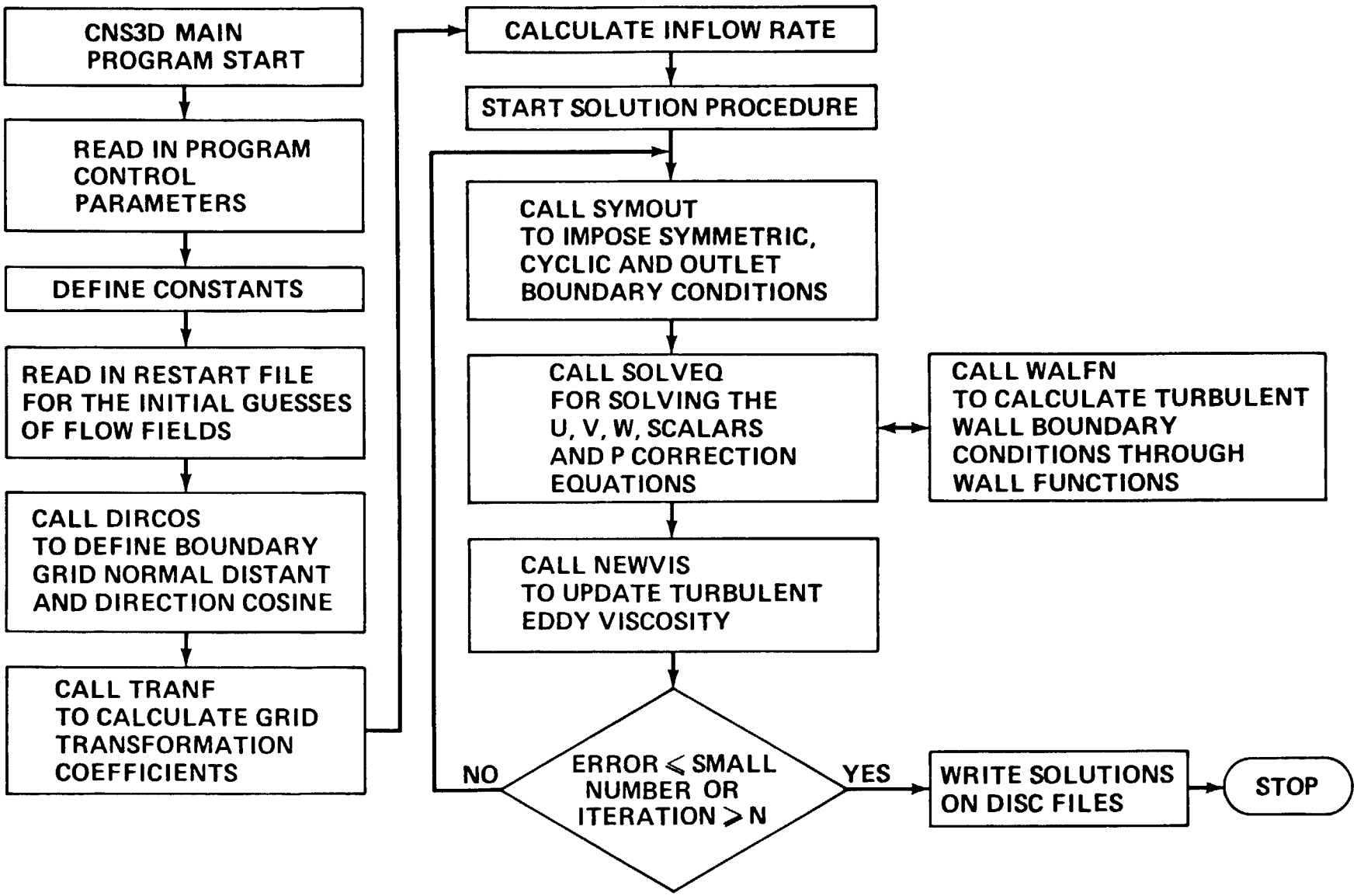


Figure A-1. Global structure of the present computer program CNS3D.

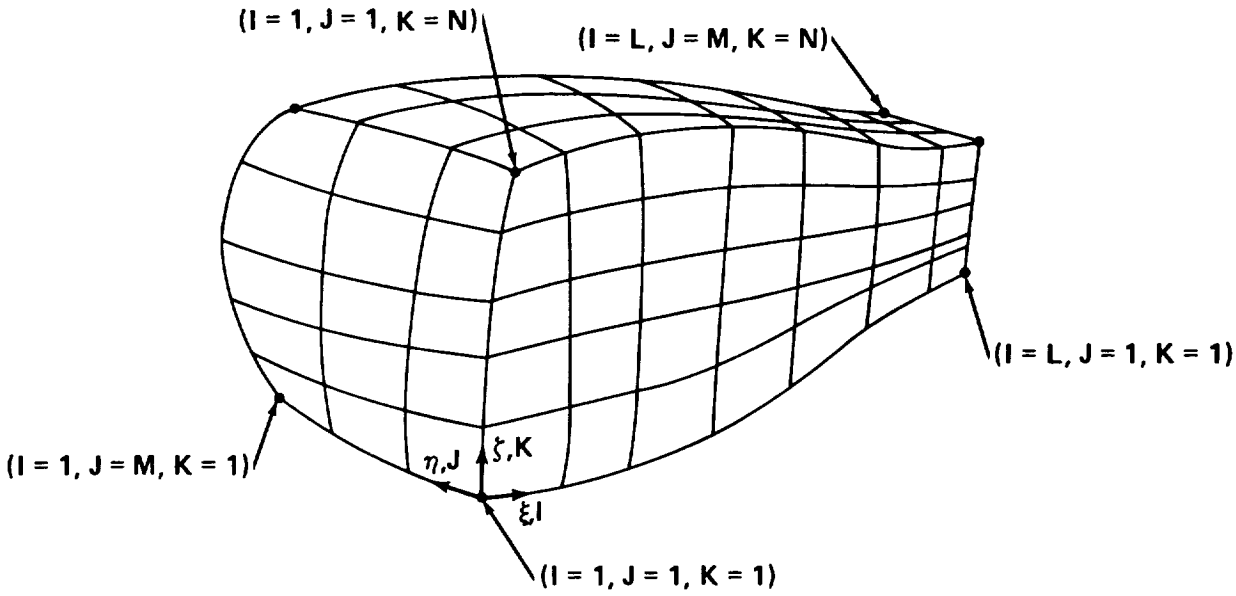


Figure A-2. Grid mesh structures for 3-D calculations.

If the flow problem involves symmetric or cyclic boundary conditions, then the user can look into the subroutine SYMOUT to specify the appropriate boundary conditions (the conditions shown in the program listing of Appendix C are for symmetric boundary conditions at K = 1). For cyclic boundary conditions at K = 1 and K = N, data at K = 2 and K = N-1 can be used to obtain boundary conditions at K = 1 and K = N by requiring same gradients across K = 1 and K = N. This method is simple but will lag the boundary conditions by one iteration. A direct method without lagging the boundary conditions can also be employed by modifying the subroutine of linear algebra solver LINERX such that the boundary conditions can be part of the solution of the TDMA (tridiagonal matrix) solver.

In case of incorporating different wall functions for turbulent flow problems (e.g., References 24, 25, and 26), subroutine BOUNC and WALFN can be modified according to the user's method of wall treatments. The set of wall functions given in the program listing of Appendix C are derived from the conventional wall law and the equilibrium turbulent kinetic energy relations [8].

When additional source terms are to be added to the transport equations due to flow problem requirements, modifications to the source term calculation section in the subroutine SOLVEQ can be carried out. Notice that in the subroutine SOLVEQ source terms for the velocities v and w are included in the u-source section. Purpose of this is to save some computing time since these source terms use similar calculation routines.

Some times it is required to solve more transport equations other than the basic ones included in Appendix C. To modify the program to incorporate more equations,

several changes are necessary. First, new variables must be added to the COMMON table (this can be easily done through the computer editor session). Then, new source term sections are added in the SOLVEQ subroutine. Finally, subroutines WALFN and SYMOUT are modified to incorporate the new variables into the boundary condition setting routines.

APPENDIX B

LIST OF FORTRAN SYMBOLS

A(K) = Matrix elements of a tridiagonal matrix

AB(I,J,K) = Link coefficients through the bottom face of a control volume

AE(I,J,K) = Link coefficients through the east face of a control volume

ALC = Underrelaxation factor for symmetry or cyclic boundary conditions

ALE = Underrelaxation factor for the ε-equation

ALK = Underrelaxation factor for the k-equation

ALP = Underrelaxation factor for the pressure correction equation

ALU = Underrelaxation factor for the u-equation

ALV = Underrelaxation factor for the v-equation

ALVIS = Underrelaxation factor for the effective viscosity

ALW = Underrelaxation factor for the w-equation

AN(I,J,K) = Link coefficient through the north face of a control volume

ANAB = Sum of the link coefficients at all faces

ANV1(I) = Modified wall boundary link coefficient for v-equation

ANW1(I) = Modified wall boundary link coefficient for w-equation

AP(I,J,K) = Sum of the link coefficients around a control volume

APO(I,J,K) = Link coefficients in time marching direction

ARDEN = Area times density across a section in physical domain

AREA = Area of a section in physical domain

AS(I,J,K) = Link coefficients through the south face of a control volume

AT(I,J,K) = Link coefficients through the top face of a control volume

AW(I,J,K) = Link coefficients through the west face of a control volume

B(K) = Matrix elements of a tridiagonal matrix

BB(I,J,K) = Coefficients in Stone's partial factorization technique

BOUNC = Subroutine for getting turbulent wall boundary conditions through wall functions

C(K) = Matrix elements of a tridiagonal matrix

C1 = Turbulence model constant, = 1.44

C2 = Turbulence model constant, = 1.92

CB = Convective flux through the bottom face of a control volume

CE = Convective flux through the east face of a control volume

CK = Von Karman constant, = 0.4

CMU = Turbulence model constant, = 0.09

CMU1 = CMU**0.25

CMU2 = CMU**0.75

CN = Convective flux through the north face of a control volume

CS = Convective flux through the south face of a control volume

CT = Convective flux through the top face of a control volume

CW = Convective flux through the west face of a control volume

CX(I,J,K) = Grid transformation coefficient, ξ_x

CY(I,J,K) = Grid transformation coefficient, ξ_y

CZ(I,J,K) = Grid transformation coefficient, ξ_z

D(K) = Matrix elements of a tridiagonal matrix

DDB = Diffusive flux through the bottom face of a control volume

DDE = Diffusive flux through the east face of a control volume

DDN = Diffusive flux through the north face of a control volume

DDS = Diffusive flux through the south face of a control volume

DDT = Diffusive flux through the top face of a control volume

DDW = Diffusive flux through the west face of a control volume

DE(I,J,K) = Turbulent kinetic energy dissipation rate, ε

DEO(I,J,K) = DE at the previous time level

DEN(I,J,K) = Density of the fluid

DENO(I,J,K) = DEN at the previous time level

DENC = Density at the center of a surface

DENIN = Initial value of density of the fluid

DIRCOS = Subroutine for calculating the boundary grid sizes and direction cosines

DITM = Wall boundary average value of dissipation rate

DK(I,J,K) = Turbulent kinetic energy, k

DKO(I,J,K) = DK at the previous time level

DTT = Time step size, Δt

DU(I,J,K) = Diffusive coefficient for the p'-equation

DV(I,J,K) = Diffusive coefficient for the p'-equation

DW(I,J,K) = Diffusive coefficient for the p'-equation

E = Wall law constant, = 9.01069

EREXT = Convergence criterion tolerance

ERRE = Maximum correction in ε

ERRF = Maximum correction of a variable

ERRK = Maximum correction in k

ERRM = Maximum correction in p

ERRU = Maximum correction in u

ERRV = Maximum correction in v

ERRW = Maximum correction in w

EX(I,J,K) = Grid transformation coefficient, η_x

EY(I,J,K) = Grid transformation coefficient, η_y

EZ(I,J,K) = Grid transformation coefficient, η_z

F(I,J,K) = Tentative variable of the transport equations

FO(I,J,K) = F at the previous time level

F1(I,J,K) = Variable quantity at the previous iteration step

FLOW = Outlet mass flow rate

FLOWIN = Inlet mass flow rate

GEN(I,J,K) = Turbulent kinetic energy production rate

HINUM = Large number, = 1.E30

I = Index along the ξ grid lines

IBC(I) = Boundary grid index

IE = Index assigned for the transport equations

IG = Problem control parameter, =1 for laminar flow and =2 for turbulent flow

IITO = Total number of wall boundary grids

IITY = Boundary grid face type

IJLO(I,J,K) = Boundary grid sequential order

INIT = Subroutine for initializing variables

INPRO = Logical parameter for updating the effective viscosity

INSOE = Logical parameter for solving the ε-equation

INSOK = Logical parameter for solving the k-equation

INSOP = Logical parameter for solving the p'-equation

INSOT = Logical parameter for solving the T-equation

INSOU = Logical parameter for solving the u-equation

INSOV = Logical parameter for solving the v-equation

INSOW = Logical parameter for solving the w-equation

IS = Starting value of I of the solution domain

ISWE = Number of sweeps for solving the ε-equation

ISWK = Number of sweeps for solving the k-equation

ISWP = Number of sweeps for solving the p'-equation

ISWU = Number of sweeps for solving the u-equation

ISWV = Number of sweeps for solving the v-equation

ISWW = Number of sweeps for solving the w-equation

IT = Last value of I of the solution domain

ITT = Number of time steps

J = Index along the η grid lines

JBC(I) = Boundary grid index

JS = Starting value of J of the solution domain

JT = Last value of J of the solution domain

K = Index along the ζ grid lines

KBC(I) = Boundary grid index

KS = Starting value of K of the solution domain

KT = Last value of K of the solution domain

L = Maximum dimension of grid system in I direction

LO = L + 1

L1 = Starting point of blockage region in I direction

L2 = Last point of blockage region in I direction

LINERX = Subroutine for solving algebraic equations

LT = L - 1

M = Maximum dimension of grid system in J direction

MO = M + 1

M1 = Starting point of blockage region in J direction

M2 = Last point of blockage region in J direction

MC(I,J,K) = Wall blockage region control parameter

MT = M - 1

N = Maximum dimension of grid system in K direction

NO = N + 1

N1 = Starting point of blockage region in K direction

N2 = Last point of blockage region in K direction

NEWVIS = Subroutine for updating the effective viscosity

NLIMT = Limit of maximum number of iterations

NT = N - 1

P = Static pressure (relative)

PCXI = Pressure gradient, P_{ξ}

PDUV = Blockage control parameter for link coefficients

PEDA = Pressure gradient, p_η

PP = Pressure correction, p'

PPBLK = Global pressure correction

PSCI = Pressure gradient, P_ζ

PTA = Wall boundary source term for the momentum equations

PW = Wall value control parameter

RENL = Reynolds number of the fluid

SIGE = Turbulence model constant, = 1.3

SIGK = Turbulence model constant, = 1.0

SINX(I) = Wall boundary direction cosine

SINY(I) = Wall boundary direction cosine

SINZ(I) = Wall boundary direction cosine

SMNUM = Small number, 1.E-30

SOC1 = Source term due to shear stress

SOC2 = Source term due to shear stress

SOC3 = Source term due to shear stress

SOLVEQ = Subroutine for solving general transport equation

SP(I,J,K) = Linear part of the source term

SPK(I,J,K) = Secondary linear part of the source term

SU(I,J,K) = Constant part of the source term

SUK(I,J,K) = Secondary constant part of the source term

SX(I,J,K) = Grid transformation coefficient, ζ_x

SY(I,J,K) = Grid transformation coefficient, ζ_y

SYMOUT = Subroutine for setting flow boundary conditions

SZ(I,J,K) = Grid transformation coefficient, ζ_z

TAUN(I) = Wall shear stress

TIMT = Total time

TJO(I,J,K) = Jacobian of metric transformation

TM(I,J,K) = Temperature

TMO(I,J,K) = TM at the previous time level

TMULT = Wall shear stress

TRANF = Subroutine for calculating the grid transformation coefficients

TXXE(I,J,K) = Metric coefficient for east face diffusive flux

TXXW(I,J,K) = Metric coefficient for west face diffusive flux

TXYN(I,J,K) = Metric coefficient for north face diffusive flux

TXYS(I,J,K) = Metric coefficient for south face diffusive flux

TXZT(I,J,K) = Metric coefficient for top face diffusive flux

TXZB(I,J,K) = Metric coefficient for bottom face diffusive flux

TYYN(I,J,K) = Metric coefficient for north face diffusive flux

TYYS(I,J,K) = Metric coefficient for south face diffusive flux

TYXE(I,J,K) = Metric coefficient for east face diffusive flux

TYXW(I,J,K) = Metric coefficient for west face diffusive flux

TYZT(I,J,K) = Metric coefficient for top face diffusive flux

TYZB(I,J,K) = Metric coefficient for bottom face diffusive flux

TZZT(I,J,K) = Metric coefficient for top face diffusive flux

TZZB(I,J,K) = Metric coefficient for bottom face diffusive flux

TZXE(I,J,K) = Metric coefficient for east face diffusive flux

TZXW(I,J,K) = Metric coefficient for west face diffusive flux

TZYN(I,J,K) = Metric coefficient for north face diffusive flux

TZYS(I,J,K) = Metric coefficient for south face diffusive flux

U(I,J,K) = U-velocity

UO(I,J,K) = U at the previous time level

UC = Velocity at the center of a surface

UCXI = U-velocity gradient, u_{ξ}

UEDA = U-velocity gradient, u_η

UINC = Velocity correction at outlet plane

USCI = U-velocity gradient, u_ζ

UX = U-velocity gradient, u_x

UY = U-velocity gradient, u_y

UZ = U-velocity gradient, u_z

V(I,J,K) = V-velocity

VO(I,J,K) = V at the previous time level

VISC = Molecular viscosity, μ

VISE(I,J,K) = Effective viscosity, μ_{eff}

VCXI = V-velocity gradient, v_ξ

VEDA = V-velocity gradient, v_η

VSCI = V-velocity gradient, v_ζ

VX = V-velocity gradient, v_x

VY = V-velocity gradient, v_y

VZ = V-velocity gradient, v_z

W(I,J,K) = W-velocity

WO(I,J,K) = W at the previous time level

WALLFN = Subroutine for calculating the wall functions

WALVAL = Subroutine for assigning wall values

WCXI = W-velocity gradient, w_ξ

WEDA = W-velocity gradient, w_η

WSCI = W-velocity gradient, w_ζ

WX = W-velocity gradient, w_x

WY = W-velocity gradient, w_y

WZ = W-velocity gradient, w_z

X(I,J,K) = X-coordinate

Y(I,J,K) = Y-coordinate

YN(I) = Wall normal distance from the last grid

YN1(I) = Wall grid volume size

YPLN(I) = Nondimensionalized YN, $y^+ = u_\tau y/\nu$

Z(I,J,K) = Z-coordinate

APPENDIX C

PROGRAM LISTING

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FORTRAN VIID: LICENSED RESTRICTED RIGHTS AS STATED IN LICENSE  **********, SEE DOCUMENTATION PACKAGE, 04-101M99.
  1  000000I          PROGRAM  CNS3D                                                                    1
  2                C      BY: Y. S. CHEN        ON: 8/13/1985                                          2
  3                C                                                                                   3
  4                C*****[ CURVILINEAR N-S CODE FOR 3-D INCOMPRESSIBLE FLOWS ]**********               4
  5                C                                                                                   5
  6  000006I           COMMON                                                                         6
  7                   1/VAR/U(21,18,10),V(21,18,10),P(21,18,10),DK(21,18,10),                        7
  8                   2  DE(21,18,10),ERRU,ERRV,ERRM,ERRK,ERRE,ERRW,                                 8
  9                   3  PP(21,18,10),W(21,18,10),TM(21,18,10)                                       9
 10                   1/PROP/ VISE(21,18,10),DEN(21,18,10),VISC,DENIN,FLOWIN                         10
 11                   1/PCOR/ DU(21,18,10),DV(21,18,10),DW(21,18,10)                                 11
 12                   1/TUR/ SIGK,SIGE,CMU,C1,C2,CMU1,CMU2,E,CK,HINUM,SMNUM,ANV1(800),               12
 13                   2 YN(800),YN1(800),SINX(800),SINY(800),SINZ(800),ANW1(800),                    13
 14                   3 YPLN(800),TAUN(800),IBC(800),JBC(800),KBC(800),IITY(800),                    14
 15                   4 TALW(800),GEN(21,18,10),MC(21,18,10),IJLO(21,18,10),IITO                     15
 16                   1/COEF/ AP(21,18,10),SU(21,18,10),SP(21,18,10),SUK(21,18,10),                  16
 17                   2   SPK(21,18,10),AE(21,18,10),AW(21,18,10),AN(21,18,10),                      17
 18                   3   AS(21,18,10),AT(21,18,10),AB(21,18,10),APO(21,18,10)                       18
 19  000006I           COMMON                                                                         19
 20                   1/TRAN/ X(21,18,10),Y(21,18,10),Z(21,18,10),TJO(21,18,10),                     20
 21                   2   CX(21,18,10),CY(21,18,10),CZ(21,18,10),                                    21
 22                   3   EX(21,18,10),EY(21,18,10),EZ(21,18,10),                                    22
 23                   3   SX(21,18,10),SY(21,18,10),SZ(21,18,10)                                     23
 24                   1/LIMT/ L,M,LT,MT,L1,L2,M1,M2,LO,MO,ISWU,ISWV,ISWP,ISWK,ISWE,                   24
 25                   2   ALU,ALV,ALP,ALK,ALE,ALVIS,ALW,N,N1,N2,NO,ISWW,IG,NT,ALC,DTT                25
 26  000006I           COMMON                                                                         26
 27                   1/TTRAN/TXXE(21,18,10),TXXW(21,18,10),TYYN(21,18,10),                          27
 28                   2   TYYS(21,18,10),TZZT(21,18,10),TZZB(21,18,10),                              28
 29                   3   TYXE(21,18,10),TYXW(21,18,10),TYZT(21,18,10),                              29
 30                   4   TYZS(21,18,10),TXYN(21,18,10),TXYS(21,18,10),                              30
 31                   5   TXZT(21,18,10),TXZB(21,18,10),TZXE(21,18,10),                              31
 32                   6   TZXW(21,18,10),TZYN(21,18,10),TZYS(21,18,10)                               32
 33                   1/UNSTDY/UO(21,18,10),VO(21,18,10),WO(21,18,10),DKO(21,18,10),                 33
 34                   2   DEO(21,18,10),DENO(21,18,10),TMO(21,18,10)                                 34
 35  000006I           LOGICAL INSOU,INSOV,INSOP,INSOK,INSOE,INPRO,INSOW,INSOT                        35
 36                C*****[ INPUT DATA GUIDE ]*************************************************       36
 37                C                                                                                  37
 38                C  NLIMT : MAXIMUM NO. OF ITERATIONS LIMIT                                         38
 39                C                                                                                  39
 40                C  IG    = 1 : LAMINAR                                                             40
 41                C           2 : TURBULENT (K-E MODEL)                                              41
 42                C                                                                                  42
 43                C  ISWP  : NO. OF SWEEPS FOR SOLVING THE P' EQUATION (PP).                         43
 44                C                                                                                  44
 45                C  ITT   : TOTAL NO. OF TIME STEPS.                                                45
 46                C                                                                                  46
 47                C  ALU,ALV,ALW,ALP,ALK,ALE,ALVIS,ALC : UNDER-RELAXATION FACTORS                     47
 48                C                                                                                  48
 49                C  RENL  : REFERENCE REYNOLDS NUMBER.                                              49
 50                C                                                                                  50
 51                C  DTT   : TIME STEP FOR UNSTEADY PROBLEMS.                                        51
 52                C                                                                                  52
 53                C*********************************************************************         53
 54                C-----INPUT DATA (PROBLEM CONTROL SETTING)                                         54
 55  000006I           READ(5,100) NLIMT,IG,ISWP,ITT                                                  55        1
 56  000030I           READ(5,200) ALU,ALV,ALW,ALP,ALK,ALE,ALVIS,ALC                                  56        2
 57  00006CI           READ(5,200) RENL,DTT                                                           57        3
```

```
 58                C-----CONSTANTS                                                                  58
 59  000090I             EREXT=1.E-3                                                               59        4
 60  00009CI             ISWU=7                                                                    60        5
 61  0000A4I             ISWV=7                                                                    61        6
 62  0000ACI             ISWW=7                                                                    62        7
 63  0000B4I             ISWK=5                                                                    63        8
 64  0000BCI             ISWE=5                                                                    64        9
 65  0000C4I             DENIN=1.0                                                                 65       10
 66  0000D0I             VISC=1./RENL                                                              66       11
 67  0000E2I             SIGL=1.0                                                                  67       12
 68  0000EEI             SIGK=1.0                                                                  68       13
 69  0000FAI             SIGE=1.3                                                                  69       14
 70  000106I             CMU=0.09                                                                  70       15
 71  000112I             C1=1.43                                                                   71       16
 72  00011EI             C2=1.92                                                                   72       17
 73  00012AI             E=9.0100?9                                                                73       18
 74  000136I             CK=0.4                                                                    74       19
 75  000142I             PI=3.141592654                                                            75       20
 76  00014EI             HINUM=1.E30                                                               76       21
 77  00015AI             SMNUM=1.E-30                                                              77       22
 78  000166I             CMU1=CMU**0.25                                                            78       23
 79  00017EI             CMU2=CMU**0.75                                                            79       24
 80  000196I             INSCU=.TRUE.                                                              80       25
 81  00019EI             INSCV=.TRUE.                                                              81       26
 82  0001A6I             INSCP=.TRUE.                                                              82       27
 83  0001AEI             INSCK=.TRUE.                                                              83       28
 84  0001B6I             INSCE=.TRUE.                                                              84       29
 85  0001BEI             INPRO=.TRUE.                                                              85       30
 86  0001C6I             INSCW=.TRUE.                                                              86       31
 87  0001CEI             INSCT=.FALSE.                                                             87       32
 88  0001D6I             IF(IG .EQ. 2) GO TO 10                                                    88       33
 89  0001ECI             INSCK=.FALSE.                                                             89       35
 90  0001F4I             INSCE=.FALSE.                                                             90       36
 91  0001FCI             INPRO=.FALSE.                                                             91       37
 92  000204I          10 CONTINUE                                                                  92       38
 93                C*****[READ IN INITIAL FLOW FIELDS FROM RESTART FILE (LU = 8)]********         93
 94  000204I             READ(8,100) L,M,N,L1,L2,M1,M2,N1,N2                                       94       39
 95  000244I             LO=L+1                                                                    95       40
 96  000252I             MO=M+1                                                                    96       41
 97  000260I             NO=N+1                                                                    97       42
 98  00026EI             LT=L-1                                                                    98       43
 99  00027CI             MT=M-1                                                                    99       44
100  00028AI             NT=N-1                                                                   100       45
101                C-----INITIALIZE VARIABLES                                                     101
102  000298I             CALL INIT                                                                102       46
103                C-----RESTART FILE                                                             103
104  0002A0I             DO 50 K=1,N                                                              104       47
105  000284I             DO 50 I=1,L                                                              105       48
106  0002C8I             READ(8,400)                                                              106       49
107  0002E4I             DO 50 J=1,M                                                              107       50
108  0002F8I             READ(8,500) X(I,J,K),Y(I,J,K),Z(I,J,K),U(I,J,K),V(I,J,K),               108       51
109                     1 W(I,J,K),P(I,J,K),TM(I,J,K),DK(I,J,K),DE(I,J,K),                       109
110                     2 VISETM,DENTM                                                           110
111  000588I             UO(I,J,K)=U(I,J,K)                                                       111       52
112  000604I             VO(I,J,K)=V(I,J,K)                                                       112       53
113  000650I             WO(I,J,K)=W(I,J,K)                                                       113       54
114  00069CI             TMO(I,J,K)=TM(I,J,K)                                                     114       55
```

```
115  0006E8I          DKO(I,J,K)=DK(I,J,K)                                                    115      56
116  000734I          DEO(I,J,K)=DE(I,J,K)                                                    116      57
117  000780I          DENO(I,J,K)=DEN(I,J,K)                                                  117      58
118  0007CCI       50 CONTINUE                                                                118      59
119               C*************************************************************************  119
120               C-----GET BOUNDARY CONTROL PARAMETERS                                       120
121  000814I          CALL DIRCOS                                                             121      60
122               C-----SET BOUNDARY TURBULENCE QUANTITIES TO ZERO                            122
123  00081CI          DO 121 I=1,L                                                            123      61
124  000830I          DO 121 J=1,M                                                            124      62
125  000844I          DO 121 K=1,N                                                            125      63
126  000858I          IF(MC(I,J,K) .NE. 0) GO TO 122                                          126      64
127  00088EI          GO TO 121                                                               127      66
128  000894I   122    DK(I,J,K)=0.0                                                           128      67
129  0008C0I          DE(I,J,K)=0.0                                                           129      68
130  0008ECI          U(I,J,K)=0.0                                                            130      69
131  000918I          V(I,J,K)=0.0                                                            131      70
132  000944I          W(I,J,K)=0.0                                                            132      71
133  000970I   121    CONTINUE                                                                133      72
134               C-----CALCULATE GRID TRANSFORMATION COEFFICIENTS                            134
135  000988I          CALL TRANF                                                              135      73
136               C-----TURBULENT VISCOSITY                                                   136
137  0009C0I          IF(INPRC) CALL NEWVIS                                                   137      74
138               C-----CALCULATE INLET MASS FLOW RATE                                        138
139  0009D4I          FLOWIN=0.0                                                              139      76
140  0009E0I          I=1                                                                     140      77
141  0009E8I          DO 45 J=2,M                                                             141      78
142  0009FCI          DO 45 K=2,N                                                             142      79
143  000A10I          UC=(U(I,J,K)+U(I,J-1,K)+U(I,J,K-1)+U(I,J-1,K-1))*0.25                   143      80
144  000AC0I          DENC=(DEN(I,J,K)+DEN(I,J-1,K)+DEN(I,J,K-1)+DEN(I,J-1,K-1))*0.25         144      81
145  000B70I          P1=(X(I,J,K)+X(I,J,K-1)-X(I,J-1,K)-X(I,J-1,K-1))*0.5                    145      82
146  000C20I          P2=(Y(I,J,K)+Y(I,J,K-1)-Y(I,J-1,K)-Y(I,J-1,K-1))*0.5                    146      83
147  000CD0I          P3=(Z(I,J,K)+Z(I,J,K-1)-Z(I,J-1,K)-Z(I,J-1,K-1))*0.5                    147      84
148  000D80I          Q1=(X(I,J,K)+X(I,J-1,K)-X(I,J,K-1)-X(I,J-1,K-1))*0.5                    148      85
149  000E30I          Q2=(Y(I,J,K)+Y(I,J-1,K)-Y(I,J,K-1)-Y(I,J-1,K-1))*0.5                    149      86
150  000EE0I          Q3=(Z(I,J,K)+Z(I,J-1,K)-Z(I,J,K-1)-Z(I,J-1,K-1))*0.5                    150      87
151  000F90I          AREA=SQRT(P1*P1+P2*P2+P3*P3)*SQRT(Q1*Q1+Q2*Q2+Q3*Q3)                     151      88
152  00102CI          FLOWIN=FLOWIN+DENC*AREA*UC                                              152      89
153  00104AI       45 CONTINUE                                                                153      90
154  00107AI          ITC=1                                                                   154      91
155               C-----TRANSIENT PROCESS                                                     155
156  001032I       2  CONTINUE                                                                156      92
157  001082I          CALL SYMOUT(3,1,2,L,2,M,2,N)                                            157      93
158  0010DCI          ITER=1                                                                  158      94
159               C-----SOLUTION PROCEDURES START                                             159
160  0010E4I       1  CONTINUE                                                                160      95
161  0010E4I          CALL SYMOUT(1,1,2,LT,2,MT,2,NT)                                         161      96
162  001140I          IF(INSCU) CALL SOLVEQ(1,ISWU,ALU,SIGU,ERRU,U,UC)                        162      97
163  001184I          IF(INSCV) CALL SOLVEQ(2,ISWV,ALV,SIGU,ERRV,V,VO)                        163      99
164  0011C8I          IF(INSCW) CALL SOLVEQ(3,ISWW,ALW,SIGU,ERRW,W,WO)                        164     101
165  00120CI          IF(INSCT) CALL SOLVEQ(4,ISWW,ALW,SIGU,ERRW,TM,TMO)                      165     103
166  001250I          IF(INSCK) CALL SOLVEQ(5,ISWK,ALK,SIGK,ERRK,DK,DKO)                      166     105
167  001294I          IF(INSOE) CALL SOLVEQ(6,ISWE,ALE,SIGE,ERRE,DE,DEO)                      167     107
168  001208I          IF(INSCP) CALL SOLVEQ(0,ISWP,ALP,SIGU,ERRM,PP,PP)                       168     109
169  00131CI          IF(INPRC) CALL NEWVIS                                                   169     111
170               C-----CONVERGENCE CHECK                                                     170
171  001330I          WRITE(6,300) ITER,ERRU,ERRV,ERRW,ERRM,ERRK,ERRE,U(7,2,6)                171     113
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```
172  00136CI          ERRMAX=ERRM+ERRU+ERRV+ERRW                                                    172      114
173  00138AI          IF(ITER .GE. 20 .AND. ERRMAX .GT. 1.E03) GO TO 99                             173      115
174  0013B2I          IF(ITER .GE. NLIMT .OR. ERRMAX .LE. EREXT) GO TO 99                           174      117
175  0013DCI          ITER=ITER+1                                                                   175      119
176  0013EAI          GO TO 1                                                                       176      120
177                C-----PRINT OUT SOLUTIONS                                                       177
178  0013EEI       99 CONTINUE                                                                      178      121
179  0013EEI          WRITE(7,100) L,M,N,L1,L2,M1,M2,N1,N2                                          179      122
180  00142CI          DO 901 K=1,N                                                                  180      123
181  001440I          DO 901 I=1,L                                                                  181      124
182  001454I          WRITE(7,400)                                                                  182      125
183  001470I          DO 902 J=1,M                                                                  183      126
184  001484I          XV=X(I,J,K)                                                                   184      127
185  0014B0I          YV=Y(I,J,K)                                                                   185      128
186  0014DCI          ZV=Z(I,J,K)                                                                   186      129
187  0015C8I          WRITE(7,500) XV,YV,ZV,U(I,J,K),V(I,J,K),W(I,J,K),P(I,J,K),                    187      130
188                1           TM(I,J,K),DK(I,J,K),DE(I,J,K),VISE(I,J,K),DEN(I,J,K)                188
189  0016E8I      902 CONTINUE                                                                     189      131
190  001700I      901 CONTINUE                                                                     190      132
191  001730I          TIMT=DTT*ITC                                                                  191      133
192  001744I          WRITE(7,300) ITC,TIMT                                                         192      134
193  001768I          IF(ITC .GE. ITT .OR. DTT .EQ. C.0) GO TO 999                                  193      135
194  001792I          ITC=ITC+1                                                                     194      137
195  0017A0I          GO TO 2                                                                       195      138
196  0017A4I      999 CONTINUE                                                                     196      139
197  0017A4I      100 FORMAT(9I5)                                                                  197
198  0017B0I      200 FORMAT(11F7.4)                                                               198
199  0017E8I      300 FORMAT(1X,I5,7E10.2)                                                         199
200  0017C0I      400 FORMAT(//)                                                                   200
201  0017DAI      500 FORMAT(3F8.4,3E12.4,2X,6E11.4)                                               201
202  0017F6I          STOP                                                                          202      140
203  0017FEI          END                                                                           203      141

NO  ERRORS:F7D  R05-01.0C  MAINPROG    CNS3C     02/21/86  09:45:56  TABLE SPACE:  11 KB
STATEMENT BUFFER:   20 LINES/1321 BYTES    STACK SPACE: 181 WORDS
SINGLE PRECISION FLOATING PT SUPPORT REQUIRED FOR EXECUTION
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 1  000000I         SUBROUTINE DIRCCS                                                  204
 2  000004I         COMMON                                                             205
 3                 1/TUR/ SIGK,SIGE,CMU,C1,C2,CMU1,CMU2,E,CK,HINUM,SMNUM,ANV1(800),   206
 4                 2 YN(800),YN1(800),SINX(800),SINY(800),SINZ(800),ANW1(800),        207
 5                 3 YPLN(800),TAUN(800),IBC(800),JBC(800),KBC(800),IITY(800),        208
 6                 4 TAUW(800),GEN(21,18,10),MC(21,18,10),IJLO(21,18,10),IITO         209
 7                 1/TRAN/ X(21,18,10),Y(21,18,10),Z(21,18,10),TJO(21,18,10),         210
 8                 2   CX(21,18,10),CY(21,18,10),CZ(21,18,10),                        211
 9                 3   EX(21,18,10),EY(21,18,10),EZ(21,18,10),                        212
10                 3   SX(21,18,10),SY(21,18,10),SZ(21,18,10)                         213
11                 1/LIMT/ L,M,LT,MT,L1,L2,M1,M2,LC,MO,ISWU,ISWV,ISWP,ISWK,ISWE,       214
12                 2   ALU,ALV,ALP,ALK,ALE,ALVIS,ALW,N,N1,N2,NO,ISWW,IG,NT,ALC,DTT    215
13           C-----SET DOMAIN BLOCKAGE CONTROL PARAMETER                               216
14           C-----SCALAR BLOCKAGE :   MC(I,J,K)=1                                     217
15           C-----PRESSURE BLOCKAGE : MC(I,J,K)=2                                     218
16  000004I       DO 10 I=1,LO                                                         219        1
17  000018I       DO 10 J=1,MO                                                         220        2
18  00002CI       DO 10 K=1,NO                                                         221        3
19  000040I       MC(I,J,K)=0                                                          222        4
20  000068I       IF(J.EQ.1.OR.J.EQ.M.OR.K.EQ.N) MC(I,J,K)=1                           223        5
21  0000C4I       IF(I.GE.L1.AND.I.LE.L2.AND.J.GE.M1.AND.J.LE.M2.AND.                  224        7
22                1   K.GE.N1.AND.K.LE.N2) MC(I,J,K)=1                                225
23  000158I       IF(I.GT.L1.AND.I.LE.L2.AND.J.GT.M1.AND.J.LE.M2.AND.                  226        9
24                1   K.GT.N1.AND.K.LE.N2) MC(I,J,K)=2                                227
25           C-----ADD BLOCKAGES AS NEEDED HERE                                        228
26  0001ECI    10 CONTINUE                                                           229       11
27           C-----CALCULATE BOUNDARY GRID SIZES AND ORIENTATIONS                      230
28  000234I       III=1                                                                231       12
29  00023CI       DO 30 I=2,LT                                                         232       13
30  000250I       DO 30 J=2,MT                                                         233       14
31  000264I       DO 30 K=2,NT                                                         234       15
32  000278I       IF(MC(I,J,K) .NE. 0) GO TO 30                                        235       16
33  0002AEI       MCT=MC(I+1,J,K)+MC(I-1,J,K)+MC(I,J+1,K)+MC(I,J-1,K)+                 236       18
34                1    MC(I,J,K+1)+MC(I,J,K-1)                                         237
35  0003ACI       IF(MCT .EQ. 0) GO TO 30                                              238       19
36  0003C2I       IF(MC(I,J+1,K) .EQ. 0) GO TO 2                                       239       21
37           C-----NORTH                                                               240
38  0003FCI       IBC(III)=I                                                           241       23
39  000410I       JBC(III)=J                                                           242       24
40  000424I       KBC(III)=K                                                           243       25
41  000438I       IITY(III)=1                                                          244       26
42  000448I       I1=I+1                                                               245       27
43  000456I       I2=I-1                                                               246       28
44  000464I       K1=K+1                                                               247       29
45  000472I       K2=K-1                                                               248       30
46  000480I       IF(I .EQ. L1) I2=I                                                   249       31
47  00049EI       IF(I .EQ. L2) I1=I                                                   250       33
48  0004BCI       IF(K .EQ. N1) K2=K                                                   251       35
49  0004DAI       IF(K .EQ. N2) K1=K                                                   252       37
50  0004F8I       J1=J-1                                                               253       39
51  000506I       J2=J-2                                                               254       40
52  000514I       P1=(Y(I1,J,K1)-Y(I1,J,K2))*(Z(I1,J,K2)-Z(I2,J,K))-                  255       41
53                1   (Z(I1,J,K1)-Z(I1,J,K2))*(Y(I1,J,K2)-Y(I2,J,K))                  256
54  000650I       P2=(Z(I1,J,K1)-Z(I1,J,K2))*(X(I1,J,K2)-X(I2,J,K))-                  257       42
55                1   (X(I1,J,K1)-X(I1,J,K2))*(Z(I1,J,K2)-Z(I2,J,K))                  258
56  00078CI       P3=(X(I1,J,K1)-X(I1,J,K2))*(Y(I1,J,K2)-Y(I2,J,K))-                  259       43
57                1   (Y(I1,J,K1)-Y(I1,J,K2))*(X(I1,J,K2)-X(I2,J,K))                  260
```

```
 58  0008C8I          PQ=SQRT(P1*P1+P2*P2+P3*P3)                                  261       44
 59  000912I          P1=P1/PQ                                                    262       45
 60  000924I          P2=P2/PQ                                                    263       46
 61  000936I          P3=P3/PQ                                                    264       47
 62  000948I          R1=(1.-P1**2)                                               265       48
 63  000964I          R2=(1.-P2**2)                                               266       49
 64  000980I          R3=(1.-P3**2)                                               267       50
 65  00099CI          SINX(III)=SQRT(R1)                                          268       51
 66  0009C2I          SINY(III)=SQRT(R2)                                          269       52
 67  0009E6I          SINZ(III)=SQRT(R3)                                          270       53
 68  000A0AI          Q1=X(I,J,K)-X(I,J1,K)                                       271       54
 69  000A5CI          Q2=Y(I,J,K)-Y(I,J1,K)                                       272       55
 70  000AAEI          Q3=Z(I,J,K)-Z(I,J1,K)                                       273       56
 71  000B00I          AA=SQRT((Q1-P1)**2+(Q2-P2)**2+(Q3-P3)**2)                   274       57
 72  000B7EI          CC=1.0                                                      275       58
 73  000B8AI          BB=SQRT(Q1*Q1+Q2*Q2+Q3*Q3)                                  276       59
 74  000BD6I          COTH=(BB*BB+CC*CC-AA*AA)/(2*BB*CC)                          277       60
 75  000C18I          YN(III)=BB*ABS(COTH)                                        278       61
 76  000C44I          Q1=X(I,J,K)-X(I,J2,K)                                       279       62
 77  000C96I          Q2=Y(I,J,K)-Y(I,J2,K)                                       280       63
 78  000CE8I          Q3=Z(I,J,K)-Z(I,J2,K)                                       281       64
 79  000D3AI          BB=SQRT(Q1*Q1+Q2*Q2+Q3*Q3)                                  282       65
 80  000D86I          AA=SQRT((Q1-P1)**2+(Q2-P2)**2+(Q3-P3)**2)                   283       66
 81  000E02I          COTH=(BB*BB+CC*CC-AA*AA)/(2*BB*CC)                          284       67
 82  000E44I          YN1(III)=(BB*ABS(COTH)+YN(III))*0.5                         285       68
 83  000E84I          IJLC(I,J,K)=III                                             286       69
 84  000E80I          III=III+1                                                   287       70
 85  000E8EI      2   CONTINUE                                                    288       71
 86  000E8EI          IF(MC(I,J-1,K) .EQ. 0) GO TO 3                              289       72
 87               C-----SOUTH                                                    290
 88  000EF8I          IBC(III)=I                                                  291       74
 89  000F0CI          JBC(III)=J                                                  292       75
 90  000F20I          KBC(III)=K                                                  293       76
 91  000F34I          IITY(III)=2                                                 294       77
 92  000F44I          I1=I+1                                                      295       78
 93  000F52I          I2=I-1                                                      296       79
 94  000F60I          K1=K+1                                                      297       80
 95  000F6EI          K2=K-1                                                      298       81
 96  000F7CI          J1=J+1                                                      299       82
 97  000F8AI          J2=J+2                                                      300       83
 98  000F98I          P1=(Y(I1,J,K1)-Y(I1,J,K2))*(Z(I1,J,K2)-Z(I2,J,K))-         301       84
 99                  1   (Z(I1,J,K1)-Z(I1,J,K2))*(Y(I1,J,K2)-Y(I2,J,K))         302
100  0010D4I          P2=(Z(I1,J,K1)-Z(I1,J,K2))*(X(I1,J,K2)-X(I2,J,K))-         303       85
101                 1   (X(I1,J,K1)-X(I1,J,K2))*(Z(I1,J,K2)-Z(I2,J,K))         304
102  001210I          P3=(X(I1,J,K1)-X(I1,J,K2))*(Y(I1,J,K2)-Y(I2,J,K))-         305       86
103                 1   (Y(I1,J,K1)-Y(I1,J,K2))*(X(I1,J,K2)-X(I2,J,K))         306
104  00134CI          PQ=SQRT(P1*P1+P2*P2+P3*P3)                                  307       87
105  001396I          P1=P1/PQ                                                    308       88
106  0013A8I          P2=P2/PQ                                                    309       89
107  0013BAI          P3=P3/PQ                                                    310       90
108  0013CCI          R1=(1.-P1**2)                                               311       91
109  0013E8I          R2=(1.-P2**2)                                               312       92
110  001404I          R3=(1.-P3**2)                                               313       93
111  001420I          SINX(III)=SQRT(R1)                                          314       94
112  001446I          SINY(III)=SQRT(R2)                                          315       95
113  00146AI          SINZ(III)=SQRT(R3)                                          316       96
114  00148EI          Q1=X(I,J,K)-X(I,J1,K)                                       317       97
```

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115  0014E0I          Q2=Y(I,J,K)-Y(I,J1,K)                                          318    98
116  001532I          Q3=Z(I,J,K)-Z(I,J1,K)                                          319    99
117  001584I          AA=SQRT((Q1-P1)**2+(Q2-P2)**2+(Q3-P3)**2)                      320   100
118  001602I          CC=1.0                                                         321   101
119  00160EI          BB=SQRT(Q1*Q1+Q2*Q2+Q3*Q3)                                     322   102
120  00165AI          COTH=(BB*BB+CC*CC-AA*AA)/(2*BB*CC)                             323   103
121  00169CI          YN(III)=BB*ABS(COTH)                                           324   104
122  0016C8I          Q1=X(I,J,K)-X(I,J2,K)                                          325   105
123  00171AI          Q2=Y(I,J,K)-Y(I,J2,K)                                          326   106
124  00176CI          Q3=Z(I,J,K)-Z(I,J2,K)                                          327   107
125  00179EI          BB=SQRT(Q1*Q1+Q2*Q2+Q3*Q3)                                     328   108
126  0017D0I          AA=SQRT((Q1-P1)**2+(Q2-P2)**2+(Q3-P3)**2)                      329   109
127  001866I          COTH=(BB*BB+CC*CC-AA*AA)/(2*BB*CC)                             330   110
128  0018C8I          YN1(III)=(BB*ABS(COTH)+YN(III))*0.5                            331   111
129  001903I          IJLC(I,J,K)=III                                                332   112
130  001934I          III=III+1                                                      333   113
131  001942I      3   CONTINUE                                                       334   114
132  001942I          IF(MC(I+1,J,K) .EQ. 0) GO TO 4                                 335   115
133               C-----EAST                                                         336
134  001970I          IBC(III)=I                                                     337   117
135  001990I          JBC(III)=J                                                     338   118
136  0019A4I          KBC(III)=K                                                     339   119
137  0019B8I          IITY(III)=3                                                    340   120
138  0019C8I          J1=J+1                                                         341   121
139  0019D6I          J2=J-1                                                         342   122
140  0019E4I          K1=K+1                                                         343   123
141  0019F2I          K2=K-1                                                         344   124
142  001A00I          IF(J .EQ. M1) J2=J                                             345   125
143  001A1EI          IF(J .EQ. M2) J1=J                                             346   127
144  001A3CI          IF(K .EQ. N1) K2=K                                             347   129
145  001A5AI          IF(K .EQ. N2) K1=K                                             348   131
146  001A78I          I1=I-1                                                         349   133
147  001A86I          I2=I-2                                                         350   134
148  001A94I          P1=(Y(I,J1,K1)-Y(I,J1,K2))*(Z(I,J1,K2)-Z(I,J2,K))-             351   135
149                  1   (Z(I,J1,K1)-Z(I,J1,K2))*(Y(I,J1,K2)-Y(I,J2,K))             352
150  001B00I          P2=(Z(I,J1,K1)-Z(I,J1,K2))*(X(I,J1,K2)-X(I,J2,K))-             353   136
151                  1   (X(I,J1,K1)-X(I,J1,K2))*(Z(I,J1,K2)-Z(I,J2,K))             354
152  001B0CI          P3=(X(I,J1,K1)-X(I,J1,K2))*(Y(I,J1,K2)-Y(I,J2,K))-             355   137
153                  1   (Y(I,J1,K1)-Y(I,J1,K2))*(X(I,J1,K2)-X(I,J2,K))             356
154  001E48I          PQ=SQRT(P1*P1+P2*P2+P3*P3)                                     357   138
155  001E92I          P1=P1/PQ                                                       358   139
156  001EA4I          P2=P2/PQ                                                       359   140
157  001EB6I          P3=P3/PQ                                                       360   141
158  001EC8I          R1=(1.-P1**2)                                                  361   142
159  001EE4I          R2=(1.-P2**2)                                                  362   143
160  001F00I          R3=(1.-P3**2)                                                  363   144
161  001F1CI          SINX(III)=SQRT(R1)                                             364   145
162  001F42I          SINY(III)=SQRT(R2)                                             365   146
163  001F66I          SINZ(III)=SQRT(R3)                                             366   147
164  001F8AI          Q1=X(I,J,K)-X(I1,J,K)                                          367   148
165  001FDCI          Q2=Y(I,J,K)-Y(I1,J,K)                                          368   149
166  00202EI          Q3=Z(I,J,K)-Z(I1,J,K)                                          369   150
167  002080I          AA=SQRT((Q1-P1)**2+(Q2-P2)**2+(Q3-P3)**2)                      370   151
168  0020FEI          CC=1.0                                                         371   152
169  00210AI          BB=SQRT(Q1*Q1+Q2*Q2+Q3*Q3)                                     372   153
170  002156I          COTH=(BB*BB+CC*CC-AA*AA)/(2*BB*CC)                             373   154
171  002198I          YN(III)=BB*ABS(COTH)                                           374   155
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172  0021C4I          Q1=X(I,J,K)-X(I2,J,K)                                          375     156
173  002216I          Q2=Y(I,J,K)-Y(I2,J,K)                                          376     157
174  002268I          Q3=Z(I,J,K)-Z(I2,J,K)                                          377     158
175  0022BAI          BB=SQRT(Q1*Q1+Q2*Q2+Q3*Q3)                                     378     159
176  002306I          AA=SQRT((Q1-P1)**2+(Q2-P2)**2+(Q3-P3)**2)                      379     160
177  002382I          COTH=(BB*BB+CC*CC-AA*AA)/(2*BB*CC)                             380     161
178  0023C4I          YN1(III)=(BB*ABS(COTH)+YN(III))*0.5                            381     162
179  002404I          IJLC(I,J,K)=III                                                382     163
180  002430I          III=III+1                                                      383     164
181  00243EI      4   CONTINUE                                                       384     165
182  00243EI          IF(MC(I-1,J,K) .EQ. 0) GO TO 5                                 385     166
183              C-----WEST                                                         386
184  002478I          I5C(III)=I                                                     387     168
185  00248CI          J8C(III)=J                                                     388     169
186  0024A0I          K8C(III)=K                                                     389     170
187  0024B4I          IITY(III)=4                                                    390     171
188  0024C4I          J1=J+1                                                         391     172
189  0024D2I          J2=J-1                                                         392     173
190  0024E0I          K1=K+1                                                         393     174
191  0024EEI          K2=K-1                                                         394     175
192  0024FCI          IF(J .EQ. M1) J2=J                                             395     176
193  00251AI          IF(J .EQ. M2) J1=J                                             396     178
194  002538I          IF(K .EQ. N1) K2=K                                             397     180
195  002556I          IF(K .EQ. N2) K1=K                                             398     182
196  002574I          I1=I+1                                                         399     184
197  002582I          I2=I+2                                                         400     185
198  002590I          P1=(Y(I,J1,K1)-Y(I,J1,K2))*(Z(I,J1,K2)-Z(I,J2,K))-             401     186
199                  1   (Z(I,J1,K1)-Z(I,J1,K2))*(Y(I,J1,K2)-Y(I,J2,K))             402
200  0026CCI          P2=(Z(I,J1,K1)-Z(I,J1,K2))*(X(I,J1,K2)-X(I,J2,K))-             403     187
201                  1   (X(I,J1,K1)-X(I,J1,K2))*(Z(I,J1,K2)-Z(I,J2,K))             404
202  0028D8I          P3=(X(I,J1,K1)-X(I,J1,K2))*(Y(I,J1,K2)-Y(I,J2,K))-             405     188
203                  1   (Y(I,J1,K1)-Y(I,J1,K2))*(X(I,J1,K2)-X(I,J2,K))             406
204  002944I          PQ=SQRT(P1*P1+P2*P2+P3*P3)                                     407     189
205  00298EI          P1=P1/PQ                                                       408     190
206  0029A0I          P2=P2/PQ                                                       409     191
207  0029B2I          P3=P3/PQ                                                       410     192
208  0029C4I          R1=1.-P1**2                                                    411     193
209  0029E0I          R2=1.-P2**2                                                    412     194
210  0029FCI          R3=1.-P3**2                                                    413     195
211  002A18I          SINX(III)=SQRT(R1)                                             414     196
212  002A3EI          SINY(III)=SQRT(R2)                                             415     197
213  002A62I          SINZ(III)=SQRT(R3)                                             416     198
214  002A86I          Q1=X(I,J,K)-X(I1,J,K)                                          417     199
215  002AD8I          Q2=Y(I,J,K)-Y(I1,J,K)                                          418     200
216  002B2AI          Q3=Z(I,J,K)-Z(I1,J,K)                                          419     201
217  002B7CI          AA=SQRT((Q1-P1)**2+(Q2-P2)**2+(Q3-P3)**2)                      420     202
218  002BFAI          CC=1.0                                                         421     203
219  002C06I          BB=SQRT(Q1*Q1+Q2*Q2+Q3*Q3)                                     422     204
220  002C52I          COTH=(BB*BB+CC*CC-AA*AA)/(2*BB*CC)                             423     205
221  002C94I          YN(III)=BB*ABS(COTH)                                           424     206
222  002CC0I          Q1=X(I,J,K)-X(I2,J,K)                                          425     207
223  002D12I          Q2=Y(I,J,K)-Y(I2,J,K)                                          426     208
224  002D64I          Q3=Z(I,J,K)-Z(I2,J,K)                                          427     209
225  002D86I          BB=SQRT(Q1*Q1+Q2*Q2+Q3*Q3)                                     428     210
226  002E02I          AA=SQRT((Q1-P1)**2+(Q2-P2)**2+(Q3-P3)**2)                      429     211
227  002E7EI          COTH=(BB*BB+CC*CC-AA*AA)/(2*BB*CC)                             430     212
228  002EC0I          YN1(III)=(BB*ABS(COTH)+YN(III))*0.5                            431     213
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229  002F00I          IJLC(I,J,K)=III                                                  432  214
230  002F2CI          III=III+1                                                        433  215
231  002F3AI        5 CONTINUE                                                         434  216
232  002F3AI          IF(MC(I,J,K+1) .EQ. 0) GO TO 6                                   435  217
233                 C-----TOP                                                        436
234  002F72I          ISC(III)=I                                                       437  219
235  002F86I          JBC(III)=J                                                       438  220
236  002F9AI          KBC(III)=K                                                       439  221
237  002FAEI          IITY(III)=5                                                      440  222
238  002FBEI          I1=I+1                                                           441  223
239  002FCCI          I2=I-1                                                           442  224
240  002FDAI          J1=J+1                                                           443  225
241  002FE8I          J2=J-1                                                           444  226
242  002FF6I          IF(I .EQ. L1) I2=I                                               445  227
243  003014I          IF(I .EQ. L2) I1=I                                               446  229
244  003032I          IF(J .EQ. M1) J2=J                                               447  231
245  003050I          IF(J .EQ. M2) J1=J                                               448  233
246  00306EI          K1=K-1                                                           449  235
247  00307CI          K2=K-2                                                           450  236
248  00308AI          P1=(Y(I1,J1,K)-Y(I1,J2,K))*(Z(I1,J2,K)-Z(I2,J,K))-               451  237
249                  1   (Z(I1,J1,K)-Z(I1,J2,K))*(Y(I1,J2,K)-Y(I2,J,K))               452
250  0031C6I          P2=(Z(I1,J1,K)-Z(I1,J2,K))*(X(I1,J2,K)-X(I2,J,K))-               453  238
251                  1   (X(I1,J1,K)-X(I1,J2,K))*(Z(I1,J2,K)-Z(I2,J,K))               454
252  003302I          P3=(X(I1,J1,K)-X(I1,J2,K))*(Y(I1,J2,K)-Y(I2,J,K))-               455  239
253                  1   (Y(I1,J1,K)-Y(I1,J2,K))*(X(I1,J2,K)-X(I2,J,K))               456
254  00343EI          PQ=SQRT(P1*P1+P2*P2+P3*P3)                                       457  240
255  00348AI          P1=P1/PQ                                                         458  241
256  00349CI          P2=P2/PQ                                                         459  242
257  0034AEI          P3=P3/PQ                                                         460  243
258  0034C0I          R1=1.-P1**2                                                      461  244
259  0034DCI          R2=1.-P2**2                                                      462  245
260  0034F8I          R3=1.-P3**2                                                      463  246
261  003514I          SINX(III)=SQRT(R1)                                               464  247
262  00353AI          SINY(III)=SQRT(R2)                                               465  248
263  00355EI          SINZ(III)=SQRT(R3)                                               466  249
264  003582I          Q1=X(I,J,K)-X(I,J,K1)                                            467  250
265  0035D4I          Q2=Y(I,J,K)-Y(I,J,K1)                                            468  251
266  003626I          Q3=Z(I,J,K)-Z(I,J,K1)                                            469  252
267  003678I          AA=SQRT((Q1-P1)**2+(Q2-P2)**2+(Q3-P3)**2)                        470  253
268  0036F6I          CC=1.0                                                           471  254
269  003702I          BB=SQRT(Q1*Q1+Q2*Q2+Q3*Q3)                                       472  255
270  00374EI          COTH=(BB*BB+CC*CC-AA*AA)/(2*BB*CC)                               473  256
271  003790I          YN(III)=BB*ABS(COTH)                                             474  257
272  0037BCI          Q1=X(I,J,K)-X(I,J,K2)                                            475  258
273  0037E0I          Q2=Y(I,J,K)-Y(I,J,K2)                                            476  259
274  003860I          Q3=Z(I,J,K)-Z(I,J,K2)                                            477  260
275  0038B2I          BB=SQRT(Q1*Q1+Q2*Q2+Q3*Q3)                                       478  261
276  0038FEI          AA=SQRT((Q1-P1)**2+(Q2-P2)**2+(Q3-P3)**2)                        479  262
277  00397AI          COTH=(BB*BB+CC*CC-AA*AA)/(2*BB*CC)                               480  263
278  0039BCI          YN1(III)=(BB*ABS(COTH)+YN(III))*0.5                              481  264
279  0039FCI          IJLC(I,J,K)=III                                                  482  265
280  003A28I          III=III+1                                                        483  266
281  003A36I        6 CONTINUE                                                         484  267
282  003A36I          IF(MC(I,J,K-1) .EQ. 0) GO TO 30                                  485  268
283                 C-----BOTTOM                                                     486
284  003A6EI          IBC(III)=I                                                       487  270
285  003A82I          JBC(III)=J                                                       488  271
```

```
286  003A96I          KBC(III)=K                                                    489     272
287  003AAAI          IITY(III)=6                                                   490     273
288  003A8AI          I1=I+1                                                        491     274
289  003AC8I          I2=I-1                                                        492     275
290  003AD6I          J1=J+1                                                        493     276
291  003AE4I          J2=J-1                                                        494     277
292  003AF2I          IF(I .EQ. L1) I2=I                                            495     278
293  003B10I          IF(I .EQ. L2) I1=I                                            496     280
294  003B2EI          IF(J .EQ. M1) J2=J                                            497     282
295  003B4CI          IF(J .EQ. M2) J1=J                                            498     284
296  003B6AI          K1=K+1                                                        499     286
297  003B78I          K2=K+2                                                        500     287
298  003B86I          P1=(Y(I1,J1,K)-Y(I1,J2,K))*(Z(I1,J2,K)-Z(I2,J,K))-           501     288
299                  1    (Z(I1,J1,K)-Z(I1,J2,K))*(Y(I1,J2,K)-Y(I2,J,K))           502
300  003CC2I          P2=(Z(I1,J1,K)-Z(I1,J2,K))*(X(I1,J2,K)-X(I2,J,K))-           503     289
301                  1    (X(I1,J1,K)-X(I1,J2,K))*(Z(I1,J2,K)-Z(I2,J,K))           504
302  003DFEI          P3=(X(I1,J1,K)-X(I1,J2,K))*(Y(I1,J2,K)-Y(I2,J,K))-           505     290
303                  1    (Y(I1,J1,K)-Y(I1,J2,K))*(X(I1,J2,K)-X(I2,J,K))           506
304  003F3AI          PQ=SQRT(P1*P1+P2*P2+P3*P3)                                    507     291
305  003F86I          P1=P1/PQ                                                      508     292
306  003F98I          P2=P2/PQ                                                      509     293
307  003FAAI          P3=P3/PQ                                                      510     294
308  003FBCI          R1=1.-P1**2                                                   511     295
309  003FD8I          R2=1.-P2**2                                                   512     296
310  003FF4I          R3=1.-P3**2                                                   513     297
311  004010I          SINX(III)=SQRT(R1)                                            514     298
312  004036I          SINY(III)=SQRT(R2)                                            515     299
313  00405AI          SINZ(III)=SQRT(R3)                                            516     300
314  00407EI          Q1=X(I,J,K)-X(I,J,K1)                                         517     301
315  004000I          Q2=Y(I,J,K)-Y(I,J,K1)                                         518     302
316  004122I          Q3=Z(I,J,K)-Z(I,J,K1)                                         519     303
317  004174I          AA=SQRT((Q1-P1)**2+(Q2-P2)**2+(Q3-P3)**2)                     520     304
318  0041F2I          CC=1.0                                                        521     305
319  0041FEI          BB=SQRT(Q1*Q1+Q2*Q2+Q3*Q3)                                    522     306
320  00424AI          COTH=(BB*BB+CC*CC-AA*AA)/(2*BB*CC)                            523     307
321  00428CI          YN(III)=BB*ABS(COTH)                                          524     308
322  004288I          Q1=X(I,J,K)-X(I,J,K2)                                         525     309
323  00430AI          Q2=Y(I,J,K)-Y(I,J,K2)                                         526     310
324  00435CI          Q3=Z(I,J,K)-Z(I,J,K2)                                         527     311
325  0043AEI          BB=SQRT(Q1*Q1+Q2*Q2+Q3*Q3)                                    528     312
326  0043FAI          AA=SQRT((Q1-P1)**2+(Q2-P2)**2+(Q3-P3)**2)                     529     313
327  004476I          COTH=(BB*BB+CC*CC-AA*AA)/(2*BB*CC)                            530     314
328  004498I          YN1(III)=(BB*ABS(COTH)+YN(III))*0.5                           531     315
329  0044F8I          IJLC(I,J,K)=III                                               532     316
330  004524I          III=III+1                                                     533     317
331  004532I     30   CONTINUE                                                      534     318
332  004580I          IITC=III-1                                                    535     319
333  00458EI          WRITE(6,100) LO,MO,NO,IITO                                    536     320
334  0045B8I    100   FORMAT(4I5)                                                   537
335  0045C4I          RETURN                                                        538     321
336  0045CCI          END                                                           539     322

NO  ERRORS:F7D  R05-01.0C  SUBROUTINE  CIRCCS    02/21/86  09:47:58  TABLE SPACE:   6 KB
STATEMENT BUFFER:   20 LINES/1321 BYTES    STACK SPACE: 203 WORDS
SINGLE PRECISION FLOATING PT SUPPORT REQUIRED FOR EXECUTION
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ORIGINAL PAGE IS
OF POOR QUALITY

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 1  000000I          SUBROUTINE TRANF                                                                  540
 2  0000004I         COMMON                                                                            541
 3                  1/VAR/U(21,18,10),V(21,18,10),P(21,18,10),DK(21,18,10),                           542
 4                  2  DE(21,18,10),ERRU,ERRV,ERRM,ERRK,ERRE,ERRW,                                   543
 5                  3  PP(21,18,10),W(21,18,10),TM(21,18,10)                                          544
 6                  1/PROP/ VISE(21,18,10),DEN(21,18,10),VISC,DENIN,FLOWIN                         545
 7                  1/PCOR/ DU(21,18,10),DV(21,18,10),DW(21,18,10)                                 546
 8                  1/TUR/ SIGK,SIGE,CMU,C1,C2,CMU1,CMU2,E,CK,HINUM,SMNUM,ANV1(800),               547
 9                  2 YN(800),YN1(800),SINX(800),SINY(300),SINZ(300),ANW1(800),                    548
10                  3 YPLN(800),TAUN(800),IBC(800),JBC(800),KBC(800),IITY(800),                    549
11                  4 TALW(800),GEN(21,18,10),MC(21,18,10),IJLO(21,18,10),IITO                     550
12                  1/COEF/ AP(21,18,10),SU(21,18,10),SP(21,18,10),SUK(21,18,10),                  551
13                  2   SPK(21,18,10),AE(21,18,10),AW(21,18,10),AN(21,18,10),                      552
14                  3   AS(21,18,10),AT(21,18,10),AB(21,18,10),APO(21,18,10)                       553
15 0000004I          COMMON                                                                         554
16                  1/TRAN/ X(21,18,10),Y(21,18,10),Z(21,18,10),TJO(21,18,10),                     555
17                  2   CX(21,18,10),CY(21,18,10),CZ(21,18,10),                                    556
18                  3   EX(21,18,10),EY(21,18,10),EZ(21,18,10),                                    557
19                  3   SX(21,18,10),SY(21,18,10),SZ(21,18,10)                                     558
20                  1/LIMT/ L,M,LT,MT,L1,L2,M1,M2,LO,MO,ISWU,ISWV,ISWP,ISWK,ISWE,                   559
21                  2   ALU,ALV,ALP,ALK,ALE,ALVIS,ALW,N,N1,N2,NO,ISWW,IG,NT,ALC,DTT                560
22 0000004I          COMMON                                                                         561
23                  1/TTRAN/TXXE(21,18,10),TXXW(21,18,10),TYYN(21,18,10),                          562
24                  2   TYYS(21,18,10),TZZT(21,18,10),TZZB(21,18,10),                              563
25                  3   TYXE(21,18,10),TYXW(21,18,10),TYZT(21,18,10),                              564
26                  4   TYZB(21,18,10),TXYN(21,18,10),TXYS(21,18,10),                              565
27                  5   TXZT(21,18,10),TXZB(21,18,10),TZXE(21,18,10),                              566
28                  6   TZXW(21,18,10),TZYN(21,18,10),TZYS(21,18,10)                               567
29                C-----CALCULATE GRID TRANSFORMATION COEFFICIENTS                                   568
30 0000004I          DO 40 I=2,L                                                                    569      1
31 0000018I          DO 40 J=2,M                                                                    570      2
32 00002CI           DO 40 K=2,N                                                                    571      3
33 000040I           P1=(X(I,J,K)+X(I,J-1,K)+X(I,J,K-1)+X(I,J-1,K-1)-X(I-1,J,K)-                    572      4
34                  1  X(I-1,J-1,K)-X(I-1,J,K-1)-X(I-1,J-1,K-1))*0.25                              573
35 0001A4I           P2=(X(I,J,K)+X(I-1,J,K)+X(I,J,K-1)+X(I-1,J,K-1)-X(I,J-1,K)-                    574      5
36                  1  X(I-1,J-1,K)-X(I,J-1,K-1)-X(I-1,J-1,K-1))*0.25                              575
37 000308I           P3=(X(I,J,K)+X(I-1,J,K)+X(I,J-1,K)+X(I-1,J-1,K)-X(I,J,K-1)-                    576      6
38                  1  X(I-1,J,K-1)-X(I,J-1,K-1)-X(I-1,J-1,K-1))*0.25                              577
39 00046CI           Q1=(Y(I,J,K)+Y(I,J-1,K)+Y(I,J,K-1)+Y(I,J-1,K-1)-Y(I-1,J,K)-                    578      7
40                  1  Y(I-1,J-1,K)-Y(I-1,J,K-1)-Y(I-1,J-1,K-1))*0.25                              579
41 0005D0I           Q2=(Y(I,J,K)+Y(I-1,J,K)+Y(I,J,K-1)+Y(I-1,J,K-1)-Y(I,J-1,K)-                    580      8
42                  1  Y(I-1,J-1,K)-Y(I,J-1,K-1)-Y(I-1,J-1,K-1))*0.25                              581
43 000734I           Q3=(Y(I,J,K)+Y(I-1,J,K)+Y(I,J-1,K)+Y(I-1,J-1,K)-Y(I,J,K-1)-                    582      9
44                  1  Y(I-1,J,K-1)-Y(I,J-1,K-1)-Y(I-1,J-1,K-1))*0.25                              583
45 0008F8I           R1=(Z(I,J,K)+Z(I,J-1,K)+Z(I,J,K-1)+Z(I,J-1,K-1)-Z(I-1,J,K)-                    584     10
46                  1  Z(I-1,J-1,K)-Z(I-1,J,K-1)-Z(I-1,J-1,K-1))*0.25                              585
47 0009FCI           R2=(Z(I,J,K)+Z(I-1,J,K)+Z(I,J,K-1)+Z(I-1,J,K-1)-Z(I,J-1,K)-                    586     11
48                  1  Z(I-1,J-1,K)-Z(I,J-1,K-1)-Z(I-1,J-1,K-1))*0.25                              587
49 000B60I           R3=(Z(I,J,K)+Z(I-1,J,K)+Z(I,J-1,K)+Z(I-1,J-1,K)-Z(I,J,K-1)-                    588     12
50                  1  Z(I-1,J,K-1)-Z(I,J-1,K-1)-Z(I-1,J-1,K-1))*0.25                              589
51 000CC4I           PTR=1.0/(P1*(Q2*R3-Q3*R2)-P2*(Q1*R3-Q3*R1)+P3*(Q1*R2-Q2*R1))                   590     13
52 000D36I           SU(I,J,K)=ABS(1.0/PTR)                                                         591     14
53 000D86I           AE(I,J,K)=PTR*(Q2*R3-Q3*R2)                                                    592     15
54 000DCCI           AW(I,J,K)=-PTR*(P2*R3-P3*R2)                                                   593     16
55 000E1AI           AN(I,J,K)=PTR*(P2*Q3-P3*Q2)                                                    594     17
56 000E60I           AS(I,J,K)=-PTR*(Q1*R3-Q3*R1)                                                   595     18
57 000EAEI           AT(I,J,K)=PTR*(P1*R3-P3*R1)                                                    596     19
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 58  000EF4I          AB(I,J,K)=-PTR*(P1*Q3-P3*Q1)                                                    597       20
 59  000F42I          DU(I,J,K)=PTR*(Q1*R2-Q2*R1)                                                     598       21
 60  000F88I          DV(I,J,K)=-PTR*(P1*R2-P2*R1)                                                    599       22
 61  000FD6I          DW(I,J,K)=PTR*(P1*Q2-P2*Q1)                                                     600       23
 62  00101CI      40  CONTINUE                                                                        601       24
 63  001064I          CALL WALVAL(1.0,2,L,2,M,2,N,AE)                                                 602       25
 64  0010BCI          CALL WALVAL(1.0,2,L,2,M,2,N,AW)                                                 603       26
 65  001114I          CALL WALVAL(1.0,2,L,2,M,2,N,AN)                                                 604       27
 66  00116CI          CALL WALVAL(1.0,2,L,2,M,2,N,AS)                                                 605       28
 67  0011C4I          CALL WALVAL(1.0,2,L,2,M,2,N,AT)                                                 606       29
 68  00121CI          CALL WALVAL(1.0,2,L,2,M,2,N,AB)                                                 607       30
 69  001274I          CALL WALVAL(1.0,2,L,2,M,2,N,DU)                                                 608       31
 70  0012CCI          CALL WALVAL(1.0,2,L,2,M,2,N,DV)                                                 609       32
 71  001324I          CALL WALVAL(1.0,2,L,2,M,2,N,DW)                                                 610       33
 72  00137CI          CALL WALVAL(1.0,2,L,2,M,2,N,SU)                                                 611       34
 73  0013D4I          DO 80 I=1,L                                                                     612       35
 74  0013E8I          DO 80 J=1,M                                                                     613       36
 75  0013FCI          DO 80 K=1,N                                                                     614       37
 76  001410I          CX(I,J,K)=(AE(I,J,K)+AE(I,J+1,K)+AE(I,J,K+1)+AE(I,J+1,K+1)+                     615       38
 77                  1 AE(I+1,J,K)+AE(I+1,J+1,K)+AE(I+1,J,K+1)+AE(I+1,J+1,K+1))*0.125                616
 78  001594I          CY(I,J,K)=(AW(I,J,K)+AW(I,J+1,K)+AW(I,J,K+1)+AW(I,J+1,K+1)+                     617       39
 79                  1 AW(I+1,J,K)+AW(I+1,J+1,K)+AW(I+1,J,K+1)+AW(I+1,J+1,K+1))*0.125                618
 80  001718I          CZ(I,J,K)=(AN(I,J,K)+AN(I,J+1,K)+AN(I,J,K+1)+AN(I,J+1,K+1)+                     619       40
 81                  1 AN(I+1,J,K)+AN(I+1,J+1,K)+AN(I+1,J,K+1)+AN(I+1,J+1,K+1))*0.125                620
 82  00189CI          EX(I,J,K)=(AS(I,J,K)+AS(I,J+1,K)+AS(I,J,K+1)+AS(I,J+1,K+1)+                     621       41
 83                  1 AS(I+1,J,K)+AS(I+1,J+1,K)+AS(I+1,J,K+1)+AS(I+1,J+1,K+1))*0.125                622
 84  001A20I          EY(I,J,K)=(AT(I,J,K)+AT(I,J+1,K)+AT(I,J,K+1)+AT(I,J+1,K+1)+                     623       42
 85                  1 AT(I+1,J,K)+AT(I+1,J+1,K)+AT(I+1,J,K+1)+AT(I+1,J+1,K+1))*0.125                624
 86  001BA4I          EZ(I,J,K)=(AB(I,J,K)+AB(I,J+1,K)+AB(I,J,K+1)+AB(I,J+1,K+1)+                     625       43
 87                  1 AB(I+1,J,K)+AB(I+1,J+1,K)+AB(I+1,J,K+1)+AB(I+1,J+1,K+1))*0.125                626
 88  001D28I          SX(I,J,K)=(DU(I,J,K)+DU(I,J+1,K)+DU(I,J,K+1)+DU(I,J+1,K+1)+                     627       44
 89                  1 DU(I+1,J,K)+DU(I+1,J+1,K)+DU(I+1,J,K+1)+DU(I+1,J+1,K+1))*0.125                628
 90  001EACI          SY(I,J,K)=(DV(I,J,K)+DV(I,J+1,K)+DV(I,J,K+1)+DV(I,J+1,K+1)+                     629       45
 91                  1 DV(I+1,J,K)+DV(I+1,J+1,K)+DV(I+1,J,K+1)+DV(I+1,J+1,K+1))*0.125                630
 92  002030I          SZ(I,J,K)=(DW(I,J,K)+DW(I,J+1,K)+DW(I,J,K+1)+DW(I,J+1,K+1)+                     631       46
 93                  1 DW(I+1,J,K)+DW(I+1,J+1,K)+DW(I+1,J,K+1)+DW(I+1,J+1,K+1))*0.125                632
 94  0021B4I          TJO(I,J,K)=(SU(I,J,K)+SU(I,J+1,K)+SU(I,J,K+1)+SU(I,J+1,K+1)+                    633       47
 95                  1 SU(I+1,J,K)+SU(I+1,J+1,K)+SU(I+1,J,K+1)+SU(I+1,J+1,K+1))*0.125                634
 96  002338I      80  CONTINUE                                                                        635       48
 97  002380I          DO 200 I=1,LO                                                                   636       49
 98  002394I          DO 200 J=1,MO                                                                   637       50
 99  0023A8I          DO 200 K=1,NO                                                                   638       51
100  0023BCI          DU(I,J,K)=0.0                                                                   639       52
101  0023E8I     200  CONTINUE                                                                        640       53
102  002430I          DO 160 I=2,LT                                                                   641       54
103  002444I          DO 160 J=2,MT                                                                   642       55
104  002458I          DO 160 K=2,NT                                                                   643       56
105  00246CI          CXE=(CX(I+1,J,K)+CX(I,J,K))*0.5                                                 644       57
106  0024C8I          CXW=(CX(I-1,J,K)+CX(I,J,K))*0.5                                                 645       58
107  002524I          CXN=(CX(I,J+1,K)+CX(I,J,K))*0.5                                                 646       59
108  002580I          CXS=(CX(I,J,K)+CX(I,J-1,K))*0.5                                                 647       60
109  0025DCI          CXT=(CX(I,J,K+1)+CX(I,J,K))*0.5                                                 648       61
110  002636I          CXB=(CX(I,J,K)+CX(I,J,K-1))*0.5                                                 649       62
111  002690I          CYE=(CY(I+1,J,K)+CY(I,J,K))*0.5                                                 650       63
112  0026ECI          CYW=(CY(I,J,K)+CY(I-1,J,K))*0.5                                                 651       64
113  002748I          CYN=(CY(I,J+1,K)+CY(I,J,K))*0.5                                                 652       65
114  0027A4I          CYS=(CY(I,J,K)+CY(I,J-1,K))*0.5                                                 653       66
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115  002800I        CYT=(CY(I,J,K+1)+CY(I,J,K))*0.5                                                                  654  67
116  00285AI        CYB=(CY(I,J,K)+CY(I,J,K-1))*0.5                                                                  655  68
117  0028B4I        CZE=(CZ(I+1,J,K)+CZ(I,J,K))*0.5                                                                  656  69
118  002910I        CZW=(CZ(I,J,K)+CZ(I-1,J,K))*0.5                                                                  657  70
119  00296CI        CZN=(CZ(I,J+1,K)+CZ(I,J,K))*0.5                                                                  658  71
120  0029C8I        CZS=(CZ(I,J,K)+CZ(I,J-1,K))*0.5                                                                  659  72
121  002A24I        CZT=(CZ(I,J,K+1)+CZ(I,J,K))*0.5                                                                  660  73
122  002A7EI        CZB=(CZ(I,J,K)+CZ(I,J,K-1))*0.5                                                                  661  74
123  002AD8I        EXE=(EX(I+1,J,K)+EX(I,J,K))*0.5                                                                  662  75
124  002B34I        EXW=(EX(I,J,K)+EX(I-1,J,K))*0.5                                                                  663  76
125  002B90I        EXN=(EX(I,J+1,K)+EX(I,J,K))*0.5                                                                  664  77
126  002BECI        EXS=(EX(I,J,K)+EX(I,J-1,K))*0.5                                                                  665  78
127  002C48I        EXT=(EX(I,J,K+1)+EX(I,J,K))*0.5                                                                  666  79
128  002CA2I        EXB=(EX(I,J,K)+EX(I,J,K-1))*0.5                                                                  667  80
129  002CFCI        EYE=(EY(I+1,J,K)+EY(I,J,K))*0.5                                                                  668  81
130  002D58I        EYW=(EY(I,J,K)+EY(I-1,J,K))*0.5                                                                  669  82
131  002DB4I        EYN=(EY(I,J+1,K)+EY(I,J,K))*0.5                                                                  670  83
132  002E10I        EYS=(EY(I,J,K)+EY(I,J-1,K))*0.5                                                                  671  84
133  002E6CI        EYT=(EY(I,J,K+1)+EY(I,J,K))*0.5                                                                  672  85
134  002EC6I        EYB=(EY(I,J,K)+EY(I,J,K-1))*0.5                                                                  673  86
135  002F20I        EZE=(EZ(I+1,J,K)+EZ(I,J,K))*0.5                                                                  674  87
136  002F7CI        EZW=(EZ(I,J,K)+EZ(I-1,J,K))*0.5                                                                  675  88
137  002FD8I        EZN=(EZ(I,J+1,K)+EZ(I,J,K))*0.5                                                                  676  89
138  003034I        EZS=(EZ(I,J,K)+EZ(I,J-1,K))*0.5                                                                  677  90
139  003090I        EZT=(EZ(I,J,K+1)+EZ(I,J,K))*0.5                                                                  678  91
140  0030EAI        EZB=(EZ(I,J,K)+EZ(I,J,K-1))*0.5                                                                  679  92
141  003144I        SXE=(SX(I+1,J,K)+SX(I,J,K))*0.5                                                                  680  93
142  0031A0I        SXW=(SX(I,J,K)+SX(I-1,J,K))*0.5                                                                  681  94
143  0031FCI        SXN=(SX(I,J+1,K)+SX(I,J,K))*0.5                                                                  682  95
144  003258I        SXS=(SX(I,J,K)+SX(I,J-1,K))*0.5                                                                  683  96
145  0032B4I        SXT=(SX(I,J,K+1)+SX(I,J,K))*0.5                                                                  684  97
146  00330EI        SXB=(SX(I,J,K)+SX(I,J,K-1))*0.5                                                                  685  98
147  003368I        SYE=(SY(I+1,J,K)+SY(I,J,K))*0.5                                                                  686  99
148  0033C4I        SYW=(SY(I,J,K)+SY(I-1,J,K))*0.5                                                                  687  100
149  003420I        SYN=(SY(I,J+1,K)+SY(I,J,K))*0.5                                                                  688  101
150  00347CI        SYS=(SY(I,J,K)+SY(I,J-1,K))*0.5                                                                  689  102
151  0034D8I        SYT=(SY(I,J,K+1)+SY(I,J,K))*0.5                                                                  690  103
152  003532I        SYB=(SY(I,J,K)+SY(I,J,K-1))*0.5                                                                  691  104
153  00358CI        SZE=(SZ(I+1,J,K)+SZ(I,J,K))*0.5                                                                  692  105
154  0035E8I        SZW=(SZ(I,J,K)+SZ(I-1,J,K))*0.5                                                                  693  106
155  003644I        SZN=(SZ(I,J+1,K)+SZ(I,J,K))*0.5                                                                  694  107
156  0036A0I        SZS=(SZ(I,J,K)+SZ(I,J-1,K))*0.5                                                                  695  108
157  0036FCI        SZT=(SZ(I,J,K+1)+SZ(I,J,K))*0.5                                                                  696  109
158  003756I        SZB=(SZ(I,J,K)+SZ(I,J,K-1))*0.5                                                                  697  110
159  0037B0I        TXXE(I,J,K)=CXE*CX(I,J,K)+CYE*CY(I,J,K)+CZE*CZ(I,J,K)                                            698  111
160  00385EI        TXXW(I,J,K)=CXW*CX(I,J,K)+CYW*CY(I,J,K)+CZW*CZ(I,J,K)                                            699  112
161  00390CI        TYYN(I,J,K)=EXN*EX(I,J,K)+EYN*EY(I,J,K)+EZN*EZ(I,J,K)                                            700  113
162  0039BAI        TYYS(I,J,K)=EXS*EX(I,J,K)+EYS*EY(I,J,K)+EZS*EZ(I,J,K)                                            701  114
163  003A68I        TZZT(I,J,K)=SXT*SX(I,J,K)+SYT*SY(I,J,K)+SZT*SZ(I,J,K)                                            702  115
164  003B16I        TZZB(I,J,K)=SXB*SX(I,J,K)+SYB*SY(I,J,K)+SZB*SZ(I,J,K)                                            703  116
165  003BC4I        TYXE(I,J,K)=(EXE*CX(I,J,K)+EYE*CY(I,J,K)+EZE*CZ(I,J,K))*0.25                                     704  117
166  003C78I        TYXW(I,J,K)=(EXW*CX(I,J,K)+EYW*CY(I,J,K)+EZW*CZ(I,J,K))*0.25                                     705  118
167  003D2CI        TYZT(I,J,K)=(EXT*SX(I,J,K)+EYT*SY(I,J,K)+EZT*SZ(I,J,K))*0.25                                     706  119
168  003DE0I        TYZB(I,J,K)=(EXB*SX(I,J,K)+EYB*SY(I,J,K)+EZB*SZ(I,J,K))*0.25                                     707  120
169  003E94I        TXYN(I,J,K)=(CXN*EX(I,J,K)+CYN*EY(I,J,K)+CZN*EZ(I,J,K))*0.25                                     708  121
170  003F48I        TXYS(I,J,K)=(CXS*EX(I,J,K)+CYS*EY(I,J,K)+CZS*EZ(I,J,K))*0.25                                     709  122
171  003FFCI        TXZT(I,J,K)=(CXT*SX(I,J,K)+CYT*SY(I,J,K)+CZT*SZ(I,J,K))*0.25                                     710  123
```

```
172  0040B0I          TXZB(I,J,K)=(CXB*SX(I,J,K)+CYB*SY(I,J,K)+CZB*SZ(I,J,K))*0.25                711     124
173  0041A4I          TZXE(I,J,K)=(SXE*CX(I,J,K)+SYE*CY(I,J,K)+SZE*CZ(I,J,K))*0.25                712     125
174  0042I8I          TZXW(I,J,K)=(SXW*CX(I,J,K)+SYW*CY(I,J,K)+SZW*CZ(I,J,K))*0.25                713     126
175  0042CCI          TZYN(I,J,K)=(SXN*EX(I,J,K)+SYN*EY(I,J,K)+SZN*EZ(I,J,K))*0.25                714     127
176  0043B0I          TZYS(I,J,K)=(SXS*EX(I,J,K)+SYS*EY(I,J,K)+SZS*EZ(I,J,K))*0.25                715     128
177  0044A4I    160   CONTINUE                                                                    716     129
178  00447CI          CALL WALVAL(1.0,2,LT,2,MT,2,NT,TXXE)                                        717     130
179  0044D4I          CALL WALVAL(1.0,2,LT,2,MT,2,NT,TXXW)                                        718     131
180  00452CI          CALL WALVAL(1.0,2,LT,2,MT,2,NT,TYYN)                                        719     132
181  0045B4I          CALL WALVAL(1.0,2,LT,2,MT,2,NT,TYYS)                                        720     133
182  00450CI          CALL WALVAL(1.0,2,LT,2,MT,2,NT,TZZT)                                        721     134
183  0046A4I          CALL WALVAL(1.0,2,LT,2,MT,2,NT,TZZB)                                        722     135
184  00468CI          CALL WALVAL(1.0,2,LT,2,MT,2,NT,TYXE)                                        723     136
185  0046E4I          CALL WALVAL(1.0,2,LT,2,MT,2,NT,TYXW)                                        724     137
186  00473CI          CALL WALVAL(1.0,2,LT,2,MT,2,NT,TYZT)                                        725     138
187  0047B4I          CALL WALVAL(1.0,2,LT,2,MT,2,NT,TYZB)                                        726     139
188  0047ECI          CALL WALVAL(1.0,2,LT,2,MT,2,NT,TXYN)                                        727     140
189  0048A4I          CALL WALVAL(1.0,2,LT,2,MT,2,NT,TXYS)                                        728     141
190  00489CI          CALL WALVAL(1.0,2,LT,2,MT,2,NT,TXZT)                                        729     142
191  0048F4I          CALL WALVAL(1.0,2,LT,2,MT,2,NT,TXZB)                                        730     143
192  00494CI          CALL WALVAL(1.0,2,LT,2,MT,2,NT,TZXE)                                        731     144
193  0049A4I          CALL WALVAL(1.0,2,LT,2,MT,2,NT,TZXW)                                        732     145
194  0049FCI          CALL WALVAL(1.0,2,LT,2,MT,2,NT,TZYN)                                        733     146
195  004A54I          CALL WALVAL(1.0,2,LT,2,MT,2,NT,TZYS)                                        734     147
196  004AACI          RETURN                                                                      735     148
197  004A94I          END                                                                         736     149

NO  ERRORS:F7D  R05-01.0C  SUBROUTINE  TRANF     02/21/86  09:50:44  TABLE SPACE:  10 KB
STATEMENT BUFFER:   20 LINES/1321 BYTES    STACK SPACE: 199 WORDS
SINGLE PRECISION FLOATING PT SUPPORT REQUIRED FOR EXECUTION
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```
  1  000000I          SUBROUTINE INIT                                                    737
  2  000004I          COMMON                                                             738
  3                  1/VAR/U(21,18,10),V(21,18,10),P(21,18,10),DK(21,18,10),            739
  4                  2  DE(21,18,10),ERRU,ERRV,ERRM,ERRK,ERRE,ERRW,                    740
  5                  3  PP(21,18,10),W(21,18,10),TM(21,18,10)                           741
  6                  1/PROP/ VISE(21,18,10),DEN(21,18,10),VISC,DENIN,FLOWIN          742
  7                  1/PCOR/ DU(21,18,10),DV(21,18,10),DW(21,18,10)                  743
  8  000004I          COMMON                                                             744
  9                  1/LIMT/ L,M,LT,MT,L1,L2,M1,M2,LC,MC,ISWU,ISWV,ISWP,ISWK,ISWE,     745
 10                  2   ALU,ALV,ALP,ALK,ALE,ALVIS,ALW,N,N1,N2,NC,ISWW,IG,NT,ALC,DTT   746
 11              C-----INITIALIZE VARIABLES                                             747
 12  000004I          DO 10 I=1,LC                                                       748        1
 13  000018I          DO 10 J=1,MC                                                       749        2
 14  00002CI          DO 10 K=1,NC                                                       750        3
 15  000040I          U(I,J,K)=0.0                                                       751        4
 16  000054I          V(I,J,K)=0.0                                                       752        5
 17  000068I          W(I,J,K)=0.0                                                       753        6
 18  00007CI          P(I,J,K)=0.0                                                       754        7
 19  000090I          PP(I,J,K)=0.0                                                      755        8
 20  0000A4I          DK(I,J,K)=0.0                                                      756        9
 21  0000B8I          DE(I,J,K)=0.0                                                      757       10
 22  0000CCI          DEN(I,J,K)=DENIN                                                   758       11
 23  0000E0I          VISE(I,J,K)=VISC                                                   759       12
 24  0000F4I       10 CONTINUE                                                           760       13
 25  000108I          RETURN                                                             761       14
 26  00011AI          END                                                                762       15

NO  ERRORS:F7D  R05-01.0C  SUBROUTINE  INIT      02/21/86  09:51:07  TABLE SPACE:    3 KB
STATEMENT BUFFER:    20 LINES/1321 BYTES     STACK SPACE: 126 WORDS
SINGLE PRECISION FLOATING PT SUPPORT REQUIRED FOR EXECUTION
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```
  1  000000I          SUBROUTINE NEWVIS                                                  763
  2  000004I          COMMON                                                             764
  3                  1/VAR/U(21,18,10),V(21,18,10),P(21,18,10),DK(21,18,10),            765
  4                  2  DE(21,18,10),ERRU,ERRV,ERRM,ERRK,ERRE,ERRW,                    766
  5                  3  PP(21,18,10),W(21,18,10),TM(21,18,10)                          767
  6                  1/PROP/ VISE(21,18,10),DEN(21,18,10),VISC,DENIN,FLOWIN             768
  7                  1/PCOR/ DU(21,18,10),DV(21,18,10),DW(21,18,10)                     769
  8                  1/TUR/ SIGK,SIGE,CMU,C1,C2,CMU1,CMU2,E,CK,HINUM,SMNUM,ANV1(800),   770
  9                  2 YN(800),YN1(800),SINX(800),SINY(800),SINZ(800),ANW1(800),        771
 10                  3 YPLN(800),TAUN(800),IBC(800),JBC(800),KBC(800),IITY(800),        772
 11                  4 TALW(800),GEN(21,18,10),MC(21,18,10),IJLO(21,18,10),IITO         773
 12  000004I          COMMON                                                             774
 13                  1/LIMT/ L,M,LT,MT,L1,L2,M1,M2,LO,MO,ISWU,ISWV,ISWP,ISWK,ISWE,      775
 14                  2   ALU,ALV,ALP,ALK,ALE,ALVIS,ALW,N,N1,N2,NO,ISWW,IG,NT,ALC,DTT    776
 15                C-----EVALUATE TURBULENT VISCOSITY                                   777
 16  000004I          DO 10 I=1,L                                                        778        1
 17  000018I          DO 10 J=1,M                                                        779        2
 18  00002CI          DO 10 K=1,N                                                        780        3
 19  000040I          IF(DK(I,J,K) .LE. SMNUM) DK(I,J,K)=SMNUM                           781        4
 20  00009EI          IF(DE(I,J,K) .LE. SMNUM) DE(I,J,K)=SMNUM                           782        6
 21  0000FCI          IF(DE(I,J,K) .LE. SMNUM) GO TO 12       -                          783        8
 22  000134I          TURVIS=DEN(I,J,K)*CMU*DK(I,J,K)**2/DE(I,J,K)+VISC                  784       10
 23  0001C2I          GO TO 14                                                           785       11
 24  0001C8I     12   TURVIS=VISC                                                        786       12
 25  0001D4I     14   CONTINUE                                                           787       13
 26  0001D4I          VISE(I,J,K)=VISE(I,J,K)+ALVIS*(TURVIS-VISE(I,J,K))                 788       14
 27  000252I     10   CONTINUE                                                           789       15
 28  00029AI          RETURN                                                             790       16
 29  0002A0I          END                                                                791       17

NO  ERRORS:F7D  R05-01.0C  SUBROUTINE  NEWVIS    02/21/86  09:51:23  TABLE SPACE:   5 KB
STATEMENT BUFFER:   20 LINES/1321 BYTES    STACK SPACE: 203 WORDS
SINGLE PRECISION FLOATING PT SUPPORT REQUIRED FOR EXECUTION
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```
      SUBROUTINE SOLVEQ(IE,ISWF,ALF,SIGF,ERRF,F,F0)                          792
      DIMENSION F(21,18,10),F1(21,18,10),F0(21,18,10)                        793
      COMMON                                                                 794
     1/VAR/U(21,18,10),V(21,18,10),P(21,18,10),QK(21,18,10),                 795
     2  QE(21,18,10),ERRU,ERRV,ERRM,ERRK,ERRE,ERRW,                          796
     3  PP(21,18,10),W(21,18,10),TM(21,18,10)                                797
     1/PROP/ VISE(21,18,10),DEN(21,18,10),VISC,DENIN,FLOWIN                 798
     1/PCOR/ DU(21,18,10),DV(21,18,10),DW(21,18,10)                         799
     1/TUR/ SIGK,SIGE,CMU,C1,C2,CMU1,CMU2,E,CK,HINUM,SMNUM,ANV1(800),        800
     2 YN(800),YN1(800),SINX(800),SINY(800),SINZ(800),ANW1(800),            801
     3 YPLN(800),TAUN(800),IBC(800),JBC(800),KBC(800),IITY(800),            802
     4 TAUW(800),GEN(21,18,10),MC(21,18,10),IJL0(21,18,10),IITO             803
     1/COEF/ AP(21,18,10),SU(21,18,10),SP(21,18,10),SUK(21,18,10),          804
     2   SPK(21,18,10),AE(21,18,10),AW(21,18,10),AN(21,18,10),              805
     3   AS(21,18,10),AT(21,18,10),AB(21,18,10),APO(21,18,10)               806
      COMMON                                                                 807
     1/TRAN/ X(21,18,10),Y(21,18,10),Z(21,18,10),TJO(21,18,10),             808
     2   CX(21,18,10),CY(21,18,10),CZ(21,18,10),                            809
     3   EX(21,18,10),EY(21,18,10),EZ(21,18,10),                            810
     3   SX(21,18,10),SY(21,18,10),SZ(21,18,10)                             811
     1/LIMT/ L,M,LT,MT,L1,L2,M1,M2,LO,MO,ISWU,ISWV,ISWP,ISWK,ISWE,           812
     2   ALU,ALV,ALP,ALK,ALE,ALVIS,ALW,N,N1,N2,NO,ISWW,IG,NT,ALC,DTT        813
      COMMON                                                                 814
     1/TTRAN/TXXE(21,18,10),TXXW(21,18,10),TYYN(21,18,10),                  815
     2   TYYS(21,18,10),TZZT(21,18,10),TZZB(21,18,10),                      816
     3   TYXE(21,18,10),TYXW(21,18,10),TYZT(21,18,10),                      817
     4   TYZB(21,18,10),TXYN(21,18,10),TXYS(21,18,10),                      818
     5   TXZT(21,18,10),TXZB(21,18,10),TZXE(21,18,10),                      819
     6   TZXW(21,18,10),TZYN(21,18,10),TZYS(21,18,10)                       820
C------TRANSPORT EQUATIONS LINEARIZATION AND SOLVER                         821
      ERRF=0.0                                                               822      1
      PI=3.141592654                                                         823      2
C-----PRESSURE CORRECTION SOLVER STARTS FROM 10                             824
      IF(IE .EQ. 0) GO TO 10                                                 825      3
C-----U, V, W, TM, K & E EQUATIONS                                          826
      IS=2                                                                   827      5
      IT=LT                                                                  828      6
      JS=2                                                                   829      7
      JT=MT                                                                  830      8
      KS=2                                                                   831      9
      KT=NT                                                                  832     10
      GO TO (21,29,29,29,29,21,21,21), IE                                    833     11
   21 CONTINUE                                                               834     12
      DO 22 I=IS,IT                                                          835     13
      DO 22 J=JS,JT                                                          836     14
      DO 22 K=KS,KT                                                          837     15
   22 F1(I,J,K)=VISE(I,J,K)/SIGF                                             838     16
C-----EVALUATE LINK COEFF. AND SOURCE TERMS                                 839
      DO 20 I=IS,IT                                                          840     17
      DO 20 J=JS,JT                                                          841     18
      DO 20 K=KS,KT                                                          842     19
      GAE=0.5*(F1(I+1,J,K)+F1(I,J,K))                                        843     20
      GAW=0.5*(F1(I-1,J,K)+F1(I,J,K))                                        844     21
      GAN=0.5*(F1(I,J+1,K)+F1(I,J,K))                                        845     22
      GAS=0.5*(F1(I,J-1,K)+F1(I,J,K))                                        846     23
      GAT=0.5*(F1(I,J,K+1)+F1(I,J,K))                                        847     24
      GAB=0.5*(F1(I,J,K-1)+F1(I,J,K))                                        848     25
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 58  003F18I          DENE=0.5*(DEN(I+1,J,K)+DEN(I,J,K))                                  849      26
 59  003F74I          DENW=0.5*(DEN(I-1,J,K)+DEN(I,J,K))                                  850      27
 60  003FD0I          DENN=0.5*(DEN(I,J+1,K)+DEN(I,J,K))                                  851      28
 61  00402CI          DENS=0.5*(DEN(I,J-1,K)+DEN(I,J,K))                                  852      29
 62  004088I          DENT=0.5*(DEN(I,J,K+1)+DEN(I,J,K))                                  853      30
 63  0040E2I          DENB=0.5*(DEN(I,J,K-1)+DEN(I,J,K))                                  854      31
 64  00413CI          UE=C.5*(U(I+1,J,K)+U(I,J,K))                                        855      32
 65  004198I          UW=C.5*(U(I-1,J,K)+U(I,J,K))                                        856      33
 66  0041F4I          UN=C.5*(U(I,J+1,K)+U(I,J,K))                                        857      34
 67  004250I          US=C.5*(U(I,J-1,K)+U(I,J,K))                                        858      35
 68  0042ACI          UT=C.5*(U(I,J,K+1)+U(I,J,K))                                        859      36
 69  004306I          UB=0.5*(U(I,J,K-1)+U(I,J,K))                                        860      37
 70  004360I          VE=0.5*(V(I+1,J,K)+V(I,J,K))                                        861      38
 71  0043BCI          VW=C.5*(V(I-1,J,K)+V(I,J,K))                                        862      39
 72  004418I          VN=C.5*(V(I,J+1,K)+V(I,J,K))                                        863      40
 73  004474I          VS=C.5*(V(I,J-1,K)+V(I,J,K))                                        864      41
 74  0044D0I          VT=C.5*(V(I,J,K+1)+V(I,J,K))                                        865      42
 75  00452AI          VB=C.5*(V(I,J,K-1)+V(I,J,K))                                        866      43
 76  004584I          WE=C.5*(W(I+1,J,K)+W(I,J,K))                                        867      44
 77  0045E0I          WW=C.5*(W(I-1,J,K)+W(I,J,K))                                        868      45
 78  00463CI          WN=C.5*(W(I,J+1,K)+W(I,J,K))                                        869      46
 79  004698I          WS=C.5*(W(I,J-1,K)+W(I,J,K))                                        870      47
 80  0046F4I          WT=C.5*(W(I,J,K+1)+W(I,J,K))                                        871      48
 81  00474EI          WB=C.5*(W(I,J,K-1)+W(I,J,K))                                        872      49
 82  0047A8I          CE=DENE*(UE*CX(I,J,K)+VE*CY(I,J,K)+WE*CZ(I,J,K))                    873      50
 83  00483CI          CW=DENW*(UW*CX(I,J,K)+VW*CY(I,J,K)+WW*CZ(I,J,K))                    874      51
 84  0048D0I          CN=DENN*(UN*EX(I,J,K)+VN*EY(I,J,K)+WN*EZ(I,J,K))                    875      52
 85  004964I          CS=DENS*(US*EX(I,J,K)+VS*EY(I,J,K)+WS*EZ(I,J,K))                    876      53
 86  0049F8I          CT=DENT*(UT*SX(I,J,K)+VT*SY(I,J,K)+WT*SZ(I,J,K))                    877      54
 87  004A8CI          CB=DENB*(UB*SX(I,J,K)+VB*SY(I,J,K)+WB*SZ(I,J,K))                    878      55
 88  004B20I          DDE=GAE*TXXE(I,J,K)+GAN*TXYN(I,J,K)-GAS*TXYS(I,J,K)+                879      56
 89                  1     GAT*TXZT(I,J,K)-GAB*TXZB(I,J,K)                               880
 90  004C0AI          DDW=GAW*TXXW(I,J,K)+GAS*TXYS(I,J,K)-GAN*TXYN(I,J,K)-                881      57
 91                  1     GAT*TXZT(I,J,K)+GAB*TXZB(I,J,K)                               882
 92  004CF4I          DDN=GAN*TYYN(I,J,K)+GAE*TYXE(I,J,K)-GAW*TYXW(I,J,K)+                883      58
 93                  1     GAT*TYZT(I,J,K)-GAB*TYZB(I,J,K)                               884
 94  004DDEI          DDS=GAS*TYYS(I,J,K)+GAW*TYXW(I,J,K)-GAE*TYXE(I,J,K)-                885      59
 95                  1     GAT*TYZT(I,J,K)+GAB*TYZB(I,J,K)                               886
 96  004EC8I          DDT=GAT*TZZT(I,J,K)+GAE*TZXE(I,J,K)-GAW*TZXW(I,J,K)+                887      60
 97                  1     GAN*TZYN(I,J,K)-GAS*TZYS(I,J,K)                               888
 98  004FB2I          DDB=GAB*TZZB(I,J,K)+GAW*TZXW(I,J,K)-GAE*TZXE(I,J,K)-                889      61
 99                  1     GAN*TZYN(I,J,K)+GAS*TZYS(I,J,K)                               890
100  00509CI          CPO=ABS(CE-CW+CN-CS+CT-CB)                                          891      62
101  0050E2I          AE(I,J,K)=(AMAX1(ABS(0.5*CE),DDE)-0.5*CE)*TJO(I,J,K)                892      63
102  005186I          AW(I,J,K)=(AMAX1(ABS(0.5*CW),DDW)+0.5*CW)*TJO(I,J,K)                893      64
103  00522AI          AN(I,J,K)=(AMAX1(ABS(0.5*CN),DDN)-0.5*CN)*TJO(I,J,K)                894      65
104  0052CEI          AS(I,J,K)=(AMAX1(ABS(0.5*CS),DDS)+0.5*CS)*TJO(I,J,K)                895      66
105  005372I          AT(I,J,K)=(AMAX1(ABS(0.5*CT),DDT)-0.5*CT)*TJO(I,J,K)                896      67
106  005416I          AB(I,J,K)=(AMAX1(ABS(0.5*CB),DDB)+0.5*CB)*TJO(I,J,K)                897      68
107  00548AI          DDNE=GAE*TYXE(I,J,K)+GAN*TXYN(I,J,K)                                898      69
108  00551AI          DDSE=-GAE*TYXE(I,J,K)-GAS*TXYS(I,J,K)                               899      70
109  005582I          DDNW=-GAW*TYXW(I,J,K)-GAN*TXYN(I,J,K)                               900      71
110  0055EAI          DDSW=GAW*TYXW(I,J,K)+GAS*TXYS(I,J,K)                                901      72
111  00564AI          DDET=GAE*TZXE(I,J,K)+GAT*TXZT(I,J,K)                                902      73
112  0056AAI          DDEB=-GAE*TZXE(I,J,K)-GAB*TXZB(I,J,K)                               903      74
113  005712I          DDWT=-GAW*TZXW(I,J,K)-GAT*TXZT(I,J,K)                               904      75
114  00577AI          DDWB=GAW*TZXW(I,J,K)+GAB*TXZB(I,J,K)                                905      76
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      DONT=GAN*TZYN(I,J,K)+GAT*TYZT(I,J,K)                                  906       77
      DONB=-GAN*TZYN(I,J,K)-GAB*TYZB(I,J,K)                                 907       78
      DOST=-GAS*TZYS(I,J,K)-GAT*TYZT(I,J,K)                                 908       79
      DOSB=GAS*TZYS(I,J,K)+GAB*TYZB(I,J,K)                                  909       80
      SU(I,J,K)=CPO*F(I,J,K)+DONE*F(I+1,J+1,K)+DOSE*F(I+1,J-1,K)+           910       81
     1          DONW*F(I-1,J+1,K)+DOSW*F(I-1,J-1,K)+                        911
     2          DOET*F(I+1,J,K+1)+DOEB*F(I+1,J,K-1)+                        912
     3          DOWT*F(I-1,J,K+1)+DOWB*F(I-1,J,K-1)+                        913
     4          DONT*F(I,J+1,K+1)+DONB*F(I,J+1,K-1)+                        914
     5          DOST*F(I,J-1,K+1)+DOSB*F(I,J-1,K-1)                         915
      DV(I,J,K)=CPO*V(I,J,K)+DONE*V(I+1,J+1,K)+DOSE*V(I+1,J-1,K)+           916       82
     1          DONW*V(I-1,J+1,K)+DOSW*V(I-1,J-1,K)+                        917
     2          DOET*V(I+1,J,K+1)+DOEB*V(I+1,J,K-1)+                        918
     3          DOWT*V(I-1,J,K+1)+DOWB*V(I-1,J,K-1)+                        919
     4          DONT*V(I,J+1,K+1)+DONB*V(I,J+1,K-1)+                        920
     5          DOST*V(I,J-1,K+1)+DOSB*V(I,J-1,K-1)                         921
      DW(I,J,K)=CPO*W(I,J,K)+DONE*W(I+1,J+1,K)+DOSE*W(I+1,J-1,K)+           922       83
     1          DONW*W(I-1,J+1,K)+DOSW*W(I-1,J-1,K)+                        923
     2          DOET*W(I+1,J,K+1)+DOEB*W(I+1,J,K-1)+                        924
     3          DOWT*W(I-1,J,K+1)+DOWB*W(I-1,J,K-1)+                        925
     4          DONT*W(I,J+1,K+1)+DONB*W(I,J+1,K-1)+                        926
     5          DOST*W(I,J-1,K+1)+DOSB*W(I,J-1,K-1)                         927
      PP(I,J,K)=CPO*DK(I,J,K)+DONE*DK(I+1,J+1,K)+DOSE*DK(I+1,J-1,K)+        928       84
     1          DONW*DK(I-1,J+1,K)+DOSW*DK(I-1,J-1,K)+                      929
     2          DOET*DK(I+1,J,K+1)+DOEB*DK(I+1,J,K-1)+                      930
     3          DOWT*DK(I-1,J,K+1)+DOWB*DK(I-1,J,K-1)+                      931
     4          DONT*DK(I,J+1,K+1)+DONB*DK(I,J+1,K-1)+                      932
     5          DOST*DK(I,J-1,K+1)+DOSB*DK(I,J-1,K-1)                       933
      SUK(I,J,K)=CPO                                                        934       85
      SPK(I,J,K)=-CPO-APC(I,J,K)                                            935       86
      UCXI=UE-UW                                                            936       87
      UEDA=UN-US                                                            937       88
      USCI=UT-UB                                                            938       89
      VCXI=VE-VW                                                            939       90
      VEDA=VN-VS                                                            940       91
      VSCI=VT-VB                                                            941       92
      WCXI=WE-WW                                                            942       93
      WEDA=WN-WS                                                            943       94
      WSCI=WT-WB                                                            944       95
      UX=UCXI*CX(I,J,K)+UEDA*EX(I,J,K)+USCI*SX(I,J,K)                       945       96
      UY=UCXI*CY(I,J,K)+UEDA*EY(I,J,K)+USCI*SY(I,J,K)                       946       97
      UZ=UCXI*CZ(I,J,K)+UEDA*EZ(I,J,K)+USCI*SZ(I,J,K)                       947       98
      VX=VCXI*CX(I,J,K)+VEDA*EX(I,J,K)+VSCI*SX(I,J,K)                       948       99
      VY=VCXI*CY(I,J,K)+VEDA*EY(I,J,K)+VSCI*SY(I,J,K)                       949      100
      VZ=VCXI*CZ(I,J,K)+VEDA*EZ(I,J,K)+VSCI*SZ(I,J,K)                       950      101
      WX=WCXI*CX(I,J,K)+WEDA*EX(I,J,K)+WSCI*SX(I,J,K)                       951      102
      WY=WCXI*CY(I,J,K)+WEDA*EY(I,J,K)+WSCI*SY(I,J,K)                       952      103
      WZ=WCXI*CZ(I,J,K)+WEDA*EZ(I,J,K)+WSCI*SZ(I,J,K)                       953      104
      GEN(I,J,K)=VISE(I,J,K)*((UY+VX)**2+(VZ+WY)**2+(WX+UZ)**2+             954      105
     1          2*(UX*UX+VY*VY+WZ*WZ))                                      955
   20 CONTINUE                                                              956      106
C-------------------------------------------------------------------------  957
   29 CONTINUE                                                              958      107
C-----CALCULATE SOURCE TERMS                                                959
      GO TO (1,2,3,4,5,6,7), IE                                             960      108
C-----U-, V-, W-SOURCES                                                     961
    1 CONTINUE                                                              962      109
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      DO 15 I=IS,IT                                                     963     110
      DO 15 J=JS,JT                                                     964     111
      DO 15 K=KS,KT                                                     965     112
      GAE=0.5*(F1(I+1,J,K)+F1(I,J,K))                                   966     113
      GAW=0.5*(F1(I-1,J,K)+F1(I,J,K))                                   967     114
      GAN=0.5*(F1(I,J+1,K)+F1(I,J,K))                                   968     115
      GAS=0.5*(F1(I,J-1,K)+F1(I,J,K))                                   969     116
      GAT=0.5*(F1(I,J,K+1)+F1(I,J,K))                                   970     117
      GAB=0.5*(F1(I,J,K-1)+F1(I,J,K))                                   971     118
      PCXI=(P(I+1,J,K)+P(I+1,J+1,K)+P(I+1,J,K+1)+P(I+1,J+1,K+1)-        972     119
     1       P(I,J,K)-P(I,J+1,K)-P(I,J,K+1)-P(I,J+1,K+1))*0.25          973
      PEDA=(P(I,J+1,K)+P(I+1,J+1,K)+P(I,J+1,K+1)+P(I+1,J+1,K+1)-        974     120
     1       P(I,J,K)-P(I+1,J,K)-P(I,J,K+1)-P(I+1,J,K+1))*0.25          975
      PSCI=(P(I,J,K+1)+P(I+1,J,K+1)+P(I,J+1,K+1)+P(I+1,J+1,K+1)-        976     121
     1       P(I,J,K)-P(I+1,J,K)-P(I,J+1,K)-P(I+1,J+1,K))*0.25          977
      UCE=U(I+1,J,K)-U(I,J,K)                                           978     122
      UCW=U(I,J,K)-U(I-1,J,K)                                           979     123
      UCN=(U(I+1,J+1,K)+U(I+1,J,K)-U(I-1,J+1,K)-U(I-1,J,K))*0.25        980     124
      UCS=(U(I+1,J,K)+U(I+1,J-1,K)-U(I-1,J,K)-U(I-1,J-1,K))*0.25        981     125
      UCT=(U(I+1,J,K+1)+U(I+1,J,K)-U(I-1,J,K+1)-U(I-1,J,K))*0.25        982     126
      UCB=(U(I+1,J,K)+U(I+1,J,K-1)-U(I-1,J,K)-U(I-1,J,K-1))*0.25        983     127
      UEE=(U(I+1,J+1,K)+U(I,J+1,K)-U(I+1,J-1,K)-U(I,J-1,K))*0.25        984     128
      UEW=(U(I,J+1,K)+U(I-1,J+1,K)-U(I,J-1,K)-U(I-1,J-1,K))*0.25        985     129
      UEN=U(I,J+1,K)-U(I,J,K)                                           986     130
      UES=U(I,J,K)-U(I,J-1,K)                                           987     131
      UET=(U(I,J+1,K+1)+U(I,J+1,K)-U(I,J-1,K+1)-U(I,J-1,K))*0.25        988     132
      UEB=(U(I,J+1,K)+U(I,J+1,K-1)-U(I,J-1,K)-U(I,J-1,K-1))*0.25        989     133
      USE=(U(I+1,J,K+1)+U(I,J,K+1)-U(I+1,J,K-1)-U(I,J,K-1))*0.25        990     134
      USW=(U(I,J,K+1)+U(I-1,J,K+1)-U(I,J,K-1)-U(I-1,J,K-1))*0.25        991     135
      USN=(U(I,J+1,K+1)+U(I,J,K+1)-U(I,J+1,K-1)-U(I,J,K-1))*0.25        992     136
      USS=(U(I,J,K+1)+U(I,J-1,K+1)-U(I,J,K-1)-U(I,J-1,K-1))*0.25        993     137
      UST=U(I,J,K+1)-U(I,J,K)                                           994     138
      USB=U(I,J,K)-U(I,J,K-1)                                           995     139
      VCE=V(I+1,J,K)-V(I,J,K)                                           996     140
      VCW=V(I,J,K)-V(I-1,J,K)                                           997     141
      VCN=(V(I+1,J+1,K)+V(I+1,J,K)-V(I-1,J+1,K)-V(I-1,J,K))*0.25        998     142
      VCS=(V(I+1,J,K)+V(I+1,J-1,K)-V(I-1,J,K)-V(I-1,J-1,K))*0.25        999     143
      VCT=(V(I+1,J,K+1)+V(I+1,J,K)-V(I-1,J,K+1)-V(I-1,J,K))*0.25        1000    144
      VCB=(V(I+1,J,K)+V(I+1,J,K-1)-V(I-1,J,K)-V(I-1,J,K-1))*0.25        1001    145
      VEE=(V(I+1,J+1,K)+V(I,J+1,K)-V(I+1,J-1,K)-V(I,J-1,K))*0.25        1002    146
      VEW=(V(I,J+1,K)+V(I-1,J+1,K)-V(I,J-1,K)-V(I-1,J-1,K))*0.25        1003    147
      VEN=V(I,J+1,K)-V(I,J,K)                                           1004    148
      VES=V(I,J,K)-V(I,J-1,K)                                           1005    149
      VET=(V(I,J+1,K+1)+V(I,J+1,K)-V(I,J-1,K+1)-V(I,J-1,K))*0.25        1006    150
      VEB=(V(I,J+1,K)+V(I,J+1,K-1)-V(I,J-1,K)-V(I,J-1,K-1))*0.25        1007    151
      VSE=(V(I+1,J,K+1)+V(I,J,K+1)-V(I+1,J,K-1)-V(I,J,K-1))*0.25        1008    152
      VSW=(V(I,J,K+1)+V(I-1,J,K+1)-V(I,J,K-1)-V(I-1,J,K-1))*0.25        1009    153
      VSN=(V(I,J+1,K+1)+V(I,J,K+1)-V(I,J+1,K-1)-V(I,J,K-1))*0.25        1010    154
      VSS=(V(I,J,K+1)+V(I,J-1,K+1)-V(I,J,K-1)-V(I,J-1,K-1))*0.25        1011    155
      VST=V(I,J,K+1)-V(I,J,K)                                           1012    156
      VSB=V(I,J,K)-V(I,J,K-1)                                           1013    157
      WCE=W(I+1,J,K)-W(I,J,K)                                           1014    158
      WCW=W(I,J,K)-W(I-1,J,K)                                           1015    159
      WCN=(W(I+1,J+1,K)+W(I+1,J,K)-W(I-1,J+1,K)-W(I-1,J,K))*0.25        1016    160
      WCS=(W(I+1,J,K)+W(I+1,J-1,K)-W(I-1,J,K)-W(I-1,J-1,K))*0.25        1017    161
      WCT=(W(I+1,J,K+1)+W(I+1,J,K)-W(I-1,J,K+1)-W(I-1,J,K))*0.25        1018    162
      WCB=(W(I+1,J,K)+W(I+1,J,K-1)-W(I-1,J,K)-W(I-1,J,K-1))*0.25        1019    163
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229  008BD2I        WEE=(W(I+1,J+1,K)+W(I,J+1,K)-W(I+1,J-1,K)-W(I,J-1,K))*0.25                                  1020    164
230  008C9EI        WEW=(W(I,J+1,K)+W(I-1,J+1,K)-W(I,J-1,K)-W(I-1,J-1,K))*0.25                                 1021    165
231  008D4AI        WEN=W(I,J+1,K)-W(I,J,K)                                                                    1022    166
232  008DA0I        WES=W(I,J,K)-W(I,J-1,K)                                                                    1023    167
233  008DF6I        WET=(W(I,J+1,K+1)+W(I,J+1,K)-W(I,J-1,K+1)-W(I,J-1,K))*0.25                                 1024    168
234  008EAEI        WEB=(W(I,J+1,K)+W(I,J+1,K-1)-W(I,J-1,K)-W(I,J-1,K-1))*0.25                                 1025    169
235  008F66I        WSE=(W(I+1,J,K+1)+W(I,J,K+1)-W(I+1,J,K-1)-W(I,J,K-1))*0.25                                 1026    170
236  00901AI        WSW=(W(I,J,K+1)+W(I-1,J,K+1)-W(I,J,K-1)-W(I-1,J,K-1))*0.25                                 1027    171
237  0090CEI        WSN=(W(I,J+1,K+1)+W(I,J,K+1)-W(I,J+1,K-1)-W(I,J,K-1))*0.25                                 1028    172
238  009182I        WSS=(W(I,J,K+1)+W(I,J-1,K+1)-W(I,J,K-1)-W(I,J-1,K-1))*0.25                                 1029    173
239  009236I        WST=W(I,J,K+1)-W(I,J,K)                                                                    1030    174
240  00928AI        WSB=W(I,J,K)-W(I,J,K-1)                                                                    1031    175
241  0092DEI        SU(I,J,K)=SU(I,J,K)-PCXI*CX(I,J,K)-PEDA*EX(I,J,K)-PSCI*SX(I,J,K)                           1032    176
242  009384I        DV(I,J,K)=DV(I,J,K)-PCXI*CY(I,J,K)-PEDA*EY(I,J,K)-PSCI*SY(I,J,K)                           1033    177
243  00948AI        DW(I,J,K)=DW(I,J,K)-PCXI*CZ(I,J,K)-PEDA*EZ(I,J,K)-PSCI*SZ(I,J,K)                           1034    178
244  009560I        CXE=(CX(I+1,J,K)+CX(I,J,K))*0.5                                                            1035    179
245  00953CI        CXW=(CX(I-1,J,K)+CX(I,J,K))*0.5                                                            1036    180
246  009618I        CXN=(CX(I,J+1,K)+CX(I,J,K))*0.5                                                            1037    181
247  009674I        CXS=(CX(I,J,K)+CX(I,J-1,K))*0.5                                                            1038    182
248  0096D0I        CXT=(CX(I,J,K+1)+CX(I,J,K))*0.5                                                            1039    183
249  00972AI        CXB=(CX(I,J,K)+CX(I,J,K-1))*0.5                                                            1040    184
250  009784I        EXE=(EX(I+1,J,K)+EX(I,J,K))*0.5                                                            1041    185
251  0097E0I        EXW=(EX(I,J,K)+EX(I-1,J,K))*0.5                                                            1042    186
252  00983CI        EXN=(EX(I,J+1,K)+EX(I,J,K))*0.5                                                            1043    187
253  009898I        EXS=(EX(I,J,K)+EX(I,J-1,K))*0.5                                                            1044    188
254  0098F4I        EXT=(EX(I,J,K+1)+EX(I,J,K))*0.5                                                            1045    189
255  00994EI        EXB=(EX(I,J,K)+EX(I,J,K-1))*0.5                                                            1046    190
256  0099A8I        SXE=(SX(I+1,J,K)+SX(I,J,K))*0.5                                                            1047    191
257  009A04I        SXW=(SX(I,J,K)+SX(I-1,J,K))*0.5                                                            1048    192
258  009A60I        SXN=(SX(I,J+1,K)+SX(I,J,K))*0.5                                                            1049    193
259  009ABCI        SXS=(SX(I,J,K)+SX(I,J-1,K))*0.5                                                            1050    194
260  009B18I        SXT=(SX(I,J,K+1)+SX(I,J,K))*0.5                                                            1051    195
261  009B72I        SXB=(SX(I,J,K)+SX(I,J,K-1))*0.5                                                            1052    196
262  009BCCI        QE=GAE*(UCE*CXE+UEE*EXE+USE*SXE)                                                           1053    197
263  009C00I        QW=GAW*(UCW*CXW+UEW*EXW+USW*SXW)                                                           1054    198
264  009C34I        QN=GAN*(UCN*CXN+UEN*EXN+USN*SXN)                                                           1055    199
265  009C68I        QS=GAS*(UCS*CXS+UES*EXS+USS*SXS)                                                           1056    200
266  009C9CI        QT=GAT*(UCT*CXT+UET*EXT+UST*SXT)                                                           1057    201
267  009CD0I        QB=GAB*(UCB*CXB+UEB*EXB+USB*SXB)                                                           1058    202
268  009D04I        SOC1=CX(I,J,K)*(QE-QW)+EX(I,J,K)*(QN-QS)+SX(I,J,K)*(QT-QB)                                 1059    203
269  009DA4I        QE=GAE*(VCE*CXE+VEE*EXE+VSE*SXE)                                                           1060    204
270  009DD8I        QW=GAW*(VCW*CXW+VEW*EXW+VSW*SXW)                                                           1061    205
271  009E0CI        QN=GAN*(VCN*CXN+VEN*EXN+VSN*SXN)                                                           1062    206
272  009E40I        QS=GAS*(VCS*CXS+VES*EXS+VSS*SXS)                                                           1063    207
273  009E74I        QT=GAT*(VCT*CXT+VET*EXT+VST*SXT)                                                           1064    208
274  009EA8I        QB=GAB*(VCB*CXB+VEB*EXB+VSB*SXB)                                                           1065    209
275  009EDCI        SOC2=CY(I,J,K)*(QE-QW)+EY(I,J,K)*(QN-QS)+SY(I,J,K)*(QT-QB)                                 1066    210
276  009F7CI        QE=GAE*(WCE*CXE+WEE*EXE+WSE*SXE)                                                           1067    211
277  009FB0I        QW=GAW*(WCW*CXW+WEW*EXW+WSW*SXW)                                                           1068    212
278  009FE4I        QN=GAN*(WCN*CXN+WEN*EXN+WSN*SXN)                                                           1069    213
279  00A018I        QS=GAS*(WCS*CXS+WES*EXS+WSS*SXS)                                                           1070    214
280  00A04CI        QT=GAT*(WCT*CXT+WET*EXT+WST*SXT)                                                           1071    215
281  00A080I        QB=GAB*(WCB*CXB+WEB*EXB+WSB*SXB)                                                           1072    216
282  00A0B4I        SOC3=CZ(I,J,K)*(QE-QW)+EZ(I,J,K)*(QN-QS)+SZ(I,J,K)*(QT-QB)                                 1073    217
283  00A154I        SU(I,J,K)=SU(I,J,K)+SOC1+SOC2+SOC3+APO(I,J,K)*FO(I,J,K)                                    1074    218
284  00A206I        CYE=(CY(I+1,J,K)+CY(I,J,K))*0.5                                                            1075    219
285  00A262I        CYW=(CY(I,J,K)+CY(I-1,J,K))*0.5                                                            1076    220
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286  00A28EI          CYN=(CY(I,J+1,K)+CY(I,J,K))*0.5                                                    1077    221
287  00A31AI          CYS=(CY(I,J,K)+CY(I,J-1,K))*0.5                                                    1078    222
288  00A376I          CYT=(CY(I,J,K+1)+CY(I,J,K))*0.5                                                    1079    223
289  00A300I          CYB=(CY(I,J,K)+CY(I,J,K-1))*0.5                                                    1080    224
290  00A42AI          EYE=(EY(I+1,J,K)+EY(I,J,K))*0.5                                                    1081    225
291  00A486I          EYW=(EY(I,J,K)+EY(I-1,J,K))*0.5                                                    1082    226
292  00A4E2I          EYN=(EY(I,J+1,K)+EY(I,J,K))*0.5                                                    1083    227
293  00A53EI          EYS=(EY(I,J,K)+EY(I,J-1,K))*0.5                                                    1084    228
294  00A59AI          EYT=(EY(I,J,K+1)+EY(I,J,K))*0.5                                                    1085    229
295  00A5F4I          EYB=(EY(I,J,K)+EY(I,J,K-1))*0.5                                                    1086    230
296  00A64EI          SYE=(SY(I+1,J,K)+SY(I,J,K))*0.5                                                    1087    231
297  00A6AAI          SYW=(SY(I,J,K)+SY(I-1,J,K))*0.5                                                    1088    232
298  00A706I          SYN=(SY(I,J+1,K)+SY(I,J,K))*0.5                                                    1089    233
299  00A762I          SYS=(SY(I,J,K)+SY(I,J-1,K))*0.5                                                    1090    234
300  00A7BEI          SYT=(SY(I,J,K+1)+SY(I,J,K))*0.5                                                    1091    235
301  00A818I          SYB=(SY(I,J,K)+SY(I,J,K-1))*0.5                                                    1092    236
302  00A872I          QE=GAE*(UCE*CYE+UEE*EYE+USE*SYE)                                                   1093    237
303  00A8A6I          QW=GAW*(UCW*CYW+UEW*EYW+USW*SYW)                                                   1094    238
304  00A8DAI          QN=GAN*(UCN*CYN+UEN*EYN+USN*SYN)                                                   1095    239
305  00A90EI          QS=GAS*(UCS*CYS+UES*EYS+USS*SYS)                                                   1096    240
306  00A942I          QT=GAT*(UCT*CYT+UET*EYT+UST*SYT)                                                   1097    241
307  00A976I          QB=GAB*(UCB*CYB+UEB*EYB+USB*SYB)                                                   1098    242
308  00A9AAI          SOC1=CX(I,J,K)*(QE-QW)+EX(I,J,K)*(QN-QS)+SX(I,J,K)*(QT-QB)                         1099    243
309  00AA4AI          QE=GAE*(VCE*CYE+VEE*EYE+VSE*SYE)                                                   1100    244
310  00AA7EI          QW=GAW*(VCW*CYW+VEW*EYW+VSW*SYW)                                                   1101    245
311  00AAB2I          QN=GAN*(VCN*CYN+VEN*EYN+VSN*SYN)                                                   1102    246
312  00AAE6I          QS=GAS*(VCS*CYS+VES*EYS+VSS*SYS)                                                   1103    247
313  00AB1AI          QT=GAT*(VCT*CYT+VET*EYT+VST*SYT)                                                   1104    248
314  00AB4EI          QB=GAB*(VCB*CYB+VEB*EYB+VSB*SYB)                                                   1105    249
315  00AB82I          SOC2=CY(I,J,K)*(QE-QW)+EY(I,J,K)*(QN-QS)+SY(I,J,K)*(QT-QB)                         1106    250
316  00AC22I          QE=GAE*(WCE*CYE+WEE*EYE+WSE*SYE)                                                   1107    251
317  00AC56I          QW=GAW*(WCW*CYW+WEW*EYW+WSW*SYW)                                                   1108    252
318  00AC8AI          QN=GAN*(WCN*CYN+WEN*EYN+WSN*SYN)                                                   1109    253
319  00ACBEI          QS=GAS*(WCS*CYS+WES*EYS+WSS*SYS)                                                   1110    254
320  00ACF2I          QT=GAT*(WCT*CYT+WET*EYT+WST*SYT)                                                   1111    255
321  00AD26I          QB=GAB*(WCB*CYB+WEB*EYB+WSB*SYB)                                                   1112    256
322  00AD5AI          SOC3=CZ(I,J,K)*(QE-QW)+EZ(I,J,K)*(QN-QS)+SZ(I,J,K)*(QT-QB)                         1113    257
323  00ADFAI          DV(I,J,K)=DV(I,J,K)+SOC1+SOC2+SOC3                                                 1114    258
324  00AE58I          CZE=(CZ(I+1,J,K)+CZ(I,J,K))*0.5                                                    1115    259
325  00AE94I          CZW=(CZ(I,J,K)+CZ(I-1,J,K))*0.5                                                    1116    260
326  00AF10I          CZN=(CZ(I,J+1,K)+CZ(I,J,K))*0.5                                                    1117    261
327  00AF6CI          CZS=(CZ(I,J,K)+CZ(I,J-1,K))*0.5                                                    1118    262
328  00AFC8I          CZT=(CZ(I,J,K+1)+CZ(I,J,K))*0.5                                                    1119    263
329  00B022I          CZB=(CZ(I,J,K)+CZ(I,J,K-1))*0.5                                                    1120    264
330  00B07CI          EZE=(EZ(I+1,J,K)+EZ(I,J,K))*0.5                                                    1121    265
331  00B0D8I          EZW=(EZ(I,J,K)+EZ(I-1,J,K))*0.5                                                    1122    266
332  00B134I          EZN=(EZ(I,J+1,K)+EZ(I,J,K))*0.5                                                    1123    267
333  00B190I          EZS=(EZ(I,J,K)+EZ(I,J-1,K))*0.5                                                    1124    268
334  00B1ECI          EZT=(EZ(I,J,K+1)+EZ(I,J,K))*0.5                                                    1125    269
335  00B246I          EZB=(EZ(I,J,K)+EZ(I,J,K-1))*0.5                                                    1126    270
336  00B2A0I          SZE=(SZ(I+1,J,K)+SZ(I,J,K))*0.5                                                    1127    271
337  00B2FCI          SZW=(SZ(I,J,K)+SZ(I-1,J,K))*0.5                                                    1128    272
338  00B358I          SZN=(SZ(I,J+1,K)+SZ(I,J,K))*0.5                                                    1129    273
339  00B394I          SZS=(SZ(I,J,K)+SZ(I,J-1,K))*0.5                                                    1130    274
340  00B410I          SZT=(SZ(I,J,K+1)+SZ(I,J,K))*0.5                                                    1131    275
341  00B46AI          SZB=(SZ(I,J,K)+SZ(I,J,K-1))*0.5                                                    1132    276
342  00B4C4I          QE=GAE*(UCE*CZE+UEE*EZE+USE*SZE)                                                   1133    277
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343  0084F8I          QW=GAW*(UCW*CZW+UEW*EZW+USW*SZW)                                    1134     278
344  00852CI          QN=GAN*(UCN*CZN+UEN*EZN+USN*SZN)                                    1135     279
345  00856CI          QS=GAS*(UCS*CZS+UES*EZS+USS*SZS)                                    1136     280
346  00859CI          QT=GAT*(UCT*CZT+UET*EZT+UST*SZT)                                    1137     281
347  0085C8I          QB=GAB*(UCB*CZB+UEB*EZB+USB*SZB)                                    1138     282
348  0085FCI          SOC1=CX(I,J,K)*(QE-QW)+EX(I,J,K)*(QN-QS)+SX(I,J,K)*(QT-QB)          1139     283
349  00869CI          QE=GAE*(VCE*CZE+VEE*EZE+VSE*SZE)                                    1140     284
350  0086D0I          QW=GAW*(VCW*CZW+VEW*EZW+VSW*SZW)                                    1141     285
351  00870CI          QN=GAN*(VCN*CZN+VEN*EZN+VSN*SZN)                                    1142     286
352  00873CI          QS=GAS*(VCS*CZS+VES*EZS+VSS*SZS)                                    1143     287
353  00876CI          QT=GAT*(VCT*CZT+VET*EZT+VST*SZT)                                    1144     288
354  0087A0I          QB=GAB*(VCB*CZB+VEB*EZB+VSB*SZB)                                    1145     289
355  0087D4I          SOC2=CY(I,J,K)*(QE-QW)+EY(I,J,K)*(QN-QS)+SY(I,J,K)*(QT-QB)          1146     290
356  00880CI          QE=GAE*(WCE*CZE+WEE*EZE+WSE*SZE)                                    1147     291
357  00884CI          QW=GAW*(WCW*CZW+WEW*EZW+WSW*SZW)                                    1148     292
358  00888CI          QN=GAN*(WCN*CZN+WEN*EZN+WSN*SZN)                                    1149     293
359  0088C0I          QS=GAS*(WCS*CZS+WES*EZS+WSS*SZS)                                    1150     294
360  0088F4I          QT=GAT*(WCT*CZT+WET*EZT+WST*SZT)                                    1151     295
361  00892BI          QB=GAB*(WCB*CZB+WEB*EZB+WSB*SZB)                                    1152     296
362  00896CI          SOC3=CZ(I,J,K)*(QE-QW)+EZ(I,J,K)*(QN-QS)+SZ(I,J,K)*(QT-QB)          1153     297
363  0089A4CI         DW(I,J,K)=DW(I,J,K)+SOC1+SOC2+SOC3                                  1154     298
364  0089AAAI         SU(I,J,K)=SU(I,J,K)*TJO(I,J,K)                                      1155     299
365  0089B1CI         SP(I,J,K)=SPK(I,J,K)*TJO(I,J,K)                                     1156     300
366  0089B8EI     15  CONTINUE                                                           1157     301
367  0089BDCI         GO TO 6C                                                            1158     302
368              C-----V-SOURCE                                                          1159
369  0089E2I      2   CONTINUE                                                          1160     303
370  0089E2I          DO 25 I=IS,IT                                                       1161     304
371  0089EAI          DO 25 J=JS,JT                                                       1162     305
372  0089C12I         DO 25 K=KS,KT                                                       1163     306
373  0089C2AI         SU(I,J,K)=(DV(I,J,K)+APC(I,J,K)*FO(I,J,K))*TJO(I,J,K)               1164     307
374  0089CEEI     25  CONTINUE                                                          1165     308
375  0089D36I         GO TO 6C                                                            1166     309
376              C-----W-SOURCE                                                          1167
377  0089D3CI     3   CONTINUE                                                          1168     310
378  0089D3CI         DO 35 I=IS,IT                                                       1169     311
379  0089D54I         DO 35 J=JS,JT                                                       1170     312
380  0089D6CI         DO 35 K=KS,KT                                                       1171     313
381  0089D84I         SU(I,J,K)=(DW(I,J,K)+APO(I,J,K)*FO(I,J,K))*TJO(I,J,K)               1172     314
382  0089E48I     35  CONTINUE                                                          1173     315
383  0089E90I         GO TO 6C                                                            1174     316
384              C-----TM-SOURCE                                                         1175
385  0089E96I     4   CONTINUE                                                          1176     317
386  0089E96I         GO TO 6C                                                            1177     318
387              C-----K-SOURCE                                                          1178
388  0089E9CI     5   CONTINUE                                                          1179     319
389  0089E9CI         DO 55 I=IS,IT                                                       1180     320
390  0089E34I         DO 55 J=JS,JT                                                       1181     321
391  0089ECCI         DO 55 K=KS,KT                                                       1182     322
392  0089EE4I         SU(I,J,K)=GEN(I,J,K)+PP(I,J,K)+APO(I,J,K)*FO(I,J,K)                 1183     323
393  0089FAAI         P1=DEN(I,J,K)**2                                                    1184     324
394  0089FDEI         SP(I,J,K)=SPK(I,J,K)-CMU*P1*F(I,J,K)/VISE(I,J,K)                    1185     325
395  008C08AI         SU(I,J,K)=SU(I,J,K)*TJO(I,J,K)                                      1186     326
396  008C0FCI         SP(I,J,K)=SP(I,J,K)*TJO(I,J,K)                                      1187     327
397  008C16EI     55  CONTINUE                                                          1188     328
398  008C186I         GO TO 6C                                                            1189     329
399              C-----E-SOURCE                                                          1190
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400  00C18CI      6   CONTINUE                                                          1191      330
401  00C18CI          DO 65 I=IS,IT                                                     1192      331
402  00C104I          DO 65 J=JS,JT                                                     1193      332
403  00C1ECI          DO 65 K=KS,KT                                                     1194      333
404  00C204I          P1=DEN(I,J,K)**2                                                  1195      334
405  00C238I          SU(I,J,K)=SU(I,J,K)+C1*CMU*GEN(I,J,K)*P1*DK(I,J,K)/               1196      335
406                  1          VISE(I,J,K)+APO(I,J,K)*FO(I,J,K)                      1197
407  00C35CI          TMDK=DK(I,J,K)+SMNUM                                              1198      336
408  00C38EI          SP(I,J,K)=SPK(I,J,K)-C2*DEN(I,J,K)*F(I,J,K)/TMDK                  1199      337
409  00C43AI          SU(I,J,K)=SU(I,J,K)*TJO(I,J,K)                                    1200      338
410  00C4ACI          SP(I,J,K)=SP(I,J,K)*TJO(I,J,K)                                    1201      339
411  00C51EI      65  CONTINUE                                                          1202      340
412  00C566I          GO TO 6C                                                          1203      341
413               C-----PF-SOURCE                                                      1204
414  00C56CI      7   CONTINUE                                                          1205      342
415  00C56CI          DO 75 I=IS,IT                                                     1206      343
416  00C584I          DO 75 J=JS,JT                                                     1207      344
417  00C59CI          DO 75 K=KS,KT                                                     1208      345
418  00C5B4I          SU(I,J,K)=SU(I,J,K)+0.85*GEN(I,J,K)+APO(I,J,K)*FO(I,J,K)          1209      346
419  00C680I          TMDK=DK(I,J,K)+SMNUM                                              1210      347
420  00C682I          SP(I,J,K)=SPK(I,J,K)-DE(I,J,K)/TMDK                               1211      348
421  00C72CI          SU(I,J,K)=SU(I,J,K)*TJO(I,J,K)                                    1212      349
422  00C79EI          SP(I,J,K)=SP(I,J,K)*TJO(I,J,K)                                    1213      350
423  00C810I      75  CONTINUE                                                          1214      351
424  00C858I      60  CONTINUE                                                          1215      352
425               C-----MODIFY WALL BOUNDARY CONDITIONS THRU WALL FUNCTIONS             1216
426  00C858I          IF(IG .NE. 2) GO TO 410                                           1217      353
427  00C86EI          CALL BOUNC(IE,F)                                                  1218      355
428  00C89CI      410 CONTINUE                                                          1219      356
429               C-----SET SYMMETRIC, CYCLIC AND EXIT LINK COEFF.                     1220
430  00C89CI          CALL SYMOUT(2,IE,IS,IT,JS,JT,KS,KT)                               1221      357
431               C-----LINK COEFF. ASSEMBLY AND BLOCKAGES                             1222
432  00C808I          DO 500 I=IS,IT                                                    1223      358
433  00C8F0I          DO 500 J=JS,JT                                                    1224      359
434  00C908I          DO 500 K=KS,KT                                                    1225      360
435  00C920I          F1(I,J,K)=F(I,J,K)                                                1226      361
436  00C972I          ANAB=AE(I,J,K)+AW(I,J,K)+AN(I,J,K)+AS(I,J,K)+AT(I,J,K)+           1227      362
437                  1     AB(I,J,K)+APO(I,J,K)                                        1228
438  00CA82I          AP(I,J,K)=ANAB-SP(I,J,K)                                          1229      363
439  00CAD4I          PDUV=1.0                                                          1230      364
440  00CAE0I          IF(MC(I,J,K) .LT. 1) GO TO 530                                    1231      365
441  00CB16I          AP(I,J,K)=ALF                                                     1232      367
442  00CB42I          AN(I,J,K)=0.0                                                     1233      368
443  00CB6EI          AS(I,J,K)=0.0                                                     1234      369
444  00CB9AI          AE(I,J,K)=0.0                                                     1235      370
445  00CBC6I          AW(I,J,K)=0.0                                                     1236      371
446  00CBF2I          AT(I,J,K)=0.0                                                     1237      372
447  00CC1EI          AB(I,J,K)=0.0                                                     1238      373
448  00CC4AI          SU(I,J,K)=F(I,J,K)                                                1239      374
449  00CC9CI          PDUV=0.C                                                          1240      375
450  00CCA8I      530 CONTINUE                                                          1241      376
451               C-----UNDER-RELAXATION                                               1242
452  00CCA8I          P1=1.2*AP(I,J,K)                                                  1243      377
453  00CCDAI          AP(I,J,K)=AP(I,J,K)/ALF                                           1244      378
454  00CD2CI          SU(I,J,K)=SU(I,J,K)+PDUV*(1.0-ALF)*AP(I,J,K)*F(I,J,K)             1245      379
455  00CDDCI          IF(IE .EQ. 1) DU(I,J,K)=TJO(I,J,K)*PDUV/(P1-ANAB)                 1246      380
456  00CE4CI      500 CONTINUE                                                          1247      382
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457                C-----LINEAR EQUATIONS SLOVER                                                    1248
458   00CE94I            DO 550 I=1,ISWF                                                          1249      383
459   00CEA8I     550    CALL LINERX(1,IS,JS,KS,IT,JT,KT,F)                                       1250      384
460                C-----CALCULATE MAXIMUM CORRECTION OF CURRENT ITERATION                         1251
461   00CF0CI            DO 555 I=IS,IT                                                           1252      385
462   00CF24I            DO 555 J=JS,JT                                                           1253      386
463   00CF3CI            DO 555 K=KS,KT                                                           1254      387
464   00CF54I            P1=ABS(F(I,J,K)-F1(I,J,K))                                               1255      388
465   00CFCAI            ERRF=AMAX1(ERRF,P1)                                                      1256      389
466   00CFEAI     555    CONTINUE                                                                 1257      390
467   00D032I            RETURN                                                                   1258      391
468                C-------------------------------------------------------------------------    1259
469                C-----P EQUATION                                                                1260
470   00D03AI       10   CONTINUE                                                                 1261      392
471   00D03AI            IS=2                                                                     1262      393
472   00D042I            IT=L                                                                     1263      394
473   00D04EI            JS=2                                                                     1264      395
474   00D056I            JT=M                                                                     1265      396
475   00D062I            KS=2                                                                     1266      397
476   00D06AI            KT=N                                                                     1267      398
477                C-----SOURCE AT PX LOCATIONS                                                    1268
478   00D076I            DO 310 I=IS,IT                                                           1269      399
479   00D08EI            DO 310 J=JS,JT                                                           1270      400
480   00D0A6I            DO 310 K=KS,KT                                                           1271      401
481   00D0BEI            F(I,J,K)=0.0                                                             1272      402
482   00D0F0I            SUK(I,J,K)=0.0                                                           1273      403
483   00D11CI            SPK(I,J,K)=0.0                                                           1274      404
484   00D148I            SP(I,J,K)=0.0                                                            1275      405
485   00D174I            DENE=DEN(I,J,K)                                                          1276      406
486   00D1A0I            DENW=DEN(I-1,J,K)                                                        1277      407
487   00D1D0I            DENN=0.25*(DEN(I,J,K)+DEN(I-1,J,K)+DEN(I,J+1,K)+DEN(I-1,J+1,K))          1278      408
488   00D284I            DENS=0.25*(DEN(I,J,K)+DEN(I-1,J,K)+DEN(I,J-1,K)+DEN(I-1,J-1,K))          1279      409
489   00D338I            DENT=0.25*(DEN(I,J,K)+DEN(I-1,J,K)+DEN(I,J,K+1)+DEN(I-1,J,K+1))          1280      410
490   00D3E8I            DENB=0.25*(DEN(I,J,K)+DEN(I-1,J,K)+DEN(I,J,K-1)+DEN(I-1,J,K-1))          1281      411
491   00D498I            UE=U(I,J,K)                                                              1282      412
492   00D4C4I            UW=U(I-1,J,K)                                                            1283      413
493   00D4F4I            UN=0.25*(U(I,J,K)+U(I-1,J,K)+U(I,J+1,K)+U(I-1,J+1,K))                    1284      414
494   00D5A8I            US=0.25*(U(I,J,K)+U(I-1,J,K)+U(I,J-1,K)+U(I-1,J-1,K))                    1285      415
495   00D65CI            UT=0.25*(U(I,J,K)+U(I-1,J,K)+U(I,J,K+1)+U(I-1,J,K+1))                    1286      416
496   00D70CI            UB=0.25*(U(I,J,K)+U(I-1,J,K)+U(I,J,K-1)+U(I-1,J,K-1))                    1287      417
497   00D7BCI            VE=V(I,J,K)                                                              1288      418
498   00D7E8I            VW=V(I-1,J,K)                                                            1289      419
499   00D818I            VN=0.25*(V(I,J,K)+V(I-1,J,K)+V(I,J+1,K)+V(I-1,J+1,K))                    1290      420
500   00D8CCI            VS=0.25*(V(I,J,K)+V(I-1,J,K)+V(I,J-1,K)+V(I-1,J-1,K))                    1291      421
501   00D980I            VT=0.25*(V(I,J,K)+V(I-1,J,K)+V(I,J,K+1)+V(I-1,J,K+1))                    1292      422
502   00DA30I            VB=0.25*(V(I,J,K)+V(I-1,J,K)+V(I,J,K-1)+V(I-1,J,K-1))                    1293      423
503   00DAE0I            WE=W(I,J,K)                                                              1294      424
504   00DB0CI            WW=W(I-1,J,K)                                                            1295      425
505   00DB3CI            WN=0.25*(W(I,J,K)+W(I-1,J,K)+W(I,J+1,K)+W(I-1,J+1,K))                    1296      426
506   00DBF0I            WS=0.25*(W(I,J,K)+W(I-1,J,K)+W(I,J-1,K)+W(I-1,J-1,K))                    1297      427
507   00DCA4I            WT=0.25*(W(I,J,K)+W(I-1,J,K)+W(I,J,K+1)+W(I-1,J,K+1))                    1298      428
508   00DD54I            WB=0.25*(W(I,J,K)+W(I-1,J,K)+W(I,J,K-1)+W(I-1,J,K-1))                    1299      429
509   00DE04I            CXQ=0.5*(CX(I,J,K)+CX(I-1,J,K))                                          1300      430
510   00DE60I            EXQ=0.5*(EX(I,J,K)+EX(I-1,J,K))                                          1301      431
511   00DE6CI            SXQ=0.5*(SX(I,J,K)+SX(I-1,J,K))                                          1302      432
512   00DF18I            CYQ=0.5*(CY(I,J,K)+CY(I-1,J,K))                                          1303      433
513   00DF74I            EYQ=0.5*(EY(I,J,K)+EY(I-1,J,K))                                          1304      434
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514  000F90I          SYQ=0.5*(SY(I,J,K)+SY(I-1,J,K))                                                                  1305     435
515  00E02CI          CZQ=0.5*(CZ(I,J,K)+CZ(I-1,J,K))                                                                  1306     436
516  00E088I          EZQ=0.5*(EZ(I,J,K)+EZ(I-1,J,K))                                                                  1307     437
517  00E0E4I          SZQ=0.5*(SZ(I,J,K)+SZ(I-1,J,K))                                                                  1308     438
518  00E140I          CE=DENE*(UE*CXQ+VE*CYQ+WE*CZQ)                                                                   1309     439
519  00E174I          CW=DENW*(UW*CXQ+VW*CYQ+WW*CZQ)                                                                   1310     440
520  00E1A8I          CN=DENN*(UN*EXQ+VN*EYQ+WN*EZQ)                                                                   1311     441
521  00E1DCI          CS=DENS*(US*EXQ+VS*EYQ+WS*EZQ)                                                                   1312     442
522  00E210I          CT=DENT*(UT*SXQ+VT*SYQ+WT*SZQ)                                                                   1313     443
523  00E244I          CB=DENB*(UB*SXQ+VB*SYQ+WB*SZQ)                                                                   1314     444
524  00E278I          SUK(I,J,K)=-(CE-CW+CN-CS+CT-CB)*(TJO(I-1,J,K)+TJO(I,J,K))*0.5                                    1315     445
525  00E322I    310   CONTINUE                                                                                         1316     446
526               C-----SOURCE AT PY LOCATIONS                                                                        1317
527  00E36AI          DO 311 I=IS,IT                                                                                   1318     447
528  00E382I          DO 311 J=JS,JT                                                                                   1319     448
529  00E39AI          DO 311 K=KS,KT                                                                                   1320     449
530  00E3B2I          DENN=DEN(I,J,K)                                                                                  1321     450
531  00E3DEI          DENS=DEN(I,J-1,K)                                                                                1322     451
532  00E40EI          DENE=0.25*(DEN(I,J,K)+DEN(I,J-1,K)+DEN(I+1,J,K)+DEN(I+1,J-1,K))                                  1323     452
533  00E4C2I          DENW=0.25*(DEN(I,J,K)+DEN(I,J-1,K)+DEN(I-1,J,K)+DEN(I-1,J-1,K))                                  1324     453
534  00E576I          DENT=0.25*(DEN(I,J,K)+DEN(I,J-1,K)+DEN(I,J,K+1)+DEN(I,J-1,K+1))                                  1325     454
535  00E626I          DENB=0.25*(DEN(I,J,K)+DEN(I,J-1,K)+DEN(I,J,K-1)+DEN(I,J-1,K-1))                                  1326     455
536  00E6D6I          UN=U(I,J,K)                                                                                      1327     456
537  00E702I          US=U(I,J-1,K)                                                                                    1328     457
538  00E732I          UE=0.25*(U(I,J,K)+U(I,J-1,K)+U(I+1,J,K)+U(I+1,J-1,K))                                            1329     458
539  00E7E6I          UW=0.25*(U(I,J,K)+U(I,J-1,K)+U(I-1,J,K)+U(I-1,J-1,K))                                            1330     459
540  00E89AI          UT=0.25*(U(I,J,K)+U(I,J-1,K)+U(I,J,K+1)+U(I,J-1,K+1))                                            1331     460
541  00E94AI          UB=0.25*(U(I,J,K)+U(I,J-1,K)+U(I,J,K-1)+U(I,J-1,K-1))                                            1332     461
542  00E9FAI          VN=V(I,J,K)                                                                                      1333     462
543  00EA26I          VS=V(I,J-1,K)                                                                                    1334     463
544  00EA56I          VE=0.25*(V(I,J,K)+V(I,J-1,K)+V(I+1,J,K)+V(I+1,J-1,K))                                            1335     464
545  00EB0AI          VW=0.25*(V(I,J,K)+V(I,J-1,K)+V(I-1,J,K)+V(I-1,J-1,K))                                            1336     465
546  00EB8EI          VT=0.25*(V(I,J,K)+V(I,J-1,K)+V(I,J,K+1)+V(I,J-1,K+1))                                            1337     466
547  00EC6EI          VB=0.25*(V(I,J,K)+V(I,J-1,K)+V(I,J,K-1)+V(I,J-1,K-1))                                            1338     467
548  00ED1EI          WN=W(I,J,K)                                                                                      1339     468
549  00ED4AI          WS=W(I,J-1,K)                                                                                    1340     469
550  00ED7AI          WE=0.25*(W(I,J,K)+W(I,J-1,K)+W(I+1,J,K)+W(I+1,J-1,K))                                            1341     470
551  00EE2EI          WW=0.25*(W(I,J,K)+W(I,J-1,K)+W(I-1,J,K)+W(I-1,J-1,K))                                            1342     471
552  00EEE2I          WT=0.25*(W(I,J,K)+W(I,J-1,K)+W(I,J,K+1)+W(I,J-1,K+1))                                            1343     472
553  00EF92I          WB=0.25*(W(I,J,K)+W(I,J-1,K)+W(I,J,K-1)+W(I,J-1,K-1))                                            1344     473
554  00F042I          CXQ=0.5*(CX(I,J,K)+CX(I,J-1,K))                                                                  1345     474
555  00F09EI          EXQ=0.5*(EX(I,J,K)+EX(I,J-1,K))                                                                  1346     475
556  00F0FAI          SXQ=0.5*(SX(I,J,K)+SX(I,J-1,K))                                                                  1347     476
557  00F156I          CYQ=0.5*(CY(I,J,K)+CY(I,J-1,K))                                                                  1348     477
558  00F192I          EYQ=0.5*(EY(I,J,K)+EY(I,J-1,K))                                                                  1349     478
559  00F20EI          SYQ=0.5*(SY(I,J,K)+SY(I,J-1,K))                                                                  1350     479
560  00F26AI          CZQ=0.5*(CZ(I,J,K)+CZ(I,J-1,K))                                                                  1351     480
561  00F2C6I          EZQ=0.5*(EZ(I,J,K)+EZ(I,J-1,K))                                                                  1352     481
562  00F322I          SZQ=0.5*(SZ(I,J,K)+SZ(I,J-1,K))                                                                  1353     482
563  00F37EI          CE=DENE*(UE*CXQ+VE*CYQ+WE*CZQ)                                                                   1354     483
564  00F3B2I          CW=DENW*(UW*CXQ+VW*CYQ+WW*CZQ)                                                                   1355     484
565  00F3E6I          CN=DENN*(UN*EXQ+VN*EYQ+WN*EZQ)                                                                   1356     485
566  00F41AI          CS=DENS*(US*EXQ+VS*EYQ+WS*EZQ)                                                                   1357     486
567  00F44EI          CT=DENT*(UT*SXQ+VT*SYQ+WT*SZQ)                                                                   1358     487
568  00F482I          CB=DENB*(UB*SXQ+VB*SYQ+WB*SZQ)                                                                   1359     488
569  00F486I          SPK(I,J,K)=-(CE-CW+CN-CS+CT-CB)*(TJO(I,J-1,K)+TJO(I,J,K))*0.5                                    1360     489
570  00F560I    311   CONTINUE                                                                                         1361     490
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571                C-----SOURCE AT PZ LOCATIONS                                                                                    1362
572   00F5A8I          DO 312 I=IS,IT                                                                                              1363      491
573   00F5C0I          DO 312 J=JS,JT                                                                                              1364      492
574   00F5D8I          DO 312 K=KS,KT                                                                                              1365      493
575   00F5F0I          DENT=DEN(I,J,K)                                                                                             1366      494
576   00F61CI          DENS=DEN(I,J,K-1)                                                                                           1367      495
577   00F64AI          DENN=0.25*(DEN(I,J,K)+DEN(I,J,K-1)+DEN(I,J+1,K)+DEN(I,J+1,K-1))                                             1368      496
578   00F6FAI          DENS=0.25*(DEN(I,J,K)+DEN(I,J,K-1)+DEN(I,J-1,K)+DEN(I,J-1,K-1))                                             1369      497
579   00F7AAI          DENE=0.25*(DEN(I,J,K)+DEN(I,J,K-1)+DEN(I+1,J,K)+DEN(I+1,J,K-1))                                             1370      498
580   00F85AI          DENW=0.25*(DEN(I,J,K)+DEN(I,J,K-1)+DEN(I-1,J,K)+DEN(I-1,J,K-1))                                             1371      499
581   00F90AI          UT=U(I,J,K)                                                                                                 1372      500
582   00F936I          UB=U(I,J,K-1)                                                                                               1373      501
583   00F964I          UN=0.25*(U(I,J,K)+U(I,J,K-1)+U(I,J+1,K)+U(I,J+1,K-1))                                                       1374      502
584   00FA14I          US=0.25*(U(I,J,K)+U(I,J,K-1)+U(I,J-1,K)+U(I,J-1,K-1))                                                       1375      503
585   00FAC4I          UE=0.25*(U(I,J,K)+U(I,J,K-1)+U(I+1,J,K)+U(I+1,J,K-1))                                                       1376      504
586   00FB74I          UW=0.25*(U(I,J,K)+U(I,J,K-1)+U(I-1,J,K)+U(I-1,J,K-1))                                                       1377      505
587   00FC24I          VT=V(I,J,K)                                                                                                 1378      506
588   00FC50I          VB=V(I,J,K-1)                                                                                               1379      507
589   00FC7EI          VN=0.25*(V(I,J,K)+V(I,J,K-1)+V(I,J+1,K)+V(I,J+1,K-1))                                                       1380      508
590   00FD2EI          VS=0.25*(V(I,J,K)+V(I,J,K-1)+V(I,J-1,K)+V(I,J-1,K-1))                                                       1381      509
591   00FDDEI          VE=0.25*(V(I,J,K)+V(I,J,K-1)+V(I+1,J,K)+V(I+1,J,K-1))                                                       1382      510
592   00FE8EI          VW=0.25*(V(I,J,K)+V(I,J,K-1)+V(I-1,J,K)+V(I-1,J,K-1))                                                       1383      511
593   00FF3EI          WT=W(I,J,K)                                                                                                 1384      512
594   00FF6AI          WB=W(I,J,K-1)                                                                                               1385      513
595   00FF98I          WN=0.25*(W(I,J,K)+W(I,J,K-1)+W(I,J+1,K)+W(I,J+1,K-1))                                                       1386      514
596   0100A8I          WS=0.25*(W(I,J,K)+W(I,J,K-1)+W(I,J-1,K)+W(I,J-1,K-1))                                                       1387      515
597   0100F8I          WE=0.25*(W(I,J,K)+W(I,J,K-1)+W(I+1,J,K)+W(I+1,J,K-1))                                                       1388      516
598   0101A8I          WW=0.25*(W(I,J,K)+W(I,J,K-1)+W(I-1,J,K)+W(I-1,J,K-1))                                                       1389      517
599   010258I          CXQ=0.5*(CX(I,J,K)+CX(I,J,K-1))                                                                             1390      518
600   0102B2I          EXQ=0.5*(EX(I,J,K)+EX(I,J,K-1))                                                                             1391      519
601   01030CI          SXQ=0.5*(SX(I,J,K)+SX(I,J,K-1))                                                                             1392      520
602   0103A6I          CYQ=0.5*(CY(I,J,K)+CY(I,J,K-1))                                                                             1393      521
603   0103C0I          EYQ=0.5*(EY(I,J,K)+EY(I,J,K-1))                                                                             1394      522
604   01041AI          SYQ=0.5*(SY(I,J,K)+SY(I,J,K-1))                                                                             1395      523
605   010474I          CZQ=0.5*(CZ(I,J,K)+CZ(I,J,K-1))                                                                             1396      524
606   0104CEI          EZQ=0.5*(EZ(I,J,K)+EZ(I,J,K-1))                                                                             1397      525
607   010528I          SZQ=0.5*(SZ(I,J,K)+SZ(I,J,K-1))                                                                             1398      526
608   0105B2I          CE=DENE*(UE*CXQ+VE*CYQ+WE*CZQ)                                                                              1399      527
609   0105B6I          CW=DENW*(UW*CXQ+VW*CYQ+WW*CZQ)                                                                              1400      528
610   0105EAI          CN=DENN*(UN*EXQ+VN*EYQ+WN*EZQ)                                                                              1401      529
611   01061EI          CS=DENS*(US*EXQ+VS*EYQ+WS*EZQ)                                                                              1402      530
612   0106B2I          CT=DENT*(UT*SXQ+VT*SYQ+WT*SZQ)                                                                              1403      531
613   0106B6I          CB=DENB*(UB*SXQ+VB*SYQ+WB*SZQ)                                                                              1404      532
614   01063AI          SP(I,J,K)=-(CE-CW+CN-CS+CT-CB)*(TJO(I,J,K-1)+TJO(I,J,K))*0.5                                                1405      533
615   0107B2I    312   CONTINUE                                                                                                    1406      534
616                C-----LINK COEFF. AND SOURCE TERM AT CELL CENTER                                                                1407
617   0107AAI          DO 100 I=IS,IT                                                                                              1408      535
618   0107C2I          DO 100 J=JS,JT                                                                                              1409      536
619   0107DAI          DO 100 K=KS,KT                                                                                              1410      537
620   0107F2I          II=I-1                                                                                                      1411      538
621   0108C0I          JJ=J-1                                                                                                      1412      539
622   01080EI          KK=K-1                                                                                                      1413      540
623   01081CI          GAE=(DU(I,J,K)+DU(I,JJ,K)+DU(I,J,KK)+DU(I,JJ,KK))*0.25                                                      1414      541
624   0108C0I          GAW=(DU(II,J,K)+DU(II,JJ,K)+DU(II,J,KK)+DU(II,JJ,KK))*0.25                                                  1415      542
625   0109B4I          GAN=(DU(I,J,K)+DU(II,J,K)+DU(I,J,KK)+DU(II,J,KK))*0.25                                                      1416      543
626   0104D8I          GAS=(DU(I,JJ,K)+DU(II,JJ,K)+DU(I,JJ,KK)+DU(II,JJ,KK))*0.25                                                  1417      544
627   010AACI          GAT=(DU(I,J,K)+DU(II,J,K)+DU(I,JJ,K)+DU(II,JJ,K))*0.25                                                      1418      545
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628  010B50I          GAB=(DU(I,J,KK)+DU(II,J,KK)+DU(I,JJ,KK)+DU(II,JJ,KK))*0.25                1419     546
629  010BF4I          TXXEQ=(TXXE(I,J,K)+TXXE(II,JJ,KK))*0.5                                   1420     547
630  010C4CI          TXXWQ=(TXXW(I,J,K)+TXXW(II,JJ,KK))*0.5                                   1421     548
631  010CA4I          TYYNQ=(TYYN(I,J,K)+TYYN(II,JJ,KK))*0.5                                   1422     549
632  010CFCI          TYYSQ=(TYYS(I,J,K)+TYYS(II,JJ,KK))*0.5                                   1423     550
633  010D54I          TZZTQ=(TZZT(I,J,K)+TZZT(II,JJ,KK))*0.5                                   1424     551
634  010DACI          TZZBQ=(TZZB(I,J,K)+TZZB(II,JJ,KK))*0.5                                   1425     552
635  010E04I          TQ=(TJC(I,J,K)+TJC(II,JJ,KK))*0.5                                        1426     553
636  010E5CI          AE(I,J,K)=TQ*(GAE*TXXEQ)                                                  1427     554
637  010E94I          AW(I,J,K)=TQ*(GAW*TXXWQ)                                                  1428     555
638  010ECCI          AN(I,J,K)=TQ*(GAN*TYYNQ)                                                  1429     556
639  010F04I          AS(I,J,K)=TQ*(GAS*TYYSQ)                                                  1430     557
640  010F3CI          AT(I,J,K)=TQ*(GAT*TZZTQ)                                                  1431     558
641  010F74I          AB(I,J,K)=TQ*(GAB*TZZBQ)                                                  1432     559
642                 C-----P' SOURCE                                                            1433
643  010FACI          P1=1.0                                                                    1434     560
644  010FB8I          P2=1.0                                                                    1435     561
645  010FC4I          P3=1.0                                                                    1436     562
646  010FD0I          P4=1.0                                                                    1437     563
647  010FDCI          P5=1.0                                                                    1438     564
648  010FE8I          P6=1.0                                                                    1439     565
649  010FF4I          IF(GAN .EQ. 0.0) P1=0.0                                                   1440     566
650  011012I          IF(GAS .EQ. 0.0) P2=0.0                                                   1441     568
651  011030I          IF(GAE .EQ. 0.0) P3=0.0                                                   1442     570
652  01104EI          IF(GAW .EQ. 0.0) P4=0.0                                                   1443     572
653  01106CI          IF(GAT .EQ. 0.0) P5=0.0                                                   1444     574
654  01108AI          IF(GAB .EQ. 0.0) P6=0.0                                                   1445     576
655  0110A8I          PQ=0.5*((P1+P2)*(P3+P4+P5+P6)+(P3+P4)*(P1+P2+P5+P6)+                      1446     578
656                  1         (P5+P6)*(P1+P2+P3+P4))                                          1447
657  01112AI          IF(PQ .EQ. 0.0) PQ=1.0                                                    1448     579
658  011148I          SU(I,J,K)=(P1*P5*SUK(I,J,K)+P5*P2*SUK(I,J-1,K)+                           1449     581
659                  1           P2*P6*SUK(I,J-1,K-1)+P6*P1*SUK(I,J,K-1)+                    1450
660                  2           P3*P5*SPK(I,J,K)+P5*P4*SPK(I-1,J,K)+                        1451
661                  3           P4*P6*SPK(I-1,J,K-1)+P6*P3*SPK(I,J,K-1)+                    1452
662                  4           P1*P3*SP(I,J,K)+P3*P2*SP(I,J-1,K)+                          1453
663                  5           P2*P4*SP(I-1,J-1,K)+P4*P1*SP(I-1,J,K))/PQ                   1454
664  01140AI      100  CONTINUE                                                              1455     582
665                 C-----LINK COEFF. ASSEMBLY AND BLOCKAGES                                1456
666  011452I          DO 870 I=IS,IT                                                            1457     583
667  01146AI          DO 870 J=JS,JT                                                            1458     584
668  011482I          DO 870 K=KS,KT                                                            1459     585
669  01149AI          ANAB=AE(I,J,K)+AW(I,J,K)+AN(I,J,K)+AS(I,J,K)+AT(I,J,K)+AB(I,J,K)          1460     586
670  0114B4I          AP(I,J,K)=ANAB                                                            1461     587
671  0114B0I          IF(MC(I,J,K) .LT. 2) GO TO 870                                            1462     588
672  0115E6I          AP(I,J,K)=1.0                                                             1463     590
673  011612I          AE(I,J,K)=0.0                                                             1464     591
674  01163EI          AW(I,J,K)=0.0                                                             1465     592
675  01166AI          AN(I,J,K)=0.0                                                             1466     593
676  011696I          AS(I,J,K)=0.0                                                             1467     594
677  0116C2I          AT(I,J,K)=0.0                                                             1468     595
678  0116EEI          AB(I,J,K)=0.0                                                             1469     596
679  01171AI          SU(I,J,K)=0.0                                                             1470     597
680  011746I      870  CONTINUE                                                              1471     598
681                 C-----LINEAR EQUATIONS SOLVER                                           1472
682  01178EI          CALL LINERX(2,IS,JS,KS,IT,JT,KT,F)                                        1473     599
683  0117DCI          DO 850 I=1,ISWF                                                           1474     600
684  0117F0I      850  CALL LINERX(1,IS,JS,KS,IT,JT,KT,F)                                    1475     601
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685                C-----PRESSURE AND VELOCITIES CORRECTIONS                                                    1476
686   011854I            PPREF=F(2,2,2)                                                                        1477      602
687   011864I            DO 880 I=IS,IT                                                                        1478      603
688   01187CI            DO 880 J=JS,JT                                                                        1479      604
689   011894I            DO 880 K=KS,KT                                                                        1480      605
690   0118ACI            IF(MC(I,J,K) .GE. 2) GO TO 880                                                        1481      606
691   0118E2I            P1=F(I,J,K)-PPREF                                                                     1482      608
692   01191AI            P(I,J,K)=P(I,J,K)+ALP*P1                                                              1483      609
693   011972I            ERRF=AMAX1(ERRF,ABS(P1))                                                              1484      610
694   0119AEI     880    CONTINUE                                                                              1485      611
695   0119F6I            DO 600 I=2,LT                                                                         1486      612
696   011A0AI            DO 600 J=2,MT                                                                         1487      613
697   011A1EI            DO 600 K=2,NT                                                                         1488      614
698   011A32I            IF(MC(I,J,K) .GE. 1) GO TO 600                                                        1489      615
699   011A68I            II=I+1                                                                                1490      617
700   011A76I            JJ=J+1                                                                                1491      618
701   011A84I            KK=K+1                                                                                1492      619
702   011A92I            PE=(F(II,J,K)+F(II,JJ,K)+F(II,J,KK)+F(II,JJ,KK))*0.25                                 1493      620
703   011B4EI            PW=(F(I,J,K)+F(I,JJ,K)+F(I,J,KK)+F(I,JJ,KK))*0.25                                     1494      621
704   011C0AI            PN=(F(I,JJ,K)+F(II,JJ,K)+F(I,JJ,KK)+F(II,JJ,KK))*0.25                                 1495      622
705   011CC6I            PS=(F(I,J,K)+F(II,J,K)+F(I,J,KK)+F(II,J,KK))*0.25                                     1496      623
706   011D82I            PT=(F(I,J,KK)+F(II,J,KK)+F(I,JJ,KK)+F(II,JJ,KK))*0.25                                 1497      624
707   011E3EI            PB=(F(I,J,K)+F(II,J,K)+F(I,JJ,K)+F(II,JJ,K))*0.25                                     1498      625
708   011EFAI            PCXI=PE-PW                                                                            1499      626
709   011F0CI            PEDA=PN-PS                                                                            1500      627
710   011F1EI            PSCI=PT-PB                                                                            1501      628
711   011F30I            CXQ=CX(I,J,K)                                                                         1502      629
712   011F5CI            EXQ=EX(I,J,K)                                                                         1503      630
713   011F88I            SXQ=SX(I,J,K)                                                                         1504      631
714   011FB4I            CYQ=CY(I,J,K)                                                                         1505      632
715   011FE0I            EYQ=EY(I,J,K)                                                                         1506      633
716   01200CI            SYQ=SY(I,J,K)                                                                         1507      634
717   012038I            CZQ=CZ(I,J,K)                                                                         1508      635
718   012064I            EZQ=EZ(I,J,K)                                                                         1509      636
719   012090I            SZQ=SZ(I,J,K)                                                                         1510      637
720   0120BCI            PXX=PCXI*CXQ+PEDA*EXQ+PSCI*SXQ                                                        1511      638
721   0120EAI            PYY=PCXI*CYQ+PEDA*EYQ+PSCI*SYQ                                                        1512      639
722   012118I            PZZ=PCXI*CZQ+PEDA*EZQ+PSCI*SZQ                                                        1513      640
723   012146I            U(I,J,K)=U(I,J,K)-DU(I,J,K)*PXX                                                       1514      641
724   0121C0I            V(I,J,K)=V(I,J,K)-DU(I,J,K)*PYY                                                       1515      642
725   01223AI            W(I,J,K)=W(I,J,K)-DU(I,J,K)*PZZ                                                       1516      643
726   0122B4I     600    CONTINUE                                                                              1517      644
727   0122FCI            RETURN                                                                                1518      645
728   012304I            END                                                                                   1519      646

NO  ERRORS:F7D  R05-01.0C  SUBROUTINE  SOLVEQ    02/21/86  09:58:31  TABLE SPACE:  18 KB
STATEMENT BUFFER:   20 LINES/1321 BYTES    STACK SPACE: 203 WORDS
SINGLE PRECISION FLOATING PT SUPPORT REQUIRED FOR EXECUTION
```

```
 1  000000I          SUBROUTINE LINERX(ISOL,IS,JS,KS,LT,MT,NT,F)                          1520
 2  000004I          DIMENSION A(20),B(20),C(20),D(20),F(21,18,10),BB(21,18,10)           1521
 3  000004I          COMMON                                                               1522
                     1/COEF/ AP(21,18,10),SU(21,18,10),SP(21,18,10),SUK(21,18,10),        1523
 5                   2   SPK(21,18,10),AE(21,18,10),AW(21,18,10),AN(21,18,10),           1524
 6                   3   AS(21,18,10),AT(21,18,10),AB(21,18,10),APO(21,18,10)            1525
 7                C-----LINEAR EQUATIONS SOLVERS                                          1526
 8  000004I          GO TO (1,2),ISOL                                                     1527        1
 9                C-----LINE-RELAXATION USING TOMA                                        1528
10  003C86I        1   CONTINUE                                                         1529        2
11  003C86I          DO 10 I=IS,LT                                                        1530        3
12  003C9EI          DO 10 J=JS,MT                                                        1531        4
13  003C86I          DO 10 K=KS,NT                                                        1532        5
14  003CCEI          BB(I,J,K)=AE(I,J,K)*F(I+1,J,K)+AN(I,J,K)*F(I,J+1,K)+                 1533        6
15                  1          SU(I,J,K)                                                 1534
16  003DC8I       10 CONTINUE                                                         1535        7
17  003E10I          A(KS-1)=0.0                                                          1536        8
18  003E24I          DO 100 I=IS,LT                                                       1537        9
19  003E3CI          DO 100 J=JS,MT                                                       1538       10
20  003E54I          C(KS-1)=F(I,J,KS-1)                                                  1539       11
21  003E90I          DO 101 K=KS,NT                                                       1540       12
22  003EA8I          A(K)=AT(I,J,K)                                                       1541       13
23  003EDAI          B(K)=AB(I,J,K)                                                       1542       14
24  003F0CI          C(K)=BB(I,J,K)+AW(I,J,K)*F(I-1,J,K)+                                 1543       15
25                  1     AS(I,J,K)*F(I,J-1,K)                                           1544
26  003FECI          D(K)=AP(I,J,K)                                                       1545       16
27  00401EI          TERM1=C(K)-B(K)*A(K-1)+1.E-30                                        1546       17
28  0040541          TERM=1.C/TERM1                                                       1547       18
29  004066I          A(K)=A(K)*TERM                                                       1548       19
30  004088I   101    C(K)=(C(K)+B(K)*C(K-1))*TERM                                         1549       20
31  0040E0I          DO 102 KK=KS,NT                                                      1550       21
32  0040F8I          K=KS+NT-KK                                                           1551       22
33  004110I   102    F(I,J,K)=A(K)*F(I,J,K+1)+C(K)                                        1552       23
34  00419EI   100    CONTINUE                                                             1553       24
35  0041CEI          RETURN                                                               1554       25
36                C-----BLOCK CORRECTION FOR PRESSURE FIELD                               1555
37  004106I     2    CONTINUE                                                             1556       26
38  004106I          DO 201 I=IS,LT                                                       1557       27
39  0041EEI          A(I)=0.C                                                             1558       28
40  004202I          B(I)=0.C                                                             1559       29
41  004216I          C(I)=0.C                                                             1560       30
42  00422AI          D(I)=0.0                                                             1561       31
43  00423EI          DO 201 J=JS,MT                                                       1562       32
44  004256I          DO 201 K=KS,NT                                                       1563       33
45  00426EI          A(I)=A(I)-AE(I,J,K)                                                  1564       34
46  004280I          B(I)=B(I)-AW(I,J,K)                                                  1565       35
47  0042F2I          C(I)=C(I)+SU(I,J,K)                                                  1566       36
48  004334I   201    CONTINUE                                                             1567       37
49  00437CI          DO 202 I=IS,LT                                                       1568       38
50  004394I          ANAB=-A(I)-B(I)                                                      1569       39
51  00433CI   202    D(I)=ANAB*1.5                                                        1570       40
52  0043EEI          DO 203 I=IS+1,LT                                                     1571       41
53  00440AI          TERM=B(I)/D(I-1)                                                     1572       42
54  00442EI          D(I)=D(I)-TERM*A(I-1)                                                1573       43
55  004462I   203    C(I)=C(I)-TERM*C(I-1)                                                1574       44
56  0044AEI          F(LT,JS,KS)=C(LT)/D(LT)                                              1575       45
57  0044F6I          DO 204 II=IS+1,LT                                                    1576       46
```

```
 58  004512I          I=IS+LT-II                                                  1577      47
 59  00452AI     204  F(I,JS,KS)=(C(I)-A(I)*F(I+1,JS,KS))/D(I)                   1578      48
 60  0045CAI          DO 205 I=IS,LT                                              1579      49
 61  0045E2I          PPBLK=F(I,JS,KS)                                            1580      50
 62  004614I          DO 205 J=JS+1,MT                                            1581      51
 63  004630I          DO 205 K=KS+1,NT                                            1582      52
 64  00464CI          F(I,J,K)=PPBLK                                              1583      53
 65  00467EI     205  CONTINUE                                                    1584      54
 66  0046C6I          RETURN                                                      1585      55
 67  0046CEI          END                                                         1586      56

NO  ERRORS:F7D  R05-01.0C  SUBROUTINE  LINERX    02/21/86  09:59:43  TABLE SPACE:    3 KB
STATEMENT BUFFER:    20 LINES/1321 BYTES    STACK SPACE: 199 WORDS
SINGLE PRECISION FLOATING PT SUPPORT REQUIRED FOR EXECUTION
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```
 1  000000I          SUBROUTINE BOUNC(IE,F)                                          1587
 2  000004I          DIMENSION F(21,18,10)                                           1588
 3  000004I          COMMON                                                          1589
 4                  1/VAR/U(21,18,10),V(21,18,10),P(21,18,10),DK(21,18,10),         1590
 5                  2  DE(21,18,10),ERRU,ERRV,ERRM,ERRK,ERRE,ERRW,                  1591
 6                  3  PP(21,18,10),W(21,18,10),TM(21,18,10)                        1592
 7                  1/PROP/ VISE(21,18,10),DEN(21,18,10),VISC,DENIN,FLOWIN          1593
 8                  1/PCOR/ DU(21,18,10),DV(21,18,10),DW(21,18,10)                  1594
 9                  1/TUR/ SIGK,SIGE,CMU,C1,C2,CMU1,CMU2,E,CK,HINUM,SMNUM,ANV1(800), 1595
10                  2 YN(800),YN1(800),SINX(800),SINY(800),SINZ(800),ANW1(800),     1596
11                  3 YPLN(800),TAUN(800),I8C(800),J8C(800),K8C(800),IITY(800),     1597
12                  4 TAUW(800),GEN(21,18,10),MC(21,18,10),IJLO(21,18,10),IITO      1598
13                  1/COEF/ AP(21,18,10),SU(21,18,10),SP(21,18,10),SUK(21,18,10),   1599
14                  2   SPK(21,18,10),AE(21,18,10),AW(21,18,10),AN(21,18,10),       1600
15                  3   AS(21,18,10),AT(21,18,10),AB(21,18,10),APO(21,18,10)        1601
16 000004I          COMMON                                                          1602
17                  1/TRAN/ X(21,18,10),Y(21,18,10),Z(21,18,10),TJO(21,18,10),      1603
18                  2   CX(21,18,10),CY(21,18,10),CZ(21,18,10),                     1604
19                  3   EX(21,18,10),EY(21,18,10),EZ(21,18,10),                     1605
20                  3   SX(21,18,10),SY(21,18,10),SZ(21,18,10)                      1606
21                  1/LIMT/ L,M,LT,MT,L1,L2,M1,M2,LO,MO,ISWU,ISWV,ISWP,ISWK,ISWE,    1607
22                  2   ALU,ALV,ALP,ALK,ALE,ALVIS,ALW,N,N1,N2,NO,ISWW,IG,NT,ALC,DTT 1608
23                C-----EVALUATE WALL BOUNDARY CONDITIONS USING WALL FUNCTIONS       1609
24 000004I          DO 150 III=1,IITO                                               1610      1
25 000024I          I=I8C(III)                                                      1611      2
26 000038I          J=J8C(III)                                                      1612      3
27 00004CI          K=K8C(III)                                                      1613      4
28 000060I          GO TO (1,2,3,4,5,6), IITY(III)                                  1614      5
29 00009EI       1  CONTINUE                                                        1615      6
30                C-----NORTH                                                       1616
31 00009EI          YP=YN(III)                                                      1617      7
32 0000B2I          YP1=YN1(III)                                                    1618      8
33 0000C6I          CDK=DK(I,J,K)                                                   1619      9
34 0000F2I          CDE=DE(I,J,K)                                                   1620     10
35 00011EI          CDEN=DEN(I,J,K)                                                 1621     11
36 00014AI          CALL WALLFN(IE,YP,YP1,CDK,CDE,CDEN,SINX(III),SINY(III),         1622     12
37                 1      SINZ(III),F(I,J+1,K),AN(I,J,K),SU(I,J,K),SP(I,J,K),      1623
38                 2      TAUN(III),YPLN(III),GEN(I,J,K),VISE(I,J,K),              1624
39                 3      U(I,J+1,K),V(I,J+1,K),W(I,J+1,K),U(I,J,K),V(I,J,K),      1625
40                 4      W(I,J,K),U(I,J-1,K),V(I,J-1,K),W(I,J-1,K),               1626
41                 5     -SPK(I,J,K),SPK(I,J,K),ANV1(III),ANW1(III),TJO(I,J,K))    1627
42 000584I          GO TO 150                                                       1628     13
43 00058AI       2  CONTINUE                                                        1629     14
44                C-----SOUTH                                                       1630
45 0005BAI          YP=YN(III)                                                      1631     15
46 0005CEI          YP1=YN1(III)                                                    1632     16
47 0005E2I          CDK=DK(I,J,K)                                                   1633     17
48 0006OEI          CDE=DE(I,J,K)                                                   1634     18
49 00063AI          CDEN=DEN(I,J,K)                                                 1635     19
50 000666I          CALL WALLFN(IE,YP,YP1,CDK,CDE,CDEN,SINX(III),SINY(III),         1636     20
51                 1      SINZ(III),F(I,J-1,K),AN(I,J,K),SU(I,J,K),SP(I,J,K),      1637
52                 2      TAUN(III),YPLN(III),GEN(I,J,K),VISE(I,J,K),              1638
53                 3      U(I,J-1,K),V(I,J-1,K),W(I,J-1,K),U(I,J,K),V(I,J,K),      1639
54                 4      W(I,J,K),U(I,J+1,K),V(I,J+1,K),W(I,J+1,K),               1640
55                 5     -SPK(I,J,K),SPK(I,J,K),ANV1(III),ANW1(III),TJO(I,J,K))    1641
56 000AD0I          GO TO 150                                                       1642     21
57 000AD6I       3  CONTINUE                                                        1643     22
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ORIGINAL PAGE IS
OF POOR QUALITY

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                C-----EAST                                                                1644
000A06I                 YP=YN(III)                                                        1645        23
000A2AI                 YP1=YN1(III)                                                      1646        24
000A4EI                 CDK=DK(I,J,K)                                                     1647        25
000B2AI                 CDE=DE(I,J,K)                                                     1648        26
000B56I                 CDEN=DEN(I,J,K)                                                   1649        27
000B82I                 CALL WALLFN(IE,YP,YP1,CDK,CDE,CDEN,SINX(III),SINY(III),           1650        28
                       1      SINZ(III),F(I+1,J,K),AN(I,J,K),SU(I,J,K),SP(I,J,K),       1651
                       2      TAUN(III),YPLN(III),GEN(I,J,K),VISE(I,J,K),               1652
                       3      U(I+1,J,K),V(I+1,J,K),W(I+1,J,K),U(I,J,K),V(I,J,K),       1653
                       4      W(I,J,K),U(I-1,J,K),V(I-1,J,K),W(I-1,J,K),                1654
                       5     -SPK(I,J,K),SPK(I,J,K),ANV1(III),ANW1(III),TJC(I,J,K))     1655
000FECI                 GO TO 150                                                         1656        29
000FF2I           4     CONTINUE                                                          1657        30
                C-----WEST                                                                1658
000FF2I                 YP=YN(III)                                                        1659        31
001006I                 YP1=YN1(III)                                                      1660        32
001014I                 CDK=DK(I,J,K)                                                     1661        33
001046I                 CDE=DE(I,J,K)                                                     1662        34
001072I                 CDEN=DEN(I,J,K)                                                   1663        35
00109EI                 CALL WALLFN(IE,YP,YP1,CDK,CDE,CDEN,SINX(III),SINY(III),           1664        36
                       1      SINZ(III),F(I-1,J,K),AN(I,J,K),SU(I,J,K),SP(I,J,K),       1665
                       2      TAUN(III),YPLN(III),GEN(I,J,K),VISE(I,J,K),               1666
                       3      U(I-1,J,K),V(I-1,J,K),W(I-1,J,K),U(I,J,K),V(I,J,K),       1667
                       4      W(I,J,K),U(I+1,J,K),V(I+1,J,K),W(I+1,J,K),                1668
                       5     -SPK(I,J,K),SPK(I,J,K),ANV1(III),ANW1(III),TJC(I,J,K))     1669
0015C8I                 GO TO 150                                                         1670        37
0015CEI           5     CONTINUE                                                          1671        38
                C-----TOP                                                                 1672
0015CEI                 YP=YN(III)                                                        1673        39
001522I                 YP1=YN1(III)                                                      1674        40
001536I                 CDK=DK(I,J,K)                                                     1675        41
001562I                 CDE=DE(I,J,K)                                                     1676        42
00158EI                 CDEN=DEN(I,J,K)                                                   1677        43
0015BAI                 CALL WALLFN(IE,YP,YP1,CDK,CDE,CDEN,SINX(III),SINY(III),           1678        44
                       1      SINZ(III),F(I,J,K+1),AN(I,J,K),SU(I,J,K),SP(I,J,K),       1679
                       2      TAUN(III),YPLN(III),GEN(I,J,K),VISE(I,J,K),               1680
                       3      U(I,J,K+1),V(I,J,K+1),W(I,J,K+1),U(I,J,K),V(I,J,K),       1681
                       4      W(I,J,K),U(I,J,K-1),V(I,J,K-1),W(I,J,K-1),                1682
                       5     -SPK(I,J,K),SPK(I,J,K),ANV1(III),ANW1(III),TJC(I,J,K))     1683
001A18I                 GO TO 150                                                         1684        45
001A1EI           6     CONTINUE                                                          1685        46
                C-----BOTTOM                                                              1686
001A1EI                 YP=YN(III)                                                        1687        47
001A32I                 YP1=YN1(III)                                                      1688        48
001A46I                 CDK=DK(I,J,K)                                                     1689        49
001A72I                 CDE=DE(I,J,K)                                                     1690        50
001A9EI                 CDEN=DEN(I,J,K)                                                   1691        51
001ACAI                 CALL WALLFN(IE,YP,YP1,CDK,CDE,CDEN,SINX(III),SINY(III),           1692        52
                       1      SINZ(III),F(I,J,K-1),AN(I,J,K),SU(I,J,K),SP(I,J,K),       1693
                       2      TAUN(III),YPLN(III),GEN(I,J,K),VISE(I,J,K),               1694
                       3      U(I,J,K-1),V(I,J,K-1),W(I,J,K-1),U(I,J,K),V(I,J,K),       1695
                       4      W(I,J,K),U(I,J,K+1),V(I,J,K+1),W(I,J,K+1),                1696
                       5     -SPK(I,J,K),SPK(I,J,K),ANV1(III),ANW1(III),TJC(I,J,K))     1697
001F28I          150  CONTINUE                                                          1698        53
001F40I                 RETURN                                                            1699        54
001F46I                 END                                                               1700        55
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NO  ERRORS:F7D  R05-01.DC  SUBROUTINE  BOUNC      02/21/86  10:00:46  TABLE SPACE:    7 KB
STATEMENT BUFFER:    20 LINES/1321 BYTES     STACK SPACE: 136 WORDS
SINGLE PRECISION FLOATING PT SUPPORT REQUIRED FOR EXECUTION
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```
      SUBROUTINE WALLFN(IE,YP,YP1,DK1,DE1,DEN1,SINX1,SINY1,                1701
     1      SINZ1,P1,AN1,SU1,SP1,                                          1702
     2      TAUN1,YPLN1,GEN1,VISE1,                                        1703
     3      U1,V1,W1,UC,VC,                                                1704
     4      WC,U2,V2,W2,                                                   1705
     5      SUK1,SPK1,ANV2,ANW2,TJC1)                                      1706
      COMMON                                                              1707
     1/PROP/ VISE(21,18,10),DEN(21,18,10),VISC,DENIN,FLOWIN               1708
     1/TUR/ SIGK,SIGE,CMU,C1,C2,CMU1,CMU2,E,CK,HINUM,SMNUM,ANV1(800),     1709
     2 YN(800),YN1(800),SINX(800),SINY(800),SINZ(800),ANW1(800),          1710
     3 YPLN(800),TAUN(800),IBC(800),JBC(800),KBC(800),IITY(800),          1711
     4 TAUW(800),GEN(21,18,10),MC(21,18,10),IJLO(21,18,10),IITO           1712
C-----WALL FUNCTIONS USING LOGARITHMIC WALL LAW                           1713
      GO TO (1,2,3,4,5,6,7), IE                                           1714        1
    1 CONTINUE                                                            1715        2
C-----U                                                                   1716
      SQRTK=SQRT(DK1)                                                     1717        3
      YPLN1=DEN1*SQRTK*CMU1*YP/VISC                                       1718        4
      IF(YPLN1 .LE. 11.63) GO TO 111                                      1719        5
      TMULT=DEN1*CMU1*SQRTK*CK/ALOG(E*YPLN1)                              1720        7
      GO TO 112                                                           1721        8
  111 TMULT=VISC/YP                                                       1722        9
  112 TAUN1=-TMULT                                                        1723       10
      PTA=TJC1*SINX1*TMULT/YP1                                            1724       11
      SP1=SP1-PTA                                                         1725       12
      SU1=SU1+PTA*P1                                                      1726       13
      P1=SQRT(1.-SINX1**2)                                                1727       14
      ANV2=AN1*SQRT(1.-SINY1**2)                                          1728       15
      ANW2=AN1*SQRT(1.-SINZ1**2)                                          1729       16
      AN1=P1*AN1                                                          1730       17
      RETURN                                                              1731       18
    2 CONTINUE                                                            1732       19
C-----V                                                                   1733
      TMULT=-TAUN1                                                        1734       20
      PTA=TJC1*SINY1*TMULT/YP1                                            1735       21
      SP1=SP1-PTA                                                         1736       22
      SU1=SU1+PTA*P1                                                      1737       23
      AN1=ANV2                                                            1738       24
      RETURN                                                              1739       25
    3 CONTINUE                                                            1740       26
C-----W                                                                   1741
      TMULT=-TAUN1                                                        1742       27
      PTA=TJC1*SINZ1*TMULT/YP1                                            1743       28
      SP1=SP1-PTA                                                         1744       29
      SU1=SU1+PTA*P1                                                      1745       30
      AN1=ANW2                                                            1746       31
      RETURN                                                              1747       32
    4 CONTINUE                                                            1748       33
C-----TM                                                                  1749
      RETURN                                                              1750       34
    5 CONTINUE                                                            1751       35
C-----K                                                                   1752
      SQRTK=SQRT(DK1)                                                     1753       36
      IF(YPLN1 .LE. 11.63) GO TO 511                                      1754       37
      DITM=DEN1*CMU2*SQRTK*ALOG(E*YPLN1)/(CK*YP)                          1755       39
      GO TO 512                                                           1756       40
  511 DITM=DEN1*CMU2*SQRTK*YPLN1/YP                                       1757       41
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 58  0003ACI     512  CONTINUE                                                                1758       42
 59  0003ACI          DDU=U0-U1                                                               1759       43
 60  0003BEI          DDV=V0-V1                                                               1760       44
 61  0003D0I          DDW=W0-W1                                                               1761       45
 62  0003E2I          P1=DDU**2+DDV**2+DDW**2                                                 1762       46
 63  0004 2AI         DDU=U2-U1                                                               1763       47
 64  0004 3CI         DDV=V2-V1                                                               1764       48
 65  0004 4EI         DDW=W2-W1                                                               1765       49
 66  0004 60I         P2=DDU**2+DDV**2+DDW**2                                                 1766       50
 67  0004A8I          YP2=2*YP1-YP                                                            1767       51
 68  0004C0I          GENR=GEN1-VISE1*P2/YP2/YP2                                              1768       52
 69  0004E6I          IF(GENR .LE. 0.0) GENR=0.0                                              1769       53
 70  0005 04I         SU1=TJO1*(SUK1*DK1*(P1*TAUN1**2/VISE1+GENR))                            1770       55
 71  0005 3EI         SP1=TJO1*(SPK1-DITM)                                                    1771       56
 72  0005 56I         AN1=0.0                                                                 1772       57
 73  0005 62I         RETURN                                                                  1773       58
 74  0005 68I       6 CONTINUE                                                                1774       59
 75                C-----E                                                                   1775
 76  0005 68I         TERM=CMU2/(CK*YP)                                                       1776       60
 77  0005 82I         SU1=HINUM*TERM*DK1**1.5                                                 1777       61
 78  0005 82I         SP1=-HINUM                                                              1778       62
 79  0005 C4I         RETURN                                                                  1779       63
 80  0005 CAI       7 CONTINUE                                                                1780       64
 81                C-----PF                                                                  1781
 82  0005 CAI         RETURN                                                                  1782       65
 83  0005 D0I         END                                                                     1783       66

NO  ERRORS:F7D  R05-01.00  SUBROUTINE  WALLFN    02/21/86  10:01:12  TABLE SPACE:    5 KB
STATEMENT BUFFER:   20 LINES/1321 BYTES    STACK SPACE: 122 WORDS
SINGLE PRECISION FLOATING PT SUPPORT REQUIRED FOR EXECUTION
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```
      SUBROUTINE SYMOUT(IC,IE,IS,IT,JS,JT,KS,KT)                                1784
      COMMON                                                                    1785
     1/VAR/U(21,18,10),V(21,18,10),P(21,18,10),DK(21,18,10),                    1786
     2  DE(21,18,10),ERRU,ERRV,ERRM,ERRK,ERRE,ERRW,                            1787
     3  PP(21,18,10),W(21,18,10),TM(21,18,10)                                  1788
     1/PROP/ VISE(21,18,10),DEN(21,18,10),VISC,DENIN,FLOWIN                     1789
     1/PCOR/ DU(21,18,10),DV(21,18,10),DW(21,18,10)                             1790
     1/COEF/ AP(21,18,10),SU(21,18,10),SP(21,18,10),SUK(21,18,10),              1791
     2   SPK(21,18,10),AE(21,18,10),AW(21,18,10),AN(21,18,10),                  1792
     3   AS(21,18,10),AT(21,18,10),AB(21,18,10),APO(21,18,10)                  1793
      COMMON                                                                    1794
     1/TRAN/ X(21,18,10),Y(21,18,10),Z(21,18,10),TJO(21,18,10),                 1795
     2   CX(21,18,10),CY(21,18,10),CZ(21,18,10),                               1796
     3   EX(21,18,10),EY(21,18,10),EZ(21,18,10),                               1797
     3   SX(21,18,10),SY(21,18,10),SZ(21,18,10)                                1798
     1/UNSTDY/UO(21,18,10),VO(21,18,10),WO(21,18,10),DKO(21,18,10),             1799
     2   DEO(21,18,10),DENO(21,18,10),TMO(21,18,10)                            1800
     1/LIMT/ L,M,LT,MT,L1,L2,M1,M2,LO,MO,ISWU,ISWV,ISWP,ISWK,ISWE,               1801
     2   ALU,ALV,ALP,ALK,ALE,ALVIS,ALW,N,N1,N2,NO,ISWW,IG,NT,ALC,DTT            1802
C-----SYMMETRIC, CYCLIC AND EXIT CONDITIONS AND LINK MODIFICATIONS               1803
      GO TO (1,2,3), IC                                                         1804        1
    1 CONTINUE                                                                  1805        2
C-----BOTTOM                                                                    1806
      K=1                                                                       1807        3
      DO 10 I=1,L                                                               1808        4
      DO 10 J=2,MT                                                              1809        5
      U(I,J,K)=U(I,J,K+1)                                                       1810        6
      V(I,J,K)=V(I,J,K+1)                                                       1811        7
      W(I,J,K)=0.0                                                              1812        8
      TM(I,J,K)=TM(I,J,K+1)                                                     1813        9
      DK(I,J,K)=DK(I,J,K+1)                                                     1814       10
      DE(I,J,K)=DE(I,J,K+1)                                                     1815       11
   10 CONTINUE                                                                  1816       12
C-----EAST OUT (BASED ON INFLOW MASS FLOW RATE)                                 1817
      I=IT                                                                      1818       13
      FLOW=0.0                                                                  1819       14
      ARDEN=0.0                                                                 1820       15
      DO 50 J=2,JT                                                              1821       16
      DO 50 K=2,KT                                                              1822       17
      UC=(V(I,J,K)+V(I,J-1,K)+V(I,J,K-1)+V(I,J-1,K-1))*0.25                     1823       18
      DENC=(DEN(I,J,K)+DEN(I,J-1,K)+DEN(I,J,K-1)+DEN(I,J-1,K-1))*0.25           1824       19
      P1=(X(I,J,K)+X(I,J,K-1)-X(I,J-1,K)-X(I,J-1,K-1))*0.5                      1825       20
      P2=(Y(I,J,K)+Y(I,J,K-1)-Y(I,J-1,K)-Y(I,J-1,K-1))*0.5                      1826       21
      P3=(Z(I,J,K)+Z(I,J,K-1)-Z(I,J-1,K)-Z(I,J-1,K-1))*0.5                      1827       22
      Q1=(X(I,J,K)+X(I,J-1,K)-X(I,J,K-1)-X(I,J-1,K-1))*0.5                      1828       23
      Q2=(Y(I,J,K)+Y(I,J-1,K)-Y(I,J,K-1)-Y(I,J-1,K-1))*0.5                      1829       24
      Q3=(Z(I,J,K)+Z(I,J-1,K)-Z(I,J,K-1)-Z(I,J-1,K-1))*0.5                      1830       25
      AREA=SQRT(P1*P1+P2*P2+P3*P3)*SQRT(Q1*Q1+Q2*Q2+Q3*Q3)                       1831       26
      FLOW=FLOW+DENC*AREA*UC                                                    1832       27
      ARDEN=ARDEN+DENC*AREA                                                     1833       28
   50 CONTINUE                                                                  1834       29
      UINC=(FLOW-FLOWIN)/ARDEN                                                  1835       30
      DO 60 J=2,JT                                                              1836       31
      DO 60 K=2,KT                                                              1837       32
      U(I+1,J,K)=U(I,J,K)                                                       1838       33
      V(I+1,J,K)=V(I,J,K)-UINC                                                  1839       34
      W(I+1,J,K)=W(I,J,K)                                                       1840       35
```

```
 58  000A44I          DK(I+1,J,K)=DK(I,J,K)                                  1841      36
 59  000A94I          DE(I+1,J,K)=DE(I,J,K)                                  1842      37
 60  000AE4I       60 CONTINUE                                               1843      38
 61  000B14I          RETURN                                                 1844      39
 62                C-----LINK COEFF. MODIFICATIONS                           1845
 63  000B1AI       2  CONTINUE                                               1846      40
 64                C-----EAST OUT                                            1847
 65  000B1AI          I=IT                                                   1848      41
 66  000B26I          DO 200 J=2,JT                                          1849      42
 67  000B3AI          DO 200 K=2,KT                                          1850      43
 68  000B4EI          AE(I,J,K)=0.0                                          1851      44
 69  000B7AI     200  CONTINUE                                               1852      45
 70  000BAAI          RETURN                                                 1853      46
 71                C-----UPDATE UNSTEADY COEFF.                              1854
 72  000BB0I       3  CONTINUE                                               1855      47
 73  000BB0I          IF(DTT .NE. 0.0) GO TO 301                             1856      48
 74  000BC8I          DO 300 I=IS,IT                                         1857      50
 75  000BE0I          DO 300 J=JS,JT                                         1858      51
 76  000BF8I          DO 300 K=KS,KT                                         1859      52
 77  000C10I     300  APO(I,J,K)=0.0                                         1860      53
 78  000C84I          RETURN                                                 1861      54
 79  000C8AI     301  CONTINUE                                               1862      55
 80  000C8AI          DO 310 I=IS,IT                                         1863      56
 81  000CA2I          DO 310 J=JS,JT                                         1864      57
 82  000CBAI          DO 310 K=KS,KT                                         1865      58
 83  000CD2I          APO(I,J,K)=DENO(I,J,K)/DTT                             1866      59
 84  000D24I          UO(I,J,K)=U(I,J,K)                                     1867      60
 85  000D70I          VO(I,J,K)=V(I,J,K)                                     1868      61
 86  000DBCI          WO(I,J,K)=W(I,J,K)                                     1869      62
 87  000E08I          TMO(I,J,K)=TM(I,J,K)                                   1870      63
 88  000E54I          DKO(I,J,K)=DK(I,J,K)                                   1871      64
 89  000EA0I          DEO(I,J,K)=DE(I,J,K)                                   1872      65
 90  000EECI     310  CONTINUE                                               1873      66
 91  000F34I          RETURN                                                 1874      67
 92  000F3AI          END                                                    1875      68

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```
  1  0000001          SUBROUTINE WALVAL(PW,IS,IT,JS,JT,KS,KT,F)                        1876
  2  0000041          DIMENSION F(21,18,10)                                            1877
  3  0000041          COMMON                                                           1878
  4                  1/TUR/ SIGK,SIGE,CMU,C1,C2,CMU1,CMU2,E,CK,HINUM,SMNUM,ANV1(800),  1879
  5                  2 YN(800),YN1(800),SINX(800),SINY(800),SINZ(800),ANW1(800),      1880
  6                  3 YPLN(800),TAUN(800),I3C(800),J3C(800),K3C(800),IITY(800),      1881
  7                  4 TAUW(800),GEN(21,18,10),MC(21,18,10),IJLO(21,18,10),IITO       1882
  8              C-----ASSIGN WALL VALUES                                             1883
  9  0000041          DO 10 J=JS,JT                                                    1884        1
 10  0000281          DO 10 K=KS,KT                                                    1885        2
 11  0000401          F(IS-1,J,K)=PW*F(IS,J,K)                                         1886        3
 12  0000A21     10   F(IT+1,J,K)=PW*F(IT,J,K)                                         1887        4
 13  0001341          DO 20 I=IS-1,IT+1                                                1888        5
 14  0001521          DO 20 K=KS,KT                                                    1889        6
 15  00016A1          F(I,JS-1,K)=PW*F(I,JS,K)                                         1890        7
 16  0001CC1     20   F(I,JT+1,K)=PW*F(I,JT,K)                                         1891        8
 17  00025E1          DO 30 I=IS-1,IT+1                                                1892        9
 18  0002701          DO 30 J=JS-1,JT+1                                                1893       10
 19  00029A1          F(I,J,KS-1)=PW*F(I,J,KS)                                         1894       11
 20  0002FA1     30   F(I,J,KT+1)=PW*F(I,J,KT)                                         1895       12
 21  00038A1          DO 40 III=1,IITO                                                 1896       13
 22  00039E1          I=I3C(III)                                                       1897       14
 23  0003B21          J=J3C(III)                                                       1898       15
 24  0003C61          K=K3C(III)                                                       1899       16
 25  0003DA1          GO TO (1,2,3,4,5,6), IITY(III)                                   1900       17
 26  0004161     1    F(I,J+1,K)=PW*F(I,J,K)                                           1901       18
 27  0004781          GO TO 4C                                                         1902       19
 28  0004781     2    F(I,J-1,K)=PW*F(I,J,K)                                           1903       20
 29  0004E01          GO TO 4C                                                         1904       21
 30  0004E61     3    F(I+1,J,K)=PW*F(I,J,K)                                           1905       22
 31  0005481          GO TO 4C                                                         1906       23
 32  0005461     4    F(I-1,J,K)=PW*F(I,J,K)                                           1907       24
 33  0005801          GO TO 4C                                                         1908       25
 34  0005861     5    F(I,J,K+1)=PW*F(I,J,K)                                           1909       26
 35  0006161          GO TO 4C                                                         1910       27
 36  00061C1     6    F(I,J,K-1)=PW*F(I,J,K)                                           1911       28
 37  00067C1     40   CONTINUE                                                         1912       29
 38  0006941          RETURN                                                           1913       30
 39  00069A1          END                                                              1914       31

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APPROVAL

A COMPUTER CODE FOR THREE-DIMENSIONAL INCOMPRESSIBLE FLOWS USING NONORTHOGONAL BODY-FITTED COORDINATE SYSTEMS

By Y. S. Chen

The information in this report has been reviewed for technical content. Review of any information concerning Department of Defense or nuclear energy activities or programs has been made by the MSFC Security Classification Officer. This report, in its entirety, has been determined to be unclassified.

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